GD2 CAR-T toxicity

Clinical and preclinical safety, intracerebroventricular delivery, and transient organ-function change relevant to concomitant therapy

Literature synthesis of 852 curated records (2003–2026)

GD2 CAR-T toxicity, with emphasis on intracerebroventricular delivery

Synthesis of **852** curated records (2003–2026, 502 with open-access full text) on GD2 CAR-T safety. Domain detail is in `notes/`; this report states the answers, the evidence behind them, and what is not known.

1. Bottom line

1. **On-target/off-tumour toxicity has not materialised clinically.** Across every published GD2 CAR-T/CAR-NKT cohort, no patient has developed the painful neuropathy that anti-GD2 antibodies cause in most children, and no radiographic or clinical evidence of normal-CNS injury has been attributed to the CAR. Autopsy in one DIPG patient found CAR transcript and lymphocytic infiltrate confined to tumour, with tumour GD2 ≫ normal brain GD2 (PMID 35130560). The mechanism is antigen-density discrimination — and it is fragile: raising scFv affinity (14G2a E101K) caused **fatal encephalitis in low-GD2 cerebellar and basal brain regions** in mice (PMID 29180536).
2. **The dominant clinical toxicity is inflammation, and its location depends on the route.** IV infusion produces CRS (dose-limiting at 3×10^6/kg in DMG: three grade 4 events) plus ICANS in ~45% of patients. ICV infusion produces almost none of either but near-universal **TIAN** on first exposure.
3. **ICV administration without lymphodepletion is markedly less toxic systemically.** In the only internal comparison (NCT04196413 arm A; 11 IV vs 62 ICV infusions): **no DLT in 62 ICV infusions**, CRS in 33.9% of ICV infusions (mostly grade 1) vs 100% of IV infusions, and **zero ICANS after ICV** despite higher CSF cytokines (PMID 39537919). The 253-dose ICV B7-H3 experience agrees: commonest AEs headache/nausea/fatigue/fever, one grade 3 hydrocephalus, no ICANS, one DLT (intratumoural haemorrhage) (PMID 39775044).
4. **The ICV-specific hazards are mechanical and hardware-related, not biochemical**: raised ICP (documented 22 and 34 mmHg episodes reversing within minutes of CSF drainage), obstructive and communicating hydrocephalus, reservoir/catheter dependence, and intratumoural haemorrhage of ambiguous attribution.
5. **Transient organ-function change is a function of systemic inflammation, and is therefore an IV problem.** No GD2 CAR-T trial reports attributed hepatic, renal or electrolyte toxicity; one trial explicitly documents normal liver enzymes, bilirubin and electrolytes acutely and long-term (PMID 40420236). In general CAR-T populations, hepatic dysfunction accompanies **64% of grade 3–4 CRS** and grade 3–4 transaminase elevation occurs in **31–64%** of severe CRS, while AKI occurs in **~18.6%** pooled and **resolves in ~79% within a month**.
6. **Cytopenias in GD2 CAR-T trials are lymphodepletion, not CAR, toxicity** — reversible in a median 4 days in one series, and occurring before cell infusion in the CAR-NKT trial. ICV re-dosing omits lymphodepletion and produced essentially no cytopenia (one grade 1 lymphocytopenia in 21 patients over 253 doses).
7. **Drug-interaction risk is real but indirect and unmeasured in this therapy.** CRS-grade inflammation suppresses CYP3A4/2C19/1A2 (midazolam clearance −65% as

CRP rose 10→300 mg/L; PBPK simvastatin AUC ×2.4 at IL-6 100 pg/mL), and IL-6 blockade reverses it (sarilumab cut simvastatin AUC by **45%**). The best transient-cytokine analogue predicts **~28% CYP3A4 suppression for about a week**. Nobody has measured a DDI in a CAR-T patient. The therapy-specific interactions that are documented are different in kind: renal function governs fludarabine exposure during conditioning, and corticosteroids blunt the CAR itself.

2. IV versus ICV, quantified

Same product, same trial, same patients (PMID 39537919; first four patients in PMID 35130560):

	IV (11 infusions)	**ICV (62 infusions)**
Lymphodepletion	Cy/Flu required	none
Dose	1×10^6/kg (MTD) or 3×10^6/kg	10–30×10^6 flat
CRS	100% of infusions; **grade 4 in 3/8 at DL2 → DLT**	33.9%; 16 grade 1, 4 grade 2, 1 grade 3 (urosepsis)
ICANS	5/11 patients (grade 3 in 1)	**0**
TIAN	91–100% of patients; grade 3–4 in 5/11	71% of infusions; grade 3–4 in 8/62
Cytopenias	grade 3–4 expected (Cy/Flu)	not reported
Hepatic/renal/electrolyte AEs	none attributed (CRS mostly ≤grade 2 at MTD)	none attributed
Haemorrhage	1 grade 5 intratumoural (vascular anomaly)	—
DLTs	yes (CRS)	**none**
Blood cytokines	higher	lower
CSF cytokines	lower	**higher**

The dissociation — higher CSF cytokines yet no ICANS — is the most informative safety observation in the collection: ICANS tracks the blood/endothelial compartment, and moving the inflammatory response into the CSF does not reproduce it. It also explains why systemic organ-function change tracks the IV route.

3. Toxicity domains and evidence grade

Domain	**GD2-specific evidence**	**Strength**	**Notes**
CRS	6 clinical cohorts; near-universal after IV; grade 4 at DL2 in DMG; mild in 95% of GD2-CART01	**Direct, consistent**	Dose- and route-driven
TIAN	Defined in the GD2 DMG trial; 100% first ICV exposure; all reversible	**Direct**	Grading not prospectively validated
ICANS	5/11 in DMG IV; grade 3 in 4/54 on GD2-CART01 (rimiducid-reversed); 0/62 ICV	**Direct**	Distinguishable from TIAN only when tumour is focal
On-target/off-tumour	Absent in all cohorts; fatal in high-affinity preclinical model		Affinity/architecture dependent

Domain	GD2-specific evidence	Strength	Notes
		Direct-negative + preclinical hazard	
Haematologic	Near-universal grade 3–4 with Cy/Flu; pooled anaemia 97%, neutropenia ≥3 93%	**Direct but confounded**	Conditioning, not CAR
Hepatic	No attributed events; explicit normal labs in one trial	**Indirect** [CART-general]	Expect grade 1–2 ALT/AST if CRS ≥3
Renal/electrolyte	No attributed events; sodium moved deliberately for TIAN	**Indirect** [CART-general]	AKI ~18.6% pooled after systemic CAR-T
Coagulopathy	Not reported in GD2 cohorts	**Indirect**	Matters before neurosurgical procedures
Cardiopulmonary	Grade 4 CRS: pressors, pulmonary oedema, intubation; capillary leak 4/10 in one trial	**Direct**	CRS-mediated
IEC-HS/HLH	None reported; high ferritin without HLH	**Indirect**; ~3.5% after CAR-T generally	The scenario where hepatic dysfunction becomes clinical
Infection	Fungal and bacterial cases; urosepsis confounded 2 CRS events	**Direct**	Stacked immunosuppression
Drug interactions	**None measured**	**Mechanistic only**	See §5
Device/hardware	Reservoir required; 1 grade 3 hydrocephalus in ICV B7-H3; haemorrhage DLT	**Direct, ICV-specific**	Under-reported as a category
Myeloid/microglial amplification	CSF scRNA-seq myeloid states differ by route; grade ≥2 TIAN correlates with CSF MCP-1/CCL2, TNF-α, IL-10	**Direct but correlative**	Mechanism untested; see `notes/08`

3a. The myeloid axis behind TIAN

TIAN correlates with myeloid, not T-cell, readouts: grade ≥2 TIAN tracks CSF **MCP-1/CCL2**, TNF-α and IL-10, and grade ≥2 CRS tracks plasma MCP-1/CCL2. CSF single-cell RNA-seq shows route-dependent myeloid states — an interferon-responsive population at peak inflammation after ICV, versus phagocytic, lipid-metabolic, immune-suppressive DAM/MDSC-like states after IV and at late timepoints. Autopsy in the first DMG patient showed dense Iba1^{+}/CD163^{+} myeloid infiltration **within tumour** and resting, non-reactive microglia in unaffected cortex. Microglia are therefore best read as **amplifiers** downstream of tumour-restricted CAR engagement, not as GD2 targets; the causal claim remains untested. Detail in `notes/08`.

4. Transient organ-function changes that could affect other therapies

Magnitude, timing and reversibility are tabulated in `notes/05`. The decision-relevant summary:

- **Liver**: laboratory transaminase elevation is common and under-reported as an AE (58% laboratory vs 8% reported in one 148-patient trial); grade ≥3 is uncommon (0–8%) unless CRS is grade 3–4, where hepatic dysfunction reaches 64%; AST>ALT pattern suggests hypoperfusion; bilirubin usually spared outside IEC-HS. Resolves with CRS.
- **Kidney**: AKI ~18.6% pooled, onset day +1 to +28 (mode day +7), mostly mild, **79% recovered within a month**; predictors are CRS, ICANS grade ≥3, pre-existing CKD; associated with worse survival even when reversible.
- **Electrolytes/fluid**: hyponatraemia from resuscitation/SIADH; **iatrogenic sodium shifts from hypertonic saline given for TIAN**; hypophosphataemia and hypokalaemia with inflammation; tumour lysis with high burden.
- **Marrow and haemostasis**: near-universal transient cytopenias with Cy/Flu, prolonged in a minority (ICAHT, predictable by CAR-HEMATOTOX); CRS coagulopathy common with low absolute bleeding rates but relevant to reservoir/EVD procedures during the inflammatory window.
- **Consequences for concomitant therapy**: nephrotoxins and contrast should be avoided in the CRS window and renally cleared drugs dosed on the current creatinine; myelosuppressive bridging or concurrent chemotherapy compounds ICAHT and should be sequenced rather than overlapped; neurosurgical procedures should be planned around coagulation parameters; steroid-sparing is preferred both for efficacy and to avoid steroid-mediated interactions.
- **After ICV dosing without lymphodepletion, none of this applies to a measurable degree** — the main systemic constraint on co-administered therapy is removed.

5. Drug-interaction assessment (explicitly graded)

What is established [mechanistic-PK]: acute inflammation suppresses CYP-mediated clearance in a CRP/IL-6-dependent way (midazolam CL −65% over CRP 10→300 mg/L; postoperative CYP3A4 −61%, CYP1A2 −53%, CYP2C19 −58%; ciclosporin Css ×3.6 at IL-6 peak; darunavir CL/F halved in COVID-19 with IL-6 the sole covariate). IL-6 blockade reverses it (sarilumab: simvastatin AUC −45%, Cmax −46%; sirukumab: midazolam AUC −30–35%). For a **transient** cytokine spike the modelled effect is **<2-fold, ~28% CYP3A4 suppression, lasting ~1 week** (blinatumomab analogue).

What is not established: any measured interaction in a CAR-T, GD2 CAR-T, or ICV CAR-T patient. No CYP-probe study exists in CRS. The magnitude in CAR-T CRS is likely to exceed the blinatumomab model (higher, longer cytokine elevation) but this is inference.

Therefore: treat CRS as a transient, bidirectional perturbation of CYP3A4-dominant clearance — levels up during CRS, down after IL-6 blockade — and manage it with therapeutic drug monitoring for narrow-index substrates (azoles, calcineurin/mTOR inhibitors, anticonvulsants, anticoagulants) rather than with pre-emptive dose changes. Do not claim a GD2-specific interaction.

Therapy-specific interactions that are documented, and matter more in practice:

- **Renal function → fludarabine exposure** (eGFR a significant covariate on clearance in the first CAR-T-specific population-PK model, PMID 41471106): a transient creatinine rise before conditioning changes lymphodepletion intensity, which changes both efficacy and toxicity.
- **Corticosteroids → CAR activity**: a pharmacodynamic interaction the trials manage explicitly (steroid-free at enrolment; response scoring deferred ≥7 days after steroids).
- **Anti-cytokine agents → infection and marrow recovery**: anakinra, tocilizumab, siltuximab, and, for IEC-HS, etoposide/ruxolitinib/emapalumab stack with lymphodepletion.
- **Hypertonic saline** for TIAN as a deliberate electrolyte intervention interacting with volume- and sodium-sensitive drugs.

6. What would change these conclusions

1. Serial laboratory and cytokine data published for an **ICV-only** cohort (arms B/C of NCT04196413) — would convert "no reported organ toxicity" into a measured claim.
2. A **CYP-probe or population-PK study around CRS and tocilizumab** in CAR-T recipients.
3. Prospective **TIAN grading** with the type 1/type 2 distinction operationalised, since BrainChild-03 could not classify most events retrospectively (PMID 41798119).
4. A larger ICV denominator for **intratumoural haemorrhage** attribution.
5. A circulating **neuro-injury biomarker** (e.g. NFL) in GD2 CAR-T patients, to test the on-target/off-tumour null result with something more sensitive than examination.
6. Any clinical use of a **higher-affinity or armoured GD2 CAR** — the preclinical hazard is specific to affinity/architecture and the current safety record should not be assumed to transfer.

7. Practical distillation

- The ICV route is the main safety lever: it converts a systemic-inflammatory toxicity problem into an intracranial-pressure and hardware problem, and removes the lymphodepletion-driven organ and marrow toxicity that otherwise constrains concomitant treatment.
- Every ICV programme needs a reservoir, ICP measurement, an osmotic-therapy and CSF-drainage pathway, and staff who will access the reservoir within minutes — not a pharmacological algorithm alone.
- Grade CRS, ICANS and TIAN separately at every infusion; fever after ICV dosing is expected.
- Assume liver enzymes and creatinine will move transiently whenever CRS reaches grade 3, monitor them daily in the acute window, and use those values (not admission values) for co-medication dosing.
- Handle drug interactions as monitoring problems in a moving inflammatory window, and say plainly that the GD2-specific interaction data do not exist.

00 — Overview: what "GD2 CAR-T toxicity" means, and why the route is the question

The three toxicity engines

GD2 CAR-T toxicity is not one thing. Three mechanisms produce overlapping syndromes and are treated differently:

1. **Systemic cytokine release** — CAR T cells activate, myeloid cells amplify (IL-1, IL-6, MCP-1), and the patient develops fever, hypotension, hypoxia, endothelial activation, transient organ dysfunction. Downstream of this sit CRS, IEC-HS/HLH, coagulopathy, hepatic and renal changes, and most of the drug-interaction risk. Dose- and route-dependent.
2. **Inflammation at the tumour site** — in CNS tumours the antitumour response happens inside a closed box. Oedema, mass effect, CSF-outflow obstruction and local electrophysiologic dysfunction produce **TIAN** (tumour inflammation-associated neurotoxicity), which is distinct from ICANS: symptoms map to tumour neuroanatomy, and the treatment is CSF drainage / hypertonic saline / anti-cytokine therapy rather than steroids alone (Majzner 2022, PMID 35130560).
3. **On-target / off-tumour engagement of normal GD2** — GD2 is expressed on peripheral nerve, some CNS neurons and melanocytes. With antibodies this reliably produces neuropathic pain; with CAR T cells it has been essentially absent clinically at standard affinity, but is fatal in preclinical models when scFv affinity is raised (Richman 2018, PMID 29180536).

A fourth, mostly non-CAR effect matters practically: **lymphodepletion**. Cyclophosphamide/fludarabine causes the grade 3–4 cytopenias reported in most GD2 CAR-T trials, and ICV re-dosing is typically given without lymphodepletion — which is a large part of why ICV looks safer.

Why the route dominates the answer

For H3K27M+ DMG the same product has now been given both ways in the same trial (NCT04196413, arm A), so the comparison is internal rather than cross-trial (Monje 2025, PMID 39537919; 11 patients, 11 IV and 62 ICV infusions):

Toxicity	IV (DL1 1×10^6/kg, DL2 3×10^6/kg)	ICV ($10–30\times10^6$ flat, no lymphodepletion)
CRS	100% of infusions; grade 4 in 3/8 at DL2 (dose-limiting)	33.9% of infusions; 16/21 grade 1, one grade 3 (in urosepsis)
ICANS	5/11 patients (1 grade 3)	0 of 62 infusions
TIAN	100% DL1, 87.5% DL2; grade 3–4 in 5/11	71% of infusions; grade 3–4 in 8/62
Dose-limiting toxicity	Yes — CRS defined DL1 as MTD	None across 62 infusions

The route changes which compartment inflames. CSF cytokines are **higher** after ICV; blood cytokines and CRS are higher after IV. Systemic organ-function toxicity and ICANS track the blood compartment; TIAN and intracranial pressure track the CSF/tumour compartment.

Evidence taxonomy used in these notes

Claims are labelled by source strength, because the temptation in this field is to import CD19-lymphoma numbers into a 30-patient pediatric CNS experience:

- **[GD2-clinical]** — from a GD2 CAR-T/CAR-NKT trial. Small n, phase 1, mostly pediatric.
- **[ICV-clinical]** — from an ICV/locoregional CAR-T trial of any target (GD2, B7-H3, IL13Rα2, EGFR806, bivalent EGFR/IL13Rα2). Route-specific but often not GD2-specific.
- **[CART-general]** — from CD19/BCMA/CD22 experience. Large n, adult-dominant, systemic route; transfers for mechanism and for lymphodepletion-related toxicity, over-predicts for ICV.
- **[GD2-antibody]** — dinutuximab/naxitamab/hu14.18/3F8. Defines the on-target-off-tumour ceiling for the antigen but not for the modality.
- **[preclinical]** — mouse/xenograft. Predicted the brainstem-oedema/hydrocephalus problem and the affinity/CNS-toxicity relationship, so it is not decorative here.
- **[mechanistic-PK]** — inflammation/cytokine effects on drug-metabolising enzymes, mostly from rheumatoid arthritis, surgery, sepsis, COVID-19 and PBPK modelling. **No GD2 CAR-T study has measured a drug–drug interaction**; this literature supports plausibility and monitoring, not a quantified GD2-specific interaction.

What the collection says in one paragraph

At standard affinity and clinically used doses, GD2 CAR T cells have not produced the on-target off-tumour neurotoxicity that preclinical high-affinity constructs predicted; the dominant clinical toxicities are CRS (IV, dose-limiting), TIAN (both routes, near-universal in DIPG but reversible with a defined algorithm), lymphodepletion-driven cytopenias, and — in neuroblastoma — grade 3 ICANS in a minority, aborted by iCasp9 activation. ICV administration without lymphodepletion largely removes systemic CRS, ICANS and measurable hepatic/renal/haematologic derangement, and concentrates risk into intracranial pressure, hydrocephalus, device-related events and intratumoural haemorrhage. Transient organ-function change — and therefore concomitant-drug risk — is essentially a function of systemic inflammation: it is real after IV infusion (hepatic dysfunction in up to 64% of grade 3–4 CRS, AKI ~18% pooled after systemic CAR-T, resolving in ~79% within a month) and near-absent after ICV dosing.

01 — GD2 CAR-T clinical toxicity, trial by trial

All datasets below are phase 1 or phase 1/2, mostly pediatric, single- or few-centre. Toxicity is reported per patient unless stated per infusion.

H3K27M+ DMG / DIPG — GD2-CAR, 4-1BB, iCasp9 (Stanford, NCT04196413)

First four patients, IV at DL1 with a second dose ICV (Majzner 2022, PMID 35130560) [GD2-clinical, ICV-clinical]

- CRS after IV: grade 1 (patient 1), grade 2 (patient 2, fluid-responsive hypotension), grade 1 (patient 3), grade 3 in the eIND spinal DMG patient (vasopressor-requiring, day +6, tocilizumab + 3 days steroids).
- TIAN after IV in all four: worsening of existing deficits mapping to tumour location — cranial nerve palsies, trismus with T2/FLAIR signal in the trigeminal nucleus, transient hearing loss (2 days), increased ataxia.
- Patient 1: day +9 acute deterioration (hemiplegia, extensor posturing) with **ICP 22 mmHg**; 10 mL CSF drained via Ommaya → return to baseline "within minutes"; anakinra started; MRI showed increased pontine oedema.
- After ICV re-dosing (30×10^6, no lymphodepletion): fever 40 °C within 24 h, then at ~48 h somnolence and new third-nerve palsy with **ICP 34 mmHg** and obstructive hydrocephalus from fourth-ventricle compression; continuous CSF drainage + hypertonic saline + anakinra + steroids; hydrocephalus resolved in 2 days, steroids stopped by day +6.
- The 5-year-old's ICV dose (12.9×10^6) caused **only mild headache**, no anti-cytokine agents, no steroids — i.e. ICV toxicity is not obligatory.
- Spinal DMG patient's first-ever ICV infusion (50×10^6 after intensified lymphodepletion): persistent fever, grade 3 encephalopathy graded as **grade 4 ICANS**, EEG with diffuse slowing and triphasic waves, no MRI change in normal brain; ICANS vs TIAN indistinguishable given thalamic/hypothalamic tumour invasion; treated with anakinra, siltuximab, IV **and ICV** corticosteroids, plus oral dasatinib to dampen CAR T-cell activity; resolved within 4 days.
- **No on-target off-tumour toxicity**: no painful neuropathy in any patient; autopsy in patient 1 showed lymphocytic infiltration and CAR transcript confined to tumour, not normal cortex, with much higher GD2 in tumour than normal brain.
- Serum LDH rose in all patients and tracked inflammation. Anakinra was detectable in CSF after IV dosing (i.e. IL-1 blockade reaches the compartment that matters).

Full arm A, 11 treated patients, 11 IV + 62 ICV infusions (Monje 2025, PMID 39537919) [GD2-clinical, ICV-clinical]

- **DLTs**: none at IV DL1; **three grade 4 CRS at IV DL2 (3×10^6/kg)** → DL1 (1×10^6/kg) is the maximally tolerated IV dose. Grade 4 CRS presentations: multi-pressor hypotension (1), pulmonary oedema with BiPAP/CPAP (1), intubation (1).

- **Zero DLTs across 62 ICV infusions.** CRS in 33.9% of ICV infusions (16 grade 1, 4 grade 2, 1 grade 3 — the grade 3 in the setting of urosepsis). **No ICANS after any ICV infusion**, despite higher CSF cytokines than after IV.
- TIAN in 91% after IV and 100% after first ICV infusion, 44/62 (71%) of ICV infusions overall; grade typically falls with successive infusions; **all TIAN reversed** under the management algorithm; no TIAN-attributed DLT. One sDMG patient had high-grade communicating hydrocephalus at peak inflammation.
- One **grade 5 intratumoural haemorrhage** in a region of known intratumoural vascular anomaly — contextualised against ~20% symptomatic intratumoural haemorrhage in DIPG by 12 months from diagnosis; causal attribution unresolved.
- One **grade 3 sensory neuropathy** (stocking-glove) after a first infusion, attributed in part to vitamin B6 toxicity and resolving after B6 cessation — the only peripheral-nerve event; **no antibody-type transient painful neuropathy in any patient**.
- Toxicity correlates: grade ≥2 CRS ↔ higher plasma MCP-1 (CCL2) and IL-2; grade ≥2 TIAN ↔ higher CSF MCP-1, IL-10, TNF.
- iCasp9 was never triggered (AP1903 not administered to any arm A patient).
- Trial design consequence: arms B and C now test **ICV-only** dosing with and without lymphodepletion — the sponsor's own read is that the systemic route is the toxic one.

GD2-expressing CNS tumours — GD2.CART ± constitutive IL-7 receptor (Baylor, NCT04099797)

(Bhat/Ahmed 2024, PMID 38771986) [GD2-clinical]

- 11 patients, IV, standard Cy/Flu lymphodepletion at all dose levels. **No DLTs.**
- GD2.CART alone (no C7R): no toxicity at all, and no durable benefit.
- C7R-GD2.CART: **TIAN grade 1 in 7/8 (88%)**, controlled with anakinra; CRS in 6/8 (75%), grade 1 in all but one, associated with rising IL-6 and IP-10.
- One patient with grade 2 CRS had heavy prior chemotherapy; subsequent DL2 patients received **split dosing** ($1.5\times10^7/m^2$ 5–7 days apart) and no further grade ≥2 CRS occurred — dose fractionation as a concrete mitigation.

Neuroblastoma — GD2-CART01 (third-generation CD28/OX40, iCasp9; Bambino Gesù)

Interim (Del Bufalo 2023, PMID 37018492) and final (Quintarelli/Locatelli 2025, PMID 40841488) [GD2-clinical]

- 27 → 54 children (35 on trial + 19 hospital exemption). Dose levels 3, 6, 10×10^6 CAR^+/kg; **no DLTs**; MTD 10×10^6/kg.
- CRS in 20/27 (74%), **mild in 19/20 (95%)**.
- **Grade 3 ICANS in 4 children, "rapidly controlled" by rimiducid activation of iCasp9** — the only clinical dataset in which the GD2 suicide switch has actually been used therapeutically, at the cost of eliminating the product.
- No new safety signals with longer follow-up; persistence ≥12 months in 64%.

Neuroblastoma — other GD2 CAR-T products

- **4SCAR-GD2 (CD28/4-1BB/CD3ζ-iCasp9), 10 patients** (Yu 2022, PMID 34724115) [GD2-clinical]: no treatment-related deaths. Grade 3–4 events were **haematologic and attributed to Cy/Flu**, fully reversible without treatment in a median of 4 days (range 0–22). CRS grade 1–2 in 9/10, acute capillary leak in 4/10, **neuropathic pain grade 1–2** transient. Peak ferritin 3417 µg/L, IL-6 1000 pg/mL, IL-10 8014 pg/mL — i.e. biochemically real inflammation with clinically mild disease.
- **Second-generation GD2-CAR with escalating lymphodepletion, 12 patients** (Straathof 2020, PMID 33239386) [GD2-clinical]: grade 2–3 CRS in 2/6 at the highest dose after Flu/Cy; **no on-target off-tumour toxicity**; no objective responses at day +28.
- **ALLO_GD2-CART01, donor-derived, 5 children** (2025, PMID 39815015) [GD2-clinical]: **all five had grade 2–3 CRS**, one grade 2 neurotoxicity, and **moderate acute GVHD in 4/5** — allogeneic products add a toxicity class that autologous GD2 CAR-T does not have.
- **GD2-CAR.15 NKT cells (IL-15 co-expressing), 12 children** (Heczey 2020 PMID 33046868; 2023 PMID 37188782) [GD2-clinical]: **no DLTs**, MTD not reached; grade 3–4 haematologic events occurred before CAR-NKT infusion and were attributed to Cy/Flu; one grade 2 CRS resolved with tocilizumab.
- **PSMA + GD2 CAR-T for refractory glioma, 6 patients** (2025, PMID 40420236) [GD2-clinical]: CRS grade 1 in 3; **no central neurotoxicity**; haematologic toxicity in 5/6 (grade ≥2 thrombocytopenia 4/6, neutropenia 3/6), 3 infections associated with marrow suppression; explicitly **"no significant abnormalities in laboratory values, including liver enzymes, bilirubin levels, or electrolytes"**, and none on long-term follow-up.
- **$GD2^+$ medulloblastoma** (2024, PMID 38551501) is preclinical/translational with iCasp9 rescue demonstrated in mice (CAR^+ cells 76.3% → 9.4% after AP1903) and murine NFL used as a neuroaxonal injury readout.

Pooled view

Systematic review/meta-analysis of 8 GD2 CAR-T neuroblastoma studies, 146 patients (2026, PMID 41530803) [GD2-clinical, pooled]:

- Most common any-grade AE: **anaemia 97.4%** (81.5–100). Most common grade ≥3: **neutropenia 93.5%** (72.7–100), then thrombocytopenia, anaemia, CRS.
- Efficacy: CR 39.6%, PR 15.8%.
- Interpretation caveat: these pooled cytopenia rates are dominated by **lymphodepleting chemotherapy**, not by GD2 target engagement, and the confidence intervals reach 100% because the studies are tiny. The correct reading is "expect near-universal transient marrow suppression whenever Cy/Flu is used", which is precisely the component ICV re-dosing omits.

Cross-trial signal for the ICV question

- Every GD2 CAR-T trial that gave cells **intravenously with lymphodepletion** reported CRS in most patients and grade 3–4 cytopenias.

- The only trial that gave cells **ICV without lymphodepletion** (Stanford arm A second and subsequent doses) reported no DLT in 62 infusions, no ICANS, mostly grade 1 CRS, and toxicity that was overwhelmingly intracranial and mechanical rather than systemic.
- No GD2 CAR-T trial has reported clinically significant transaminitis, hyperbilirubinaemia, AKI or electrolyte derangement as an attributed toxicity; where laboratory values were explicitly reported normal, the route was ICV or the CRS was low grade.

02 — Preclinical toxicity and on-target / off-tumour GD2 biology

The affinity trap

Richman 2018 (PMID 29180536) [preclinical] is the reference hazard for this antigen. Variants of the 14G2a scFv engineered for improved stability/affinity, including **E101K**, gave better antitumour activity against GD2⁺ neuroblastoma xenografts **and lethal CNS toxicity**: extensive CAR T-cell infiltration and proliferation in brain with neuronal destruction, an encephalitis localised to **cerebellum and basal brain regions that express only low amounts of GD2**.

Three conclusions that constrain every subsequent GD2 CAR design:

1. Toxicity was not driven by high-antigen tissue — it appeared where GD2 is low, i.e. affinity raised the CAR above the antigen-density threshold that normally protects normal tissue.
2. Antibody safety does not transfer to CAR safety at the same epitope: dinutuximab is approved, yet the same binder in a high-affinity CAR was fatal in mice.
3. The clinically used GD2 CARs deliberately keep **standard 14g2a affinity**, which is why the density-discrimination argument (Majzner 2022 autopsy: tumour GD2 ≫ normal brain GD2) is load bearing rather than rhetorical. Affinity-tuning literature in the collection (PMID 34966954) makes the same point generically.

Brainstem inflammation predicted the clinical problem

Preclinical H3K27M DMG models with GD2-CAR T cells (Mount 2018, PMID 29662203) [preclinical] showed clearance of tumour **and** brainstem oedema in a fraction of mice, progressing to obstructive hydrocephalus because of the neuroanatomy. This is the direct ancestor of the clinical TIAN management algorithm: the Stanford protocol excluded bulky thalamic and cerebellar tumours, mandated an Ommaya for ICP monitoring in DIPG, and pre-specified CSF removal, hypertonic saline, anti-cytokine agents and corticosteroids (Majzner 2022, PMID 35130560).

So the preclinical signal that mattered was **not** on-target off-tumour neurotoxicity — it was on-tumour inflammation in a space-limited compartment.

What normal-tissue GD2 does with antibodies

[GD2-antibody] evidence sets the expectation for target engagement in normal nerve:

- Anti-GD2 antibodies cause **dose-limiting on-target/off-tumour neuropathic pain** in most children (ACCELERATE Paediatric Strategy Forum, PMID 41240535; mechanism review PMID 34884452).
- Rat models reproduce anti-GD2-induced allodynia and it is reducible by pretreatment (DFMO, PMID 32697811); dinutuximab variants engineered for reduced neuropathy

induction exist (PMID 35792784); complement/Fc engineering and IgA formats are being used to dissociate efficacy from pain (PMID 37479484).
- FAERS disproportionality analysis of dinutuximab/dinutuximab β/naxitamab (PMID 38153627) confirms a broad real-world toxicity spectrum including signals absent from labels.

The clinical GD2 CAR-T experience does **not** reproduce this: no painful neuropathy in the Stanford DMG cohort (Monje 2025), none attributed in Straathof 2020, only transient grade 1–2 neuropathic pain in 4SCAR-GD2 (PMID 34724115). Plausible reasons — different effector mechanism (no complement activation on nerve), lower and non-repeated peak target occupancy on nerve, and antigen-density thresholding — but this is inference, not a demonstrated mechanism.

Preclinical work on making it safer

- **iCasp9/AP1903** eliminates activated CAR T cells preferentially and is built into GD2-CART01, 4SCAR-GD2 and the Stanford construct (PMID 25389405; medulloblastoma model PMID 38551501 showing CAR^+ fraction 76.3% → 9.4% after AP1903).
- **Costimulation-only / logic-gated designs** to avoid activation on normal tissue (PMID 28341563, ALK AND-gate PMID 36396552).
- **Microenvironment-actuated and density-dependent ("velcro-like") CARs** claiming efficacy without toxicity (PMID 39841845, PMID 41005308); sonogenetic spatial control (PMID 40179881).
- Murine **neurofilament light chain** as a circulating neuroaxonal injury biomarker in CAR-T models (PMID 38551501) — the obvious translational readout for a "did we injure nerve?" question that clinical GD2 trials currently answer only by examination.

Transfer caveat

Xenograft models are lymphodepleted, immunodeficient and small; they cannot model human CRS (no functional myeloid compartment), and the mouse brainstem has different compliance and CSF dynamics. They have proven predictive for **spatial** risk (where inflammation is dangerous) and for **affinity/architecture** risk, and predictive for nothing about hepatic, renal or drug-interaction behaviour.

03 — The ICV route: what it removes, what it adds, and TIAN

What ICV administration removes

From the internal IV-vs-ICV comparison in NCT04196413 arm A (Monje 2025, PMID 39537919) [GD2-clinical, ICV-clinical]:

- **Dose-limiting CRS**: grade 4 CRS was the IV DLT at 3×10^6/kg; across 62 ICV infusions there was no DLT and CRS was absent in 41/62 infusions, grade 1 in 16.
- **ICANS entirely**: 0/62 ICV infusions, versus 5/11 patients after IV. This was explicitly unexpected because CSF cytokines are higher after ICV — consistent with ICANS being driven by systemic inflammation and endothelial/BBB injury from the blood side rather than by CSF cytokine concentration per se.
- **Lymphodepletion**, and with it the grade 3–4 cytopenias, febrile neutropenia and infection risk that dominate the AE tables of IV GD2 CAR-T trials. ICV re-dosing in arm A used no lymphodepletion.
- **Measurable systemic organ derangement**: no attributed hepatic, renal or electrolyte toxicity in the ICV experience; in the ICV-only B7-H3 trial the sole haematologic AE recorded during the DLT window was one grade 1 lymphocytopenia (Vitanza 2025, PMID 39775044).

What ICV administration adds or keeps

1. **TIAN** — near-universal on first ICV exposure in DIPG, then attenuating.
2. **Intracranial pressure and CSF-outflow risk** — the dangerous, minutes-to-hours toxicity.
3. **Hardware** — Ommaya/Rickham reservoir or CNS catheter as an eligibility requirement, with its own infection, malposition and revision burden; ICV trials require a catheter before enrolment (Vitanza 2025).
4. **Intratumoural haemorrhage** — the single DLT in ICV B7-H3 (grade 4, PMID 39775044) and the one grade 5 event in GD2 arm A (PMID 39537919). Both cohorts argue background DIPG haemorrhage rates (~20% by 12 months) make attribution ambiguous; neither can exclude a contribution.
5. **Repeat-dosing exposure** — ICV programmes give many infusions (arm A median 4, range 1–17; BrainChild-03 median 9, range 1–81 over up to 37.5 months), so cumulative and late toxicity, and anti-CAR immunity, become the relevant questions rather than single-infusion peak toxicity.

TIAN: definition, incidence, management

TIAN is **not** ICANS. It is neurotoxicity attributable to inflammation at the tumour site, graded by CTCAE, and split into two types (Majzner 2022, PMID 35130560; retrospective application in BrainChild-03, Ronsley 2026, PMID 41798119):

- **Type 1 — mechanical**: oedema, mass effect, obstruction of CSF flow → raised ICP, obstructive or communicating hydrocephalus, herniation risk. Treatment is **CSF removal via reservoir/EVD, hypertonic saline**, then anti-cytokine agents and corticosteroids.
- **Type 2 — electrophysiologic**: transient worsening of existing focal deficits without a mechanical cause (cranial-nerve dysfunction, trismus, hearing loss, ataxia, weakness), mapping to tumour neuroanatomy and often with corresponding T2/FLAIR change.

Incidence:

Cohort	Route	TIAN
GD2-CAR DMG arm A (n=11)	IV then ICV	91% of patients after IV; 100% after first ICV; 44/62 (71%) of ICV infusions; grade 3–4 in 8/62 ICV, 5/11 after IV; all reversible, no DLT
C7R-GD2.CART CNS tumours (n=8 C7R)	IV	grade 1 in 7/8 (88%), anakinra-controlled
B7-H3 CAR-T DIPG, BrainChild-03 (n=21)	ICV	16/21 (76%) met TIAN criteria; 49/152 infusions (32%); 34 grade 1, 14 grade 2, 1 grade 3; 1 needed CSF diversion; 13 had worsening pre-existing deficits

Practical points that recur across these cohorts:

- Symptoms are **predictable from tumour location** — pontine tumours give cranial-nerve and bulbar dysfunction; thalamic/cerebellar bulk was an exclusion criterion on the GD2 trial precisely because the preclinical models showed lethal hydrocephalus there.
- The therapeutic window is **short**: ICP 22 and 34 mmHg episodes in Majzner 2022 resolved "within minutes" of CSF drainage. This is why reservoir access, not pharmacology, is the primary intervention, and why ICP monitoring capability is an eligibility requirement rather than a nicety.
- **Fever after an ICV infusion is expected** and is not by itself CRS: BrainChild-03 explicitly interpreted fever as evidence of local immune activation (PMID 39775044).
- TIAN grade **falls with successive infusions** in both GD2 and B7-H3 cohorts — the first exposure is the risky one.
- Corticosteroids are used, but the algorithms put CSF diversion and IL-1/IL-6 blockade first, partly to protect CAR T-cell activity. Anakinra crosses into CSF after IV dosing (measurable CSF IL-1RA, PMID 35130560); ICV dexamethasone has been used in a refractory case (PMID 41479887 documents early intrathecal dexamethasone/ methotrexate for ICANS), and Majzner 2022 used ICV steroids plus dasatinib as an off-label CAR "off-switch".

Comparator locoregional CAR-T trials (route-specific safety)

- **ICV B7-H3, DIPG, BrainChild-03 arm C** (PMID 39775044): 21 patients, **253 ICV doses**, up to 10×10^7 cells/dose declared the maximally tolerated dose regimen;

commonest AEs headache 81%, nausea/vomiting 81%, fatigue 62%, fever 57%, hiccups 29%; mostly grade 1–2; **hydrocephalus in 1 (grade 3)**; no ICANS; single DLT = grade 4 intratumoural haemorrhage. Multiyear repeat dosing was tolerable.

- **ICV B7-H3 for non-pontine DMG / other pediatric CNS tumours** (PMID 42503899) and **intracranial B7-H3 for recurrent GBM** (PMID 42562965) extend the same safety pattern.
- **Intraventricular/intrathecal bivalent EGFR + IL13Rα2 CAR-T in recurrent GBM** (PMID 38480922, PMID 40451950) and **CARv3-TEAM-E** — locoregional CNS CAR-T with early inflammatory neurotoxicity managed without dose-limiting systemic toxicity.
- **EGFR806-CAR locoregional in pediatric CNS tumours** (PMID 40070357) and **IL13Rα2 intraventricular** (BrainChild-01) complete the picture: across targets, the locoregional route consistently converts the toxicity problem from systemic-inflammatory to intracranial-mechanical.
- **Lymphodepletion in solid/brain-tumour CAR-T** (PMID 39542517) reviews whether the systemic conditioning that drives most measurable organ toxicity is even needed for locoregional delivery — the question arms B/C of NCT04196413 are now testing prospectively.

Route-specific unknowns

- Whether repeated ICV dosing causes cumulative ependymal/leptomeningeal injury, or CSF-flow impairment, is unmeasured; nobody reports longitudinal CSF dynamics.
- Device-related infection rates specific to CAR-T ICV programmes are not reported separately from general reservoir experience; the collection does contain post-CAR-T CNS infection reports (e.g. carbapenem-resistant Klebsiella CNS infection, PMID 41511674).
- Whether ICV dosing can cause CRS at higher cell doses than tested ($>10\times10^7$ per dose) is unknown — the grade 3 CRS in arm A ICV occurred with concurrent urosepsis, and CSF-to-blood cell egress is documented.

04 — CRS, ICANS and IEC-HS/HLH

CRS mechanism relevant to GD2 products

CRS is myeloid-amplified, not purely T-cell-derived: monocyte/macrophage **IL-1 and IL-6** drive the syndrome and IL-1 blockade separates CRS from neurotoxicity in models (Norelli 2018, PMID 29808007; Giavridis 2018, PMID 29808005). This is the mechanistic justification for anakinra being used alongside or ahead of tocilizumab in the GD2 CNS trials, and for anakinra's usefulness in TIAN, since IL-1RA reaches the CSF (PMID 35130560).

Grading is ASTCT consensus (Lee 2019, PMID 30592986); pediatric management follows dedicated guidelines (PMID 30082906). Endothelial activation and BBB disruption underlie severe neurotoxicity (Gust 2017 and related records in the `neurotoxicity` category), which is why endothelial-activation indices (EASIX and variants) recur in the collection as predictors of coagulopathy and ICANS.

Incidence in GD2 products

Product / route	CRS	Severe CRS	Neurotoxicity
GD2-CAR DMG, IV DL1 (n=3)	3/3	none	ICANS 1 (grade 2); TIAN 3/3
GD2-CAR DMG, IV DL2 (n=8)	8/8	**3 grade 4 (DLT)**	ICANS 4/8 (1 grade 3); TIAN 7/8
GD2-CAR DMG, ICV (62 infusions)	21 (33.9%)	1 grade 3 (with urosepsis)	**ICANS 0**; TIAN 44 (71%)
GD2-CART01 neuroblastoma (n=27)	20 (74%)	1/20 non-mild	**grade 3 ICANS in 4/54 overall, reversed by rimiducid/iCasp9**
C7R-GD2.CART CNS (n=8)	6 (75%)	1 grade 2	TIAN grade 1 in 7/8
4SCAR-GD2 neuroblastoma (n=10)	9 (90%)	none >grade 2	grade 1–2 neuropathic pain
GD2-CAR.15 NKT (n=12)	1 (grade 2)	none	none
ALLO_GD2-CART01 (n=5)	5	grade 2–3 in all	1 grade 2

Reading: CRS is essentially universal with IV administration of an active GD2 product, but severity is dose-driven and mostly low grade; the two high-severity signals are **IV dose escalation** (grade 4 CRS at 3×10^6/kg in DMG) and **allogeneic products** (grade 2–3 CRS in 5/5 plus GVHD).

Anti-cytokine therapy actually used

- **Tocilizumab** (IL-6R) — first line for CRS in every GD2 trial.
- **Anakinra** (IL-1R) — used early and pre-emptively in the CNS trials, including for TIAN; the only agent with documented CSF penetration in this setting.
- **Siltuximab** (IL-6) — used in the refractory grade 4 ICANS case (PMID 35130560).

- **Corticosteroids** — IV and, in one case, ICV.
- **Dasatinib** — off-label pharmacological "pause switch" on CAR T-cell signalling (PMID 35130560); itacitinib (JAK1) has been evaluated for CRS cytokine reduction in GD2 CAR-T contexts (PMID 32998963).
- **iCasp9 + rimiducid/AP1903** — definitive but destroys the product; used clinically in 4 children with grade 3 ICANS on GD2-CART01 (PMID 40841488).

IEC-HS / carHLH

[CART-general]. Immune effector cell-associated HLH-like syndrome (IEC-HS, formerly carHLH) is the hyperinflammatory extreme: overall incidence after CAR-T ~3.5%, with ferritin elevation as the most reliable feature and **grade 3–4 ALT/AST elevation as part of the diagnostic picture** (PMID 41345813 review; ASTCT/consensus description PMID 36906275; carHLH after CD19 PMID 34263927; CD22 HLH-like CRS variant PMID 34525183; management PMID 41779733, PMID 41311365).

Why it matters for GD2 CAR-T:

- GD2 CAR-T trials report high ferritin (peak 3417 µg/L in 4SCAR-GD2) without formal HLH, so the syndrome is on the differential whenever fever persists with rising ferritin, falling fibrinogen and transaminitis.
- IEC-HS is **the** CAR-T scenario in which hepatic dysfunction becomes clinically meaningful rather than a laboratory finding, and where etoposide, ruxolitinib, anakinra or emapalumab (IFN-γ blockade, PMID 35917457, PMID 42136039) enter the picture — each with its own interaction and myelosuppression profile.
- H-Score and HLH-2004 criteria perform poorly in this population (PMID 35045791); use the IEC-HS framework instead.

No case of IEC-HS has been reported after ICV GD2 CAR-T in the published cohorts.

ICANS vs TIAN: the practical discrimination

- **ICANS**: encephalopathy, aphasia, tremor, seizures, decreased level of consciousness; diffuse EEG slowing; symptoms **not** confined to tumour neuroanatomy; usually follows systemic CRS by days; responds to steroids ± anakinra.
- **TIAN**: deficits map to tumour; may occur without systemic CRS; type 1 needs CSF drainage/osmotic therapy urgently.
- They are **not always separable**: in the patient with extensive thalamic/hypothalamic/insular tumour invasion, grade 4 ICANS and TIAN could not be distinguished (PMID 35130560). Where tumour is widespread, treat for both.

05 — Transient organ-function change: hepatic, renal/electrolyte, haematologic, coagulation, cardiopulmonary

This note answers the "what changes, by how much, for how long, and does it matter for other drugs?" question. Almost all quantitative data are [CART-general] (CD19/BCMA/CD22, IV route, lymphodepleted adults); GD2-specific data are sparse and reassuring. The bridge between them is **systemic inflammation grade**, not the target antigen: if CRS is grade ≥3 the organ changes below should be expected regardless of what the CAR binds.

Hepatic

Best single source: **Hepatic effects and potential DILI of CAR-T therapy** (2026, PMID 41345813), a systematic review of liver safety across CAR-T trials [CART-general].

- Grade ≥3 ALT elevation (5–20× ULN) ranged **0–8%** across 19 studies, with **11 of 19 reporting no grade ≥3 elevations at all**; AST behaves similarly.
- Reported AE incidence badly understates laboratory reality: in ZUMA-5 (n=148) increased ALT was reported as an AE in **8%** while the actual laboratory abnormality frequency was **58%**.
- **Hepatic dysfunction is the most common organ dysfunction in grade 3–4 CRS: 64% of patients** (vs cardiac dysfunction 36%). Another cohort reported "liver toxicity" in 31%.
- Grade 3–4 ALT/AST elevation occurs in **31–64%** of patients with severe CRS. **AST > ALT** is typical, suggesting an ischaemic/hypoperfusion contribution rather than direct hepatocellular toxicity.
- Bilirubin rise is uncommon and usually mild; in one pooled analysis of 137 patients, 12 (11%) had grade ≥3 total bilirubin and 11 of those 12 (92%) had CRS — i.e. bilirubin elevation is a marker of CRS severity, not of an independent hepatic syndrome.
- In IEC-HS, grade 3–4 transaminase elevation is very common and is part of the diagnostic construct; bilirubin >ULN in >50% and >2× ULN in ~33%.
- Rarer, genuinely hepatic entities do occur: acute liver failure without CRS/IEC-HS within 2 months, sinusoidal obstruction syndrome, early immune-related hepatitis, isolated grade ≥3 transaminitis, and late (beyond the acute window) transaminase elevation.

Pattern: transaminase elevation is common, tracks CRS grade, peaks with the inflammatory peak (days ~5–14 after IV infusion), is usually asymptomatic and resolves as CRS resolves. Bilirubin and synthetic function are usually preserved outside IEC-HS.

GD2-specific: no GD2 CAR-T trial reports hepatic toxicity as an attributed AE. The PSMA+GD2 glioma trial explicitly recorded **no abnormalities of liver enzymes or bilirubin**, acutely or on long-term follow-up (PMID 40420236). Anti-GD2 antibody regimens do include hepatic events (e.g. in dinutuximab-containing induction, PMID 37760578, and ^{131}I-MIBG + dinutuximab, PMID 40549985), but those are chemotherapy/radiopharmaceutical-containing combinations.

Renal and electrolyte

- Pooled incidence of AKI after CAR-T: **18.6%** across 22 cohort studies / 3,376 patients (17% in adults), with CRS incidence 75.4% in the same set (Kanduri systematic review, cited in PMID 39506012) [CART-general].
- **Transience is the rule**: in a 24-patient AKI series, onset was day +1 to +28 (13 of 22 post-infusion cases at day +7), and **renal function recovered in 19/24 (79%) within the first month** (PMID 38500492). Comparable series: AKI in 30% of 46 NHL patients (Gutgarts), 22% of 67 DLBCL patients, both "mild with fast recovery"; no excess versus auto-HCT in the largest comparison (232 CAR-T vs 414 auto-HCT).
- Risk factors: male sex, pre-existing CKD, **CRS**, and **ICANS grade ≥3** — i.e. AKI is a hypoperfusion/inflammation phenomenon, aggravated by nephrotoxic co-medication and contrast.
- AKI after CAR-T is associated with inferior survival in some series (PMID 40589093) — it is a severity marker even when it resolves.
- Mechanisms and management are reviewed from the nephrology side (PMID 39781479, PMID 40107382, PMID 41536632, ADQI consensus PMID 41361704), including CAR-T in dialysis-dependent and transplant recipients (PMID 41342049, PMID 34401733).
- Electrolyte disorders accompany CRS and immunotherapy generally (PMID 35063968): hyponatraemia (fluid resuscitation, SIADH, and — specific to this population — **hypertonic saline used for TIAN deliberately moves sodium**), hypophosphataemia, hypokalaemia, and tumour lysis where disease burden is high. FGF23-driven hypophosphataemia is described in HLH (PMID 42379468).
- **GD2/ICV-specific**: no attributed renal or electrolyte toxicity in any GD2 CAR-T trial; the ICV B7-H3 AE table contains no renal or electrolyte events at all (PMID 39775044). But **sodium management is an active intervention in TIAN**, so sodium, osmolality and fluid balance are on the monitoring list for reasons unrelated to nephrotoxicity.

Haematologic and coagulation

- In GD2 CAR-T trials the grade 3–4 haematologic events are attributed to **Cy/Flu lymphodepletion**, not to the CAR: fully reversible in a median 4 days (range 0–22) in 4SCAR-GD2 (PMID 34724115); in CAR-NKT the grade 3–4 cytopenias occurred before cell infusion (PMID 33046868).
- Pooled GD2 neuroblastoma data: any-grade anaemia 97%, grade ≥3 neutropenia 93% (PMID 41530803) — dominated by conditioning, tiny studies, very wide CIs.
- [CART-general] the durable problem is **ICAHT** (immune effector cell-associated haematotoxicity): biphasic and sometimes prolonged cytopenias with clonal-haematopoiesis and marrow-infiltration correlates (mechanism/management PMID 39973361; EHA/EBMT grading applied PMID 38181508; T-ICAHT for thrombocytopenia PMID 40258181; "How I treat cytopenias after CAR-T" PMID 36800563). **CAR-HEMATOTOX** predicts it (PMID 34166502; population validation PMID 40668622; comparative validation PMID 41229978), and ICAHT predicts infection and mortality.
- Coagulopathy: CRS-related coagulopathy is common with low absolute bleeding/ thrombosis rates (PMID 41228344, PMID 34595461); DIC occurs and is under-recognised (PMID 36794498, national cohort PMID 42633171); endothelial activation predicts DIC/CRS/ICANS (PMID 36503182); bleeding and thrombosis meta-analysis

PMID 38574863. Practically: fibrinogen, D-dimer, platelets and INR move during CRS, which matters if the patient is anticoagulated or needs a neurosurgical procedure (reservoir placement, EVD, biopsy) during the inflammatory window.

- **ICV consequence**: without lymphodepletion, ICV cohorts show essentially no cytopenia (one grade 1 lymphocytopenia in 21 patients / 253 doses, PMID 39775044). That removes the transfusion, neutropenic-fever, antimicrobial-prophylaxis and coagulation-management burden almost entirely — and it removes the main constraint on giving concurrent myelosuppressive therapy.

Cardiopulmonary

The severe events in GD2 CAR-T are CRS-mediated: multi-pressor hypotension, pulmonary oedema requiring BiPAP/CPAP, and intubation — all three grade 4 CRS presentations at IV DL2 in the DMG trial (PMID 39537919). Capillary leak occurred in 4/10 in 4SCAR-GD2. [CART-general] cardiac dysfunction accompanies 36% of grade 3–4 CRS. No cardiopulmonary events were reported after ICV dosing beyond fever and one hypotension/hypertension pair in the B7-H3 AE table.

Infection

Lymphodepletion + steroids + anti-cytokine agents + prolonged cytopenias = infection risk, and infection confounds every toxicity attribution (the single grade 3 CRS after ICV GD2 dosing occurred with urosepsis; a grade 3 CRS-equivalent in an sDMG patient likewise). Infectious complications across novel CAR targets including GD2 are catalogued in PMID 34619768; fungal disease in the setting of GD2 CAR-T plus immunosuppression in PMID 33509115; CNS infection after CAR-T in PMID 41511674.

Summary table — transient change, magnitude, timing

Parameter	Direction/magnitude	Timing	Reversible?	Driver
ALT/AST	grade 1–2 common; grade ≥3 in 0–8% overall, 31–64% if CRS grade 3–4; lab abnormality rate ≫ reported AE rate	with CRS peak, ~d5–14 IV	Yes, with CRS	Inflammation/ hypoperfusion
Bilirubin	usually normal; grade ≥3 ~11% in severe-CRS cohorts, 92% of those with CRS	CRS peak	Yes	CRS severity marker
Creatinine	AKI ~18.6% pooled, mostly KDIGO 1	day +1 to +28, mode day +7	79% recover ≤1 month	CRS, hypoperfusion, nephrotoxins
Sodium	hyponatraemia from fluids/ SIADH; **iatrogenic hypernatraemia from hypertonic saline for TIAN**	during CRS/ TIAN	Yes	Treatment as much as disease
ANC/platelets/Hb	grade 3–4 near-universal with Cy/Flu; prolonged in a minority (ICAHT)	days 0–28, may extend months	Mostly	Lymphodepletion ≫ CAR

Parameter	Direction/magnitude	Timing	Reversible?	Driver
Fibrinogen/D-dimer/ INR	CRS coagulopathy common, bleeding uncommon	CRS peak	Yes	Endothelial activation
LDH, ferritin, CRP	rise in all responders; ferritin to thousands	inflammation peak	Yes	Tumour lysis + inflammation
All of the above after ICV without lymphodepletion	essentially unchanged	—	—	Compartmentalised inflammation

06 — Drug interactions and pharmacokinetics: what transient inflammation does to concomitant medications

Framing. No GD2 CAR-T study has measured a drug–drug interaction. What exists is a well-quantified mechanistic literature showing that (a) acute inflammation suppresses cytochrome P450 activity, and (b) blocking IL-6 reverses that suppression. Both directions are relevant, because CAR-T patients receive an inflammatory insult and then, often within hours, an IL-6 blocker. Everything in this note is therefore **[mechanistic-PK]**: a strong basis for monitoring and for anticipating dose changes, not a measured GD2-specific interaction.

Direction 1 — CRS suppresses CYP-mediated clearance (drug levels rise)

- Human endotoxin/inflammation studies: antipyrine clearance **−35%** (95% CI 18–48) by day 2, hexobarbital −20%, theophylline −20% (reviewed in PMID 36059001).
- Surgery-induced acute inflammation: metabolic ratios fell by **CYP1A2 −53%, CYP2C19 −58%, CYP3A4 −61%** within 1–3 days postoperatively (same review).
- Midazolam clearance fell **65.4%** as CRP rose from 10 to 300 mg/L; CRP and IL-6 were covariates on midazolam CL.
- COVID-19 vs HIV: darunavir CL/F 4.1 vs 10.3 mL/h and AUC 161,387 vs 75,727 ng·h/mL, with **IL-6 the only significant covariate** (PMID 32856282).
- Bone-marrow-transplant engraftment: cyclosporine steady-state concentration rose **3.6-fold**, peaking concurrently with IL-6.
- PBPK, IL-6 10 → 100 pg/mL: simvastatin AUC **×2.36**, midazolam ×2.08, omeprazole ×1.97, dextromethorphan ×1.37, warfarin ×1.29, caffeine ×1.07 — CYP3A4 substrates most sensitive (PMID 34496043, PMID 38044486, PMID 40733104).
- **The closest analogue to CRS**: modelling of transient IL-6 elevation after **blinatumomab** gave maximal suppression of CYP3A4 **28% at 48 h lasting ~7 days**, CYP1A2 9%, CYP2C9 17%, with AUC ratios of 1.9 (simvastatin) and 1.7 (midazolam) — i.e. for a short cytokine spike, expect **<2-fold** exposure changes lasting about a week, magnitude driven by duration of cytokine elevation (PMID 38044486).
- Clinical corroboration in other settings: voriconazole exposure rises with inflammation and can produce phenoconversion of CYP2C19 genotype (PMID 39010708, PMID 41766069, PMID 42300127); pediatric-specific systematic review of inflammation on CYP activity (PMID 34462878).

Direction 2 — IL-6 blockade removes the suppression (drug levels fall)

- **Sarilumab 200 mg single dose + simvastatin** in RA: simvastatin Cmax **−46%**, **AUC∞ −45%**; β-hydroxy-simvastatin acid Cmax −35%, AUC −36%; CRP fell >7-fold over the same week (PMID 27722854).
- Tocilizumab reduces simvastatin Cmax and AUC by the same mechanism (cited in the same paper); sirukumab decreased midazolam AUC 30–35%, omeprazole 37–45%, S-warfarin 18–19% while increasing caffeine AUC 20–34%.
- Continuous IL-6R antagonist therapy in RA changes CYP activity measurably (PMID 37831074); olokizumab shows the same class effect (PMID 38984761).
- Regulatory-style assessments exist for many pro-inflammatory-pathway biologics (PMID 36890677, PMID 27391296, PMID 38050329 for mosunetuzumab, PMID 38383116 for cytokine–CYP trends).

Net effect in a CAR-T patient: exposure to CYP3A4/2C19/1A2 substrates can rise during CRS and then fall abruptly when tocilizumab/siltuximab is given — a two-way, few-days-scale moving target rather than a steady shift.

Concomitant drugs where this plausibly matters in a GD2 CAR-T patient

Prioritised by likelihood of co-prescription in this population and narrowness of therapeutic index:

Drug class	Concern	Practical response
Antiseizure medications (levetiracetam mostly non-CYP; but phenytoin, carbamazepine, valproate, lacosamide)	Seizure prophylaxis is standard in CNS CAR-T; CYP substrates/inducers shift with inflammation, and steroids interact with several	Prefer non-CYP agents (levetiracetam); level-monitor if using phenytoin/carbamazepine/valproate
Azole antifungals (voriconazole, posaconazole, itraconazole)	Both victims (inflammation-driven overexposure, documented phenoconversion) and perpetrators (CYP3A4 inhibition)	Therapeutic drug monitoring during and after CRS; avoid assuming genotype-based dosing holds
Calcineurin/mTOR inhibitors (tacrolimus, sirolimus, ciclosporin)	3.6-fold ciclosporin rise documented with IL-6 peaks; used for GVHD in allogeneic GD2 products	Frequent trough monitoring, especially in ALLO_GD2-CART01-type settings with GVHD
Dexamethasone / methylprednisolone	CYP3A4 substrate and inducer; used at variable doses for TIAN/ICANS; also the drug most likely to blunt CAR efficacy	Expect variable exposure; use the minimum effective duration per the trial algorithms
Dasatinib (used as a CAR "off-switch")	CYP3A4 substrate — exposure will move with both CRS and tocilizumab	Not formally studied in this use
Opioids, midazolam, other sedatives	CYP3A4-dependent; children with brainstem disease already at risk of hypoventilation	Titrate to effect; anticipate reduced clearance during CRS
Anticoagulants (warfarin, DOACs)		

Drug class	Concern	Practical response
	CYP/coagulation double hit: warfarin S-enantiomer clearance changes plus CRS coagulopathy	INR/anti-Xa monitoring through the inflammatory window
Statins	Largest documented swing in either direction (×2.36 up with IL-6; −45% with sarilumab)	Consider holding during the acute window; the clinical consequence is usually small in this population
Chemotherapy given concurrently or as bridging	Hepatic/renal changes and cytopenias limit dosing; bridging intensity worsens haematopoietic recovery (PMID 40267180)	Sequence rather than overlap where possible
Nephrotoxins (aminoglycosides, amphotericin, contrast, NSAIDs)	AKI risk multiplies with CRS hypoperfusion	Avoid in the CRS window; dose to renal function daily, not on admission creatinine

Interactions specific to this therapy, not mediated by CYP

1. **Lymphodepletion dosing depends on renal function.** Fludarabine is renally eliminated; the first population-PK model built exclusively in CAR-T recipients found weight, **eGFR** and CAR construct type as significant covariates on clearance, with reduced clearance below eGFR 120 mL/min/1.73 m^2 (PMID 41471106). Fludarabine over-exposure is linked to treatment-related mortality in HSCT, and under-exposure to weaker lymphodepletion. So a transient pre-infusion creatinine rise is not a cosmetic finding — it changes the conditioning exposure. ICV re-dosing without lymphodepletion sidesteps this entirely.
2. **Corticosteroids and CAR T-cell activity.** Steroid exposure before or during infusion is an exclusion/deferral criterion in the GD2 DMG trial and CIS scoring was deferred until ≥7 days after steroid discontinuation (PMID 39537919). The interaction of interest is pharmacodynamic — steroids impair the therapy — and this is why the TIAN algorithm reaches for CSF drainage, hypertonic saline and IL-1/IL-6 blockade first.
3. **Hypertonic saline** for TIAN alters sodium and free-water handling, which changes the distribution/level of drugs sensitive to volume status and affects concurrent antiseizure agents.
4. **Anti-cytokine agents are themselves immunosuppressive** (anakinra, tocilizumab, siltuximab, emapalumab, ruxolitinib, etoposide for IEC-HS), stacking with lymphodepletion for infection risk and with cytopenias for marrow recovery.
5. **Therapeutic proteins do not have classic CYP interactions themselves** — the interaction is always via the cytokine milieu (PMID 36890677, PMID 27391296). Do not look for a "CAR T-cell DDI"; look for an inflammation-mediated one.

What is genuinely unknown

- Whether **ICV** administration, which raises CSF cytokines but not plasma cytokines, produces any systemic PK effect at all. Mechanistically it should be minimal; nobody has measured it.

- The magnitude and duration of CYP suppression during CAR-T CRS specifically — the blinatumomab transient-IL-6 model (~28% CYP3A4 suppression for a week) is the best available proxy but CAR-T CRS is longer and more intense than blinatumomab's cytokine spike.
- Whether inflammation-driven changes in **renal glomerular filtration** during CRS (modelled in PMID 37715056) are large enough to require dose adjustment of renally cleared co-medications beyond routine creatinine-based dosing.

07 — Monitoring and mitigation

Everything here is derived from what the published GD2 and locoregional CNS CAR-T protocols actually did, plus general CAR-T guidance (ASTCT grading PMID 30592986; pediatric management PMID 30082906; ICAHT grading PMID 38181508; IEC-HS management PMID 41779733). It is a synthesis of published practice, not a protocol.

Structural safeguards used in the GD2 CNS programme

- **Eligibility geography**: exclusion of bulky thalamic or cerebellar tumours (preclinical lethal hydrocephalus at those sites) and of clinically significant dysphagia as a marker of medullary dysfunction (PMID 39537919).
- **No corticosteroids at enrolment**, and CIS scoring deferred ≥7 days after steroid discontinuation.
- **Ommaya/Rickham reservoir placed before treatment** in DIPG, for ICP measurement and therapeutic CSF removal — routine plus symptom-prompted measurements. ICV trials of other targets require a CNS catheter as an eligibility criterion (PMID 39775044).
- **Pre-specified TIAN algorithm**: CSF removal → hypertonic saline → anti-cytokine agents → corticosteroids, with neurocritical-care availability.
- **iCasp9 suicide gene** in the construct, with rimiducid/AP1903 available (used in 4 children with grade 3 ICANS on GD2-CART01, PMID 40841488; never needed in DMG arm A).
- **Dose separation of route**: IV DL1 (1×10^6/kg) as MTD, ICV flat doses $10–30 \times 10^6$ without lymphodepletion, ≥21–28 days between infusions, requiring circulating CAR <5% and toxicity resolved to <grade 2 before re-dosing.

Monitoring that the literature justifies

Neurologic / intracranial (both routes; the priority for ICV)

- Structured neurological examination at fixed timepoints, with a quantified improvement score so that worsening is detectable against a moving baseline.
- ICP measurement via reservoir — scheduled and on symptoms. Documented episodes: 22 mmHg and 34 mmHg with resolution within minutes of drainage.
- Low threshold for MRI (T2/FLAIR at the tumour site correlates with focal TIAN symptoms) and for EEG when encephalopathy is present (triphasic waves/diffuse slowing seen in the grade 4 ICANS case).
- Explicitly grade CRS, ICANS and TIAN **separately** at every infusion; do not collapse fever after ICV dosing into "CRS".

Systemic inflammation (IV route, and any febrile ICV patient)

- CRP, ferritin, fibrinogen, D-dimer, LDH, triglycerides; cytokine panel where available. Rising ferritin with falling fibrinogen and transaminitis → evaluate for IEC-HS.
- Blood counts; anticipate Cy/Flu nadir; use CAR-HEMATOTOX-type baseline risk assessment where the patient is heavily pretreated.

- Liver panel (ALT, AST, bilirubin, albumin, INR) through the CRS window — noting that grade 1–2 transaminase change is expected and does not by itself require intervention, while AST>ALT with hypotension suggests hypoperfusion, and bilirubin rise flags CRS severity or IEC-HS.
- Creatinine/eGFR **daily during the acute window**, with electrolytes including sodium, potassium, phosphate, magnesium; urine output and fluid balance. Sodium and osmolality are mandatory if hypertonic saline is used.
- Culture-directed infection workup for every fever, since infection both mimics and precipitates CRS (the two grade 3 CRS events in the ICV/sDMG experience were confounded by urosepsis).

Concomitant medications

- Reconcile the medication list against CYP3A4/2C19/1A2 dependence at baseline (see note 06) and again when tocilizumab/siltuximab is given.
- Therapeutic drug monitoring during and for ~1 week after the inflammatory peak for azoles, calcineurin/mTOR inhibitors, and any narrow-index CYP substrate; INR/anti-Xa if anticoagulated.
- Recalculate renally cleared drug doses on the current creatinine, not the admission value; this includes fludarabine exposure planning before conditioning (PMID 41471106).

Mitigation strategies with evidence behind them

Strategy	Evidence	Note
Use the ICV route for re-dosing, without lymphodepletion	62 ICV infusions, no DLT, no ICANS, CRS mostly grade 1, no organ-function signal (PMID 39537919); 253 ICV doses of B7-H3 with one DLT (PMID 39775044)	The single largest toxicity reduction available
Cap the IV dose	Grade 4 CRS at 3×10^6/kg vs none at 1×10^6/kg in DMG	MTD is route-specific
Split/fractionate dosing	No further grade ≥2 CRS after switching to $1.5\times10^7/m^2$ split 5–7 days apart (PMID 38771986)	Cheap, no engineering needed
Keep scFv affinity standard	Fatal encephalitis with high-affinity E101K in mice (PMID 29180536)	Antigen-density discrimination is the safety mechanism
Exclude high-risk tumour geography	Preclinical hydrocephalus in thalamic/cerebellar disease	Now standard eligibility
Pre-emptive IL-1 blockade (anakinra)	Used prophylactically/early for TIAN and CRS in GD2 CNS trials; CSF-penetrant	IL-1 mechanism separates CRS from neurotoxicity (PMID 29808007, PMID 29808005)
CSF drainage + hypertonic saline first for type 1 TIAN	Minutes-scale reversal of ICP events	Preserves CAR activity vs steroids
iCasp9 + rimiducid	Reversed grade 3 ICANS in 4/54 children (PMID 40841488)	Definitive but ablates the therapy
Pharmacological CAR pausing (dasatinib)	Used in refractory grade 4 ICANS (PMID 35130560); itacitinib explored (PMID 32998963)	Reversible alternative to iCasp9; not formally trialled

Strategy	Evidence	Note
Engineering: logic gates, affinity tuning, microenvironment-actuated CARs	Preclinical only (PMID 28341563, PMID 36396552, PMID 39841845, PMID 41005308)	Not yet in GD2 patients

Gaps worth closing

1. **No prospective organ-function dataset for ICV GD2 CAR-T.** The claim "ICV spares the liver and kidneys" rests on absence of reported AEs, not on published serial laboratory values.
2. **No measured DDI or CYP-probe study in any CAR-T CRS cohort**, let alone GD2 or ICV. A midazolam or 4-drug-cocktail probe study around CRS and tocilizumab administration would settle a question currently answered by PBPK extrapolation.
3. **No standardised TIAN grading validated prospectively.** BrainChild-03's retrospective application of TIAN criteria (PMID 41798119) showed most events could not be classified as type 1 vs type 2 — the distinction that determines whether CSF diversion is indicated.
4. **Repeat-dosing late toxicity** (ependymal injury, CSF-flow changes, anti-CAR immunity, device complications) is unreported despite cohorts receiving up to 81 doses over >3 years.
5. **No circulating neuro-injury biomarker** in clinical GD2 CAR-T; NFL is used in mouse models (PMID 38551501) and would directly address the residual on-target off-tumour question.
6. **Attribution of intratumoural haemorrhage** — two cohorts, one grade 5 and one grade 4 event, each argued to be background DIPG natural history. Only a larger denominator resolves this.

08 — The myeloid/microglial axis: recruitment, states, and its relation to TIAN

The clinical toxicity of GD2 CAR-T in the CNS correlates better with **myeloid** readouts than with T-cell readouts. This note collects what is actually measured, because it is the mechanistic bridge between "CAR T cells engage tumour" and "the patient's brainstem swells".

What is measured in GD2 CAR-T patients

Tumour tissue (autopsy, DIPG-1; PMID 35130560) [GD2-clinical]

- Prominent **microglial and other myeloid infiltration of tumour**: $CD163^{+}$ myeloid cells throughout tumour tissue, and $Iba1^{+}$ myeloid cells densely infiltrating $H3K27M^{+}$ regions on confocal microscopy.
- In the **unaffected cortex** of the same patient, CD163 staining showed only microglia in their resting perivascular position — **no reactive microglia/macrophages**. Combined with CAR transcript and lymphocytic infiltrate confined to tumour, and tumour GD2 ≫ normal-brain GD2, this is the strongest single piece of human evidence that the inflammatory response is spatially restricted to tumour rather than provoked diffusely in normal brain.

CSF single-cell RNA-seq across routes (PMID 35130560) [GD2-clinical, ICV-clinical]

- CSF myeloid cells resolved into **seven clusters** — monocytes, microglia and macrophages with distinct functional signatures, plus a proliferating myeloid cluster (myeloid identity called on AIF1, CSF1R, CX3CR1, CD14, CD68, CD163).
- **Route-dependent myeloid states**:
- after **ICV** administration, at peak inflammation, an **interferon-response myeloid population** dominates;
- after **IV** administration and at late timepoints, myeloid cells instead express **phagocytosis and lipid-metabolism programmes** with a strongly **immune-suppressive** profile, aligning with **disease-associated microglia (DAM)**, **MDSC** and **axon-tract-associated microglia** signatures.
- The patient with the most severe intracranial events (DIPG-1) had **differentially increased interleukin, chemokine and neutrophil-degranulation pathways** in the CSF myeloid fraction relative to the other patients at peak inflammation.

So the ICV route does not merely relocate inflammation — it elicits a **different myeloid state** (IFN-responsive/activating) than the IV route (suppressive/DAM-like). Whether the ICV state is mechanistically responsible for the higher CSF cytokines with lower systemic toxicity is untested.

Cytokine correlates of toxicity (PMID 39537919) [GD2-clinical]

- Grade ≥2 **CRS** correlated with higher **plasma MCP-1/CCL2** — a monocyte-recruiting chemokine — plus a trend for IL-2.
- Grade ≥2 **TIAN** correlated with higher **CSF MCP-1/CCL2**, IL-10 and TNF-α.

- The authors' own reading: the association of severe TIAN with myeloid-response chemokines "raises the prospect that **myeloid cells may contribute to neurological symptoms** induced by CAR T cell therapy for CNS tumours, a hypothesis that requires testing in future studies."

That is the current state of the evidence: **correlative and explicitly hypothesis-generating**. No GD2 CAR-T study has depleted or blocked myeloid recruitment and measured the effect on TIAN.

Why a myeloid mechanism is plausible

[CART-general] mechanism, transferable because it is antigen-agnostic:

- CRS itself is **macrophage-mediated**: monocyte/macrophage IL-1 and IL-6 drive the syndrome, and **IL-1 blockade abates CRS while separating it from neurotoxicity** (Giavridis 2018, PMID 29808005; Norelli 2018, PMID 29808007). Monocyte depletion has been explored as a CRS intervention (PMID 34570442 review; monocyte-depletion preprint, DOI 10.1101/2020.11.22.20232801).
- In ICANS, **$CD14^+$ myeloid cells are found in CSF**, and the proposed cascade is endothelial activation → BBB disruption → immunocyte infiltration → astrocyte injury and **microglial activation** → neuronal dysfunction (PMID 35871757). Autopsy in CAR-T cerebral oedema showed microglial activation, extensive in one case and perivascular in another (cited in PMID 35871757).
- Cytokine-level detail (PMID 33391257): **GM-CSF** secreted by CNS-infiltrating helper T cells activates microglia, and GM-CSF neutralisation reduced brain contrast enhancement in a CAR-T xenograft; **IFN-γ** drives microglial activation and antigen presentation; **IL-6** induces microglial proliferation; **IL-15** induces microglial cytokine production, and increased astrocytic IL-15 worsened cerebral oedema in mice — relevant because IL-15-armoured products exist in the GD2 space (GD2-CAR.15 NKT). Microglia also retain **epigenetic memory** of inflammatory stimuli, a proposed substrate for prolonged neurologic sequelae.
- Preclinically in GD2/DMG models, tumour clearance produced **brainstem oedema and obstructive hydrocephalus** (PMID 29662203) — the mass-effect endpoint that TIAN type 1 describes.

Is microglia a target or an amplifier?

An important distinction for the off-target question: nothing in this corpus shows GD2 CAR T cells **targeting** microglia. GD2 in normal CNS is described on neurons, astrocytes and peripheral nerve fibres, not as a microglial marker, and the only demonstrated GD2 CAR-mediated normal-CNS injury is the high-affinity **E101K** encephalitis, where CAR T cells infiltrated and proliferated in low-GD2 cerebellar and basal brain regions with neuronal destruction (PMID 29180536) — a T-cell-driven lesion, with myeloid involvement not separately characterised in that report.

Practical framing:

- **Amplifier (evidenced)**: myeloid recruitment and microglial activation downstream of CAR-T engagement, correlating with CRS (plasma CCL2) and TIAN (CSF CCL2/TNF/IL-10).
- **Target (no evidence)**: GD2 CAR-T killing microglia.
- **Consequence**: interventions aimed at the myeloid arm — anakinra (IL-1R), tocilizumab/siltuximab (IL-6), and in the hyperinflammatory extreme emapalumab (IFN-γ) or ruxolitinib — are the mechanistically-matched drugs for TIAN and CRS, whereas corticosteroids act broadly and also blunt the CAR. This is exactly the order the GD2 CNS protocols use.

Open questions this would answer if measured

1. Does blocking **CCL2/CCR2** or GM-CSF (lenzilumab-type) reduce TIAN without reducing antitumour effect? Untested in GD2 CAR-T.
2. Do the **DAM/MDSC-like suppressive myeloid states** seen after IV dosing and at late timepoints act as a resistance mechanism as well as a toxicity mechanism?
3. Does the attenuation of TIAN over successive ICV infusions reflect **myeloid-state adaptation** (tolerisation) or simply less tumour left to inflame?
4. Is there measurable **microglial injury** (e.g. sTREM2, GFAP, NfL in CSF) after ICV GD2 CAR-T? No GD2 trial reports neuroglial-injury biomarkers, although CSF/serum **GFAP and NfL** track neurotoxicity and endothelial dysfunction after CD19 CAR-T (PMID 41553539) and murine NfL has been used preclinically (PMID 38551501).

Appendix: full reference list

All 852 curated records, grouped by primary toxicity domain and ordered by year (descending). Records without a PMID are conference abstracts.

neurotoxicity (199)

- Ahmad R, Wild M, Friesen T, Hall M, Kouba S, Maliske S. Encephalopathy Post CAR-T Cell Therapy in Relapsed/Refractory Multiple Myeloma: Beyond ICANS Journal of investigative medicine high impact case reports 2026. PMID 42575512.
- Almashayekh A, Sulaiman S, Alaarag A, Omaish R, Mekuria Z, Balts N, Henson JC, Vellanki S, Trikannad A, Spears G, Mohan-Lal B, Pandey T, Odueke A, Schinke C, Gentille-Sanchez C, Jain N, Alshawayat A. Hypophosphatemia as a Marker for Immune Effector Cell-Associated Neurotoxicity Syndrome in Chimeric Antigen Receptor T Cell Recipients Hematology/oncology and stem cell therapy 2026. PMID 42169662.
- Alqaisi O, Dibas M, Tai P, Dibas A, Souied O, Al-Ghabeesh S. CAR T-Cell Therapy in Neurology: A Scoping Review of Neuro-Oncology, Autoimmune Diseases & Neurotoxicity Annals of clinical and translational neurology 2026. PMID 42046195.
- Angleitner AC, Maulhardt M, Wulf G, Büntzel J. Viable CAR T-cells remain detectable in cerebrospinal fluid in patients with grade ≥3 ICANS despite corticosteroid therapy Frontiers in oncology 2026. PMID 42370159.
- Arai H, Sasada S, Kazahari H, Homma Y, Kaburaki K. Life-Threatening Prolonged ICANS Following Tarlatamab Treatment in Extensive-Stage Small Cell Lung Cancer With Brain Metastases: A Case Report Respirology case reports 2026. PMID 42146875.
- Asioli GM, Pondrelli F, Spinardi L, Faccioli L, Vornetti G, Maffei M, Zinzani PL, Casadei B, Maffini E, Bonifazi F, Guarino M. Prospective observational study of magnetic resonance imaging in anti-CD19 CAR T-cell-associated neurotoxicity Journal of neurology 2026. PMID 41615499.
- Aziz F, Chakraborty A, Saha D, Dutta P. Immune effector cell-associated neurotoxicity syndrome: integrative mechanisms, predictive biomarkers, and translational pathways for prevention in CAR T-cell therapy Frontiers in neurology 2026. PMID 41859418.
- Diorio C, Myers RM, Li Y, Liu H, Belfer S, DiNofia A, Leahy AB, Newman H, Dolan J, Martinez Z, Elhachimi A, Lindell RB, Mullen C, Li A, Kim J, Teachey DT, Maude SL, Grupp SA, McGuire JL. Preinfusion risk factors for the development of severe neurotoxicity following CART therapy in pediatric patients HemaSphere 2026. PMID 41959729.
- Eldaya RW, Parsons MS. Post-Treatment Causes of Encephalopathy Neuroimaging clinics of North America 2026. PMID 42399024.
- Favero A, Mohamed S, Battista ML, Cenacchi V, Sartori A, Mazzon G, Bratina A, Tomaselli M, Patriarca F, Zaja F, Manganotti P, Benussi A. Immune Effector Cell-Associated Neurotoxicity Delayed Relapse After Chimeric Antigen Receptor T-Cell Therapy: A Case Report Neurology(R) neuroimmunology & neuroinflammation 2026. PMID 41248447.
- Ferri GM, Frank A, Li D, Anand P, Abdallah M, Avalone V, Mark Sloan J, Sanchorawala V, Lerner A, Szalat RE, Petrocca F, Edwards CV, Bell BN. When ICAN(S) Becomes ICAN'T: Clinician and Staff Perspectives on In-Hospital Neurotoxicity Grading Clinical lymphoma, myeloma & leukemia 2026. PMID 41015712.
- Freund BE, Safa A, Colella FE, Chiriboga G, Betiku T, Shar A, Drees C, Britton J, Feyissa AM, Tatum WO. Is EEG useful in predicting neurotoxicity and outcomes related to chimeric antigen receptor (CAR) T-cell therapy? Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology 2026. PMID 41494410.
- Fridberg G, Amit O, Sherf Y, Perry C, Herishanu Y, Joffe E, Shragai T, Vitkon R, Preis M, Perk S, Sarid N, Barouch SB, Gold R, Glait-Santar C, Avivi I, Ram R. ICANS mitigation in high-risk elderly patients treated with CD28 co-stimulatory anti-CD19 CAR-T cells using a standardization protocol: a pilot study Haematologica 2026. PMID 41676895.
- Friedman AR, Gil-Perotin S, Flanagan Wesley S. CAR T cell therapy: Autoimmune neurological uses and neurotoxicities Best practice & research. Clinical rheumatology 2026. PMID 41934064.
- Frisch A, Marino L, Alsaadi D, Kasarabada A, Hua G, Ajebo G, Medlin S, Gul Z. Cyclophosphamide for the Treatment of Refractory Immune Effector Cell-Associated Neurotoxicity Syndrome Following CD19-Targeted CAR T-Cell Therapy Journal of medical cases 2026. PMID 41487336.
- Galli E, Modoni A, Battistini L, Rossi M, Guerrera G, Pansini I, Corrente A, Hohaus S, Chiusolo P, Sorà F, Calabresi P, Sica S. Neuronal injury biomarkers GFAP and neurofilament light chains (NfL) are associated with

neurotoxicity and endothelial dysfunction in adult patients treated with antiCD19 CART cells Clinical and experimental medicine 2026. PMID 41553539.

- Griesinger AM, Widener M, Donson AM, Mettetal A, Hankinson TC, Handler M, Mulcahy Levy JM, Lindsay HB, Macy ME, Foreman NK, Dorris K. Molecular biology and phase 1 study of GM-CSF and intrathecal trastuzumab in children with recurrent -posterior fossa ependymoma Neuro-oncology 2026. PMID 41920921.
- Guo X, Wei Y, Fang Y, Lin R. Blinatumomab-induced catastrophic immune effector cell-associated neurotoxicity with diffuse cerebral edema in a patient with ALL: Case report International journal of clinical pharmacology and therapeutics 2026. PMID 42638547.
- Hansen DK, Castaneda Puglianini OA, Grajales-Cruz A, Nagar SP, Ghosh S, Alegria V, De Braganca KC, Lengil T, Sharma M, Pai H, Perciavalle M, Maitland J, Emond B, Bixby T, Qureshi ZP, Janakiram M. Real-World Incidence and Management of Non-Immune Effector Cell-Associated Neurotoxicity Syndrome Neurologic Events Following Ciltacabtagene Autoleucel in Multiple Myeloma Clinical epidemiology 2026. PMID 42227005.
- Hayashino K, Fujii N, Kitamura W, Kamoi C, Kondo T, Ikegawa S, Seike K, Fujiwara H, Asada N, Ennishi D, Fujii K, Maeda Y. The Association Between Hyponatremia and Immune Effector Cell-Associated Neurotoxicity Syndrome After Chimeric Antigen Receptor T Cell Therapy Transplantation and cellular therapy 2026. PMID 42364733.
- Hino K, Edahiro T, Muta T, Ureshino H, Nakatani S, Nunomura T, Okatani T, Yoshida N, Chishaki R, Yoshida T, Shindo T, Ichinohe T. Tumor Inflammation-associated Neurotoxicity Following CD19-targeted Chimeric Antigen Receptor T-cell Therapy in a Patient with High-grade B-cell Lymphoma Internal medicine (Tokyo, Japan) 2026. PMID 41033998.
- Kato K, Takigawa K, Miyawaki K, Taniguchi S, Shimo M, Mori K, Terasaki T, Sakoda T, Jinnouchi F, Yamauchi T, Shima T, Kikushige Y, Mori Y, Akashi K. Direct Flow Cytometric Assessment of Cerebrospinal Fluid CAR-T Cells for ICANS Diagnosis Research Square 2026. DOI 10.21203/rs.3.rs-9396828/v1.
- Kazzi C, Griffith SP, Ko KY, Pearce DJ, Wronski M, Seery N, Wesselingh R, Tan TH, Simpson T, Abbott C, Li J, Cole E, Westworth M, Inam S, Rushen T, Parsons S, Melitsis L, Wong SM, Butzkueven H, Van Der Walt A, Nandurkar H, Tam CS, Fleming S, O'Brien TJ, Alpitsis R, Spencer A, Malpas CB, Monif M. Repeatable Battery for the Assessment of Neuropsychological Status: Preliminary Utility for Subacute Cognitive Monitoring after Immune Effector Cell-Associated Neurotoxicity Syndrome after Chimeric Antigen Receptor T-Cell Therapy Transplantation and cellular therapy 2026. PMID 41314392.
- Khatib H, Parizat A, Halloun J, Shouval R, Shibli N, Alalouf O, Zuckerman T, Halamish I, Sapir D, Yehudai-Ofir D, Ringelstein-Harlev S, Horowitz NA, Lavi N, Horesh N, Fineman R, Tzoran I, Levi T, Dann EJ, Mruwat A, Brenner B, Hoffman R, Rozenberg A, Shechtman Y, Beyar-Katz O. Platelet and Fibrinogen Dynamics After CAR-T Cell Therapy in Relapsed/Refractory B-Cell Lymphoma May Predict ICANS Onset European journal of haematology 2026. PMID 41681048.
- Khoudari H, Shoaib A, Nashatizadeh M, Ahmed N, Lutfi F, Mushtaq M, Shune L, Singh A, Abhyankar S, McGuirk J, Abdelhakim H. Delayed fatal neurotoxicity in post CAR-T cell therapy for multiple myeloma, a case report Leukemia research reports 2026. PMID 41853755.
- Lapucci C, Gambella M, Cipriano E, Raiola AM, Varaldo R, Ghiso A, Centanaro M, Capello E, Schenone A, Castellan L, Barletta L, Angelucci E, Inglese M. Brain Microstructural Damage as Potential Biomarker of Immune Cell-Associated Neurotoxicity Syndrome Journal of neuroimaging : official journal of the American Society of Neuroimaging 2026. PMID 41546434.
- Lasocki A, Harrison SJ, Dickinson M, Kalincik T. The role of neuroimaging in neurotoxicity after chimeric antigen receptor T-cell therapy Therapeutic advances in neurological disorders 2026. PMID 41537162.
- Ma Y, Chang L, Mi R, Wang L, Chen L, Wei X. Case Report: Safety analysis of blinatumomab consolidation therapy in two cases of acute B-cell lymphoblastic leukemia in complete remission complicated with immune effector cell-associated neurotoxicity syndrome Frontiers in oncology 2026. PMID 41809830.
- Miller DM. Nursing Considerations for CAR T-Cell Therapy: Management of Cytokine Release Syndrome and Immune Effector Cell-Associated Neurotoxicity Syndrome Clinical journal of oncology nursing 2026. PMID 42462106.
- Nicholas MA, Abu-Haimed A, Abushullaih F, Davidson EN, Fenton GA, Kehm JW, Kocoglu H, Yared JA, Fromowitz A, Hardy NM, Rapoport AP, Atanackovic D, Pham L, Badros A, Ahmad H. Beyond the typical immune effector cell-associated neurotoxicity syndrome: Characterizing non-immune effector cell-associated neurotoxicity syndrome neurological complications in chimeric antigen receptor T-cell therapy-A retrospective analysis Neuro-oncology advances 2026. PMID 42621663.
- Nishihara H, Jinnouchi F, Ishihara D, Imanaga H, Sasaki K, Sakoda T, Yamauchi T, Miyawaki K, Shima T, Shimo M, Henzan T, Kunisaki Y, Kikushige Y, Mori Y, Akashi K, Kato K. Early prediction of ICANS using CRS characteristics in B-cell lymphoma patients receiving CAR-T therapy International journal of hematology 2026. PMID 42489952.
- Nishiyama R, Makita S. [Management of ICANS in Patient Treated with Anti-CD19 CAR T-Cell Therapy] Gan to kagaku ryoho. Cancer & chemotherapy 2026. PMID 42551936.
- Pagliuca S, Jacquet C, Rubio MT, Sorrentino P. The neurotoxic legacy of CAR-T cells: where do we stand? Therapeutic advances in neurological disorders 2026. PMID 42152959.

- Parizat A, Alalouf O, Sapir D, Shibli N, Perets R, Aran D, Katz OB, Shechtman Y. Early Detection of CAR-T-Associated Neurotoxicity via Cytokine Monitoring in Serum medRxiv 2026. DOI 10.64898/2026.03.03.26347491.
- Riedel A, Stanger AMP, Kimmich LM, Hug S, Fehn AE, Schroeder JC, Besemer B, Casadei N, Demidov G, Thapa PJ, Faul C, Hensen L, Lengerke C, Bethge W. Extensive non-clonal CAR T-cell expansion causing fatal hyperinflammatory syndrome with immune effector cell-associated hemophagocytic lymphohistiocytosis, cytokine release syndrome, and neurotoxicity syndrome Haematologica 2026. PMID 40874337.
- Rodríguez-Otero P, Gállego Pérez-Larraya J, Lin Y, Dietrich J, van de Donk NWCJ, Mason XL, Dhakal B, Jakubowiak AJ, Lendvai N, Li H, Yeh TM, Lee E, Marquez L, Sharma M, Montes de Oca R, Wang F, Plaks V, Henry K, Taraseviciute A, Jethava Y, Jablow L, Afifi S, De Braganca KC, Madduri D, Zhu J, Koneru M, Akram M, Patel N, Cohen AD, Santomasso BD. Non-ICANS Neurologic Events in Patients With Relapsed/Refractory Multiple Myeloma Treated With Ciltacabtagene Autoleucel: Clinical Presentation and Management in CARTITUDE Studies Clinical lymphoma, myeloma & leukemia 2026. PMID 42161729.
- Sato T, Chida K, Gandhi S, Takabe K. Neurotoxicity of Immunotherapy: Immune Checkpoint Inhibitor-Related Encephalitis vs. Immune Effector Cell-Associated Neurotoxicity Syndrome World journal of oncology 2026. PMID 41488284.
- Schmid T, Shaforostova I, Bacher U, Seipel K, Kronig MN, Pabst T. Prophylactic Anakinra to Prevent Neurotoxicity After CAR T-Cell Therapy in Aggressive B-Cell Lymphomas: A Single-Center Real-World Experience Cancers 2026. PMID 42279369.
- Schroeder T, Chitadze G, Dargvainiene J, Martens T, Gebauer N, Frimmel J, Brüggemann M, Valerius T, Schub N, Baldus CD, Pott C, Leypoldt F, Stürner K, Stölzel F. Non-ICANS neurotoxicities CD19-directed CAR T-cell therapy and the emergence of movement and neurocognitive treatment-emergent adverse events: a case report Frontiers in immunology 2026. PMID 41777880.
- Sidana S, Nagar SP, Ghosh S, Fan L, Alegria V, De Braganca KC, Lengil T, Sharma M, Pai H, Perciavalle M, Maitland J, Emond B, Bixby T, Qureshi ZP, Fonseca R. Real-World Description of Non-ICANS Neurologic Events Among Patients with Relapsed or Refractory Multiple Myeloma Treated with Ciltacabtagene Autoleucel Using Two Large US Databases Oncology and therapy 2026. PMID 42000954.
- Song D, Han J, Dong C, Wang Q, Zhu X, Li J, Rao M, Liu H, Ma Z, Teng Y, Dou C, Xiao M. Identification and Preclinical Evaluation of a Novel GD2 Antibody-Drug Conjugate for Solid Tumors in Children and Adults Molecular cancer therapeutics 2026. PMID 41609209.
- Tashiro Y, Jo T, Kitawaki T, Yoshinaga N, Sakamoto T, Shirakawa K, Kanda J, Nishikori M, Yamashita K, Nagao M, Takaori-Kondo A, Arai Y. Low reticulocyte count at infusion is a risk factor for ICANS in CAR-T cell therapy Cytotherapy 2026. PMID 41762956.
- Thirugnanam A, Donthineni K, Mammi M, Hundel J, Munteh D, D'Amore F, Bunevicius A, Mekary RA. Chimeric antigen receptor T-cell therapy and immune effector cell-associated neurotoxicity syndrome: A meta-analysis Critical reviews in oncology/hematology 2026. PMID 41448297.
- Torasawa M, Shukuya T, Huang J, Nagai R, Ayame N, Arai Y, Hayakawa D, Hayashi T, Nagaoka T, Takahashi K. Tumor Inflammation-Associated Neurotoxicity Masquerading as Severe Immune Effector Cell-Associated Neurotoxicity Syndrome in a Patient With SCLC Treated With Tarlatamab: A Case Report Journal of thoracic oncology : official publication of the International Association for the Study of Lung Cancer 2026. PMID 41848463.
- Tun AM, Patel R, St-Pierre F, Ouchveridze J, Niu A, Thordardottir T, Obasi J, Rosenthal A, Pophali PA, Fenske T, Karmali R, Johnston PB, Ahmed S. Neurotoxicity in Older Patients with Relapsed or Refractory Large B-Cell Lymphoma Treated with Chimeric Antigen Receptor T-Cell Therapy: A Multicenter Study Transplantation and cellular therapy 2026. PMID 41935576.
- Vagos-Mata A, Castro P, Valade S, Carpentier AF, Gallo de Moraes A, Maamar A, Picard M, Moreau AS, Chawla S, Wallet F, Klouche K, Metaxa V, Boissel N, Van De Louw A, Seguin A, Mokart D, Böll B, Di Blasi R, Thieblemont C, Issa N, Fernandez S, Munshi L, Bauer PR, Joannidis M, Moreno-Gonzalez G, Zafrani L, Darmon M, Azoulay E, Nine-I investigators. The clinical picture of ICANS after CAR-T cell therapy in critically ill patients: A retrospective clustering analysis of a multicentre multinational cohort study Annals of intensive care 2026. PMID 42592119.
- Völker L, Müller-Jensen L, Carl S, Nay S, Möllenkamp TM, Schultze-Florey C, Grote-Levi L, Jendretzky KF, Konen FF, Könecke C, Eder M, Panagiota V, Heidel FH, Bullinger L, Damm F, Frick M, Penack O, Ludwig R, Buß EA, Boehmerle W, Endres M, Gudi V, Huehnchen P, Skripuletz T, Möhn N. The Berlin-Hannover ICANS severity assessment-a novel bedside test to evaluate CAR T-cell-associated neurotoxicity Frontiers in neurology 2026. PMID 41648666.
- Werner JM, Gödel P, Galldiks N, Krone C, Schroeter M, Holtick U, Flümann R, Wunderlich G, Hallek M, Fink GR, Scheid C, Onur OA, Richardson T. Non-ICANS neurotoxicity after BCMA-directed CAR-T therapy: Clinical spectrum, outcomes, and a framework for neurology-oncology co-management HemaSphere 2026. PMID 42306065.

- Yamamoto Y, Masauji T, Shimasaki T. Immunotherapy-associated dysgraphia as an early neurocognitive manifestation of immune effector cell-associated neurotoxicity syndrome: clinical characteristics, mechanistic insights, and assessment challenges Frontiers in immunology 2026. PMID 42136681.
- Yoshihara M, Yuyama S, Yasumuro O, Funakoshi R. Syndrome of inappropriate antidiuretic hormone secretion following immune effector cell-associated neurotoxicity syndrome during blinatumomab therapy: a case report Journal of pharmaceutical health care and sciences 2026. PMID 42432807.
- Yoshihara M, Yuyama S, Yasumuro O, Funakoshi R. Syndrome of inappropriate antidiuretic hormone secretion (SIADH) following immune effector cell-associated neurotoxicity syndrome (ICANS) during blinatumomab therapy: A case report Research Square 2026. DOI 10.21203/rs.3.rs-9674917/v1.
- Zahedi A, Ifejeokwu OV, Griffin SP, Dean EA. Repurposing mitochondrial-targeting drugs for management of ICANS in CAR T-cell therapy: a novel steroid-sparing approach Frontiers in pharmacology 2026. PMID 42582333.
- van Besien H, Ozkan G, Easton N, Tix T, Alhomoud M, Shouval R, Rejeski K, Yamshon S. Non-ICANS neurologic toxicity after BCMA CAR-T therapy: a systematic review and meta-analysis of 4630 patients with multiple myeloma Blood advances 2026. PMID 41886632.
- Aboubakr O, Houillier C, Choquet S, Dupont S, Hoang-Xuan K, Mathon B. Epileptic seizures in patients with primary central nervous system lymphoma: A systematic review Revue neurologique 2025. PMID 38042665.
- Agarwal RD, Kunwor R, Sudanagunta K. Posterior reversible encephalopathy syndrome after CD19 chimeric antigen receptor therapy for B-acute lymphoblastic leukemia: case report Frontiers in oncology 2025. PMID 40989690.
- Angelillo P, Alarcon Tomas A, Campodonico E, Novak U, Gabellier L, Blau IW, Kwon M, Ram R, Forcade E, Besley C, Vucinic V, Kröger N, Corral LL, Vydra J, Ferrari S, von Tresckow B, Rubio MT, Hernani R, Sica S, Stölzel F, Rovira M, Wagner-Drouet E, Pérez-Simón JA, Castilla-Llorente C, Ferreri AJM, Hoogenboom JD, Sanchez Ortega I, Malard F, Kuball J, Ruggeri A. Severe ICANS after CAR T-cell therapy and assessment of prevention with levetiracetam for seizure prophylaxis following CAR T-cell for DLBCL & PMBCL in Europe: a survey on behalf of the Cellular Therapy & Immunobiology Working Party (CTIWP) of the EBMT Bone marrow transplantation 2025. PMID 39455896.
- Belloni S, Giacon C, Magon A, Girardi D, Arcidiacono MA, Ghizzardi G, Conte G, Caruso R, Arrigoni C. Factors Associated with Immune Effector Cell-Associated Neurotoxicity Syndrome in Adults with Hematological Malignancies Undergoing Chimeric Antigen Receptor T-Cell Therapy: A Systematic Review Seminars in oncology nursing 2025. PMID 40628565.
- Blumenberg V, Puliafito BR, Graham CE, Leick MB, Chowdhury MR, King M, Harris DL, Raje NS, Branagan AR, Yee AJ, Cirstea D, Gallagher KME, Dietrich J, Maus MV, Frigault MJ. Cyclophosphamide mitigates non-ICANS neurotoxicities following ciltacabtagene autoleucel treatment Blood 2025. PMID 40163799.
- Buciuc AG, Tran S, Weber M, Padilla V, Rueda-Lara M, Espinel Z. Immune Effector Cell-Associated Neurotoxicity Syndrome After CAR T-Cell Therapy and Other Psychiatric Manifestations: A Review and Case Series Journal of clinical medicine 2025. PMID 40094890.
- Cao L, Zhou M, Wei Z, Zhu R, Chen H, Zhou L, Hu S, Bai Z, Wu S. Frontline therapy combining with intrathecal dexamethasone effectively alleviate immune effector cell-associated neurotoxicity syndrome in pediatric relapsed B-ALL receiving pCAR-19B-cases report BMC pediatrics 2025. PMID 41039358.
- De Philippis C, Giacomel A, Pensato U, Pinton C, Taurino D, Mannina D, Mariotti J, Sarina B, Marcheselli S, Timofeeva I, Capizzuto R, Santoro A, Bramanti S. Late-onset relapsing neurotoxicity after Brexucabtagene autoleucel associated with high chimeric antigen receptor T cells in cerebrospinal fluid Cytotherapy 2025. PMID 39152952.
- Fatahichegeni M, Ansarian MA, Wang Y, Ren J, Zhang T, Wang X. Immune effector cell-associated neurotoxicity syndrome following CAR T-cell therapy: a review of recent advances Journal of translational medicine 2025. PMID 41454396.
- Freund BE, Feyissa AM, Betiku OE, Shar A, Drees C, Sherman W, Qin H, Britton JW, Barrios MS, Quinones-Hinojosa A, Tatum WO. Acute Symptomatic Seizures During CAR T-Cell Therapy for Hematologic Malignancies: Tri-Site Mayo Clinic Experience Neurology 2025. PMID 40215424.
- Fukui D, Sakakibara-Konishi J, Azuma R, Yoshida Y, Morinaga D, Ito S, Furuta M, Takasihma Y, Kitai H, Konno S. Rapid disease progression and hydrocephalus following tarlatamab therapy initiation: A case report Respiratory medicine case reports 2025. PMID 41311375.
- Giesler S, Biavasco F, Wertheimer T, Zeiser R, Vinnakota JM. Biology-driven therapies of CAR T immune effector cell-associated neurotoxicity syndrome Immunotherapy 2025. PMID 40438973.
- Gomez-Llobell M, Serra Smith C, Gómez-Costas D, Gómez-Centurión I, García Domínguez JM, Escudero-Vilaplana V, Fernández Bullido Y, Carbonell D, Pion M, Bailén R, Gil-Perotin S, Martínez Ginés ML, Revuelta-Herrero JL, Fernández-Caldas P, Fernández VAP, Collado-Borrell R, Bastos-Oreiro M, Villanueva-Bueno C, García-Sanz R, Kwon M. ICANS risk model in CD19 CAR-T therapy: insights from serum and CSF cytokine profiling Bone marrow transplantation 2025. PMID 40707782.

- Goto H. CRS and ICANS in CAR T-cell therapy for large B-cell lymphoma Journal of clinical and experimental hematopathology : JCEH 2025. PMID 41443883.
- Hassankhani A, Bagley SJ, Dahmoush H, Parsons MS, Nabavizadeh A. Role of Neuroimaging in Cancer-Treatment Neurotoxicity AJR. American journal of roentgenology 2025. PMID 40557987.
- He J, Connors J, Meador A, Xu S, Meador H, Jiang H, Fueyo J, Gomez-Manzano C, Friedman GK, Zaky W, Sadighi Z, Slopis JM, Ahmad AH. Immunotherapy-related neurotoxicity in the central nervous system of children with cancer Neuro-oncology 2025. PMID 39535217.
- Jonokuchi AJ, Yu AW, Kaur G, Satwani P, Reshef R, Wesley SF. Surviving Grade 4 ICANS: A Case Report and Discussion of Emerging Management Strategies Journal of child neurology 2025. PMID 40370190.
- Kato K, Nakagaki H, Yamauchi T, Mukaino T, Sakata A, Watanabe E, Watanabe M, Ishihara D, Imanaga H, Sasaki K, Sakoda T, Jinnouchi F, Miyawaki K, Shima T, Kikushige Y, Mori Y, Hotta T, Kunisaki Y, Shigeto H, Isobe N, Akashi K. Pre-CAR-T GTE Scoring of Electroencephalogram Abnormalities as a Predictive Biomarker for ICANS Research Square 2025. DOI 10.21203/rs.3.rs-6087514/v1.
- Kaulen LD, Martinez-Lage M, Abramson JS, Karschnia P, Doubrovinskaia S, Shankar GM, Choi BD, Ramundo CM, Ehret F, Barnes JA, El-Jawahri A, Hochberg EP, Johnson PC, Soumerai JD, Plotkin SR, Batchelor TT, Wick W, Maus MV, Chen YB, Frigault MJ, Dietrich J. Clinical presentation, management, and outcome of TIAN in CNS lymphoma treated with CD19-CAR T-cell therapy Blood 2025. PMID 40663771.
- Kazzi C, Simpson T, Wronski M, Seery N, Wesselingh R, Tan TH, Ko KY, Pearce DJ, Abbott C, Li J, Griffith SP, Wong SM, Westworth M, Inam S, Tam CS, Fleming S, Van Der Walt A, O'Brien TJ, Alpitsis R, Spencer A, Malpas CB, Sanfilippo P, Monif M. CLOX and neurotox: Utility of the clock drawing task in monitoring for immune effector cell-associated neurotoxicity syndrome following chimeric antigen receptor T-cell therapy British journal of haematology 2025. PMID 40717546.
- Kazzi C, Simpson T, Abbott C, Wronski M, Seery N, Tan TH, Wesselingh R, Ko K, Wong SM, Inam S, Tam C, Fleming S, O'Brien TJ, Alpitsis R, Spencer A, Malpas C, Monif M. Monitoring the neurological complications of chimeric antigen receptor (CAR) T-cell therapy in patients with sensory and physical impairments and non-native-speaking backgrounds using modified immune effector cell-associated encephalopathy (ICE) scores: a case series BMJ neurology open 2025. PMID 39950093.
- Lazzari ZT, Gandh AS, Sannananja B, Belagaje SR, Hutto SK. Progressive Multiple Cranial Neuropathies as a Manifestation of CAR-T Neurotoxicity The Neurohospitalist 2025. PMID 41147017.
- Lim KJC, Tan M, Parrondo R, Chhabra S, Dooley K, De Menezes Silva Corraes A, Carabenciov D, Gertz M, Hwa L, Haily S, Kapoor P, Kourelis T, Warsame R, Cook J, Binder M, Bergsagel PL, Yadav U, Wiedmeier-Nutor E, Geyer S, Ailawadhi S, Fonseca R, Kumar S, Zekeridou A, Lin Y. Clinical course, risk factors and mitigating strategies for Immune effector cell-associated late onset neurotoxicities after ciltacabtagene autoleucel CAR-T in multiple myeloma Blood cancer journal 2025. PMID 41476076.
- Liu B, Dai Y, Liu L, Wang Q, Guan G. The timing of concurrent intrathecal chemotherapy during blinatumomab infusion influences neurotoxicity in pediatric acute lymphoblastic leukemia Frontiers in immunology 2025. PMID 41550943.
- Llata-Ortega J, Pau-Parra A, Alonso-Martínez C, García-Roche A, Carpio C, Barba P, Murillo-Florez I, Iacoboni G. Use of Dasatinib for the Treatment of Refractory Immune Effector Cell-Associated Neurotoxicity Syndrome After Axicabtagene Ciloleucel Infusion: A Case Report European journal of haematology 2025. PMID 40954138.
- Lu IN, Müller-Miny L, Krekeler C, Cheung PF, Antonopoulou G, Jeibmann A, Schulte-Mecklenbeck A, Kerl K, Call S, Reicherts C, Bleckmann A, Stelljes M, Lenz G, Wiendl H, Meyer Zu Hörste G, Grauer OM. The CXCL16/CXCR6 axis is linked to immune effector cell-associated neurotoxicity in chimeric antigen receptor (CAR) T cell therapy Genome medicine 2025. PMID 40588764.
- Mancuso K, Talarico M, Manzato E, Barbato S, Tacchetti P, Zamagni E, Cavo M. T-Cell Redirecting Therapies in Multiple Myeloma: Pathogenesis and Management of Toxicities Beyond CRS and ICANS Cancers 2025. PMID 40361441.
- Marino MD, Mader JP, Kardasis W, Murphy M, Mahmoud SY. Evolving Magnetic Resonance Imaging (MRI) Findings in Immune Effector Cell-Associated Neurotoxicity Syndrome Cureus 2025. PMID 40539162.
- McGuire JL, Pinto S, Erdogan EN, Li Y, Bhatia A, Oztek MA, Vossough A, Wright JN, Shah R, Carapia NT, Shams N, Westermann C, Taraseviciute A, Yates B, Naik S, Gardner R, Annesley C, Hsieh E, Diorio C, Myers R, Epperly R, Talleur A, Shalabi H, Shah N, Gust J. Neuroimaging Findings in Children and Young Adults With Neurotoxicity After CAR T-Cell Therapy for B-Cell Malignancies Neurology 2025. PMID 40921024.
- Modoni A, Vollono C, Galli E, Capriati L, Sorà F, Hohaus S, Servidei S, Piccirillo N, Calabresi P, Sica S. Predictors of Neurotoxicity in a Large Cohort of Italian Patients Undergoing Anti-CD19 Chimeric Antigen Receptor (CAR) T-Cell Therapy Brain and behavior 2025. PMID 40984814.
- Mokhtari S, Asquith JM, Kareem SS, Bachmeier CA, Pina Y, Faramand RG, Kim Y, Peguero EN, Sahebjam S, Jaffer MH, Iacono DP, Jain MD, Vogelbaum MA, Davila ML, Forsyth PA, Locke FL, Lazaryan A. Intravenous Immunoglobulin (IVIG) for Patients with Severe Neurotoxicity Associated with Chimeric Antigen Receptor T-Cell (CAR-T) Therapy International journal of molecular sciences 2025. PMID 40332772.

- Nomiyama T, Setoyama D, Yamanaka I, Shimo M, Miyawaki K, Yamauchi T, Jinnouchi F, Sakoda T, Sasaki K, Shima T, Kikushige Y, Mori Y, Akashi K, Kato K, Kunisaki Y. Cerebrospinal fluid proteomics exerts predictive potential for immune effector cell-associated neurotoxicity syndrome (ICANS) in CAR-T cell therapy Leukemia 2025. PMID 40065076.
- Olaya Vargas A, Salazar-Rosales H, Caballero-Palacios M, Ramírez-Uribe N, López-Hernández G, Morales-Martinez E, Cervantes-Delgado M, Velasco Hidalgo L, Castellanos-Toledo A, Cárdenas-Cardos R, Niembro-Zúñiga A, Rivera-Luna R, Olaya-Nieto A, Ponce-Cruz J, Shalkow-Klincovstein J, Melchor-Vidal Y, Díaz-Machorro R. High-risk neuroblastoma in Mexico: from multimodal treatment to immunotherapy. Regarding the first case treated with naxitamab Frontiers in oncology 2025. PMID 40606987.
- Ow KV. CAR T-Cell Therapy Unveiled: Navigating Beyond CRS and ICANS to Address Delayed Complications and Optimize Management Strategies Journal of the advanced practitioner in oncology 2025. PMID 39990042.
- Park L, Tsai Y, Hu R, Lim H, Faulhaber LD, Burleigh K, Bose M, Faulhaber EM, Draper IH, Smith SE, Shih AY, Hirayama AY, Turtle CJ, Annesley CE, Gardner RA, Gustafson HH, Zheng Y, Gust J. Cytokines mediate increased endothelial-leukocyte interaction and brain capillary plugging during CAR T cell neurotoxicity bioRxiv 2025. DOI 10.1101/2025.02.19.638920.
- Pensato U, Muccioli L, Taurino D, Pondrelli F, Asioli GM, de Philippis C, Mannina D, Bagnato G, Marcheselli S, Zinzani PL, Bonifazi F, Bramanti S, Guarino M. Impact of neurotoxicity and steroid therapy on cancer progression-free survival in lymphoma patients treated with anti-CD19 CAR T cells Neuro-oncology practice 2025. PMID 40487586.
- Pondrelli F, Muccioli L, Mason F, Zenesini C, Ferri L, Asioli GM, Rossi S, Rinaldi R, Rondelli F, Nicodemo M, D'Angelo R, Barone V, Sambati L, Pensato U, Zinzani PL, Casadei B, Bonifazi F, Maffini E, Pierucci E, Cortelli P, Tinuper P, Bisulli F, Guarino M. EEG as a predictive biomarker of neurotoxicity in anti-CD19 CAR T-cell therapy Journal of neurology 2025. PMID 40278905.
- Rozenberg A, Shelly S, Winer R, Fineman R, Haller S, Ringelstein-Harlev S, Henig I, Yehudai-Ofir D, Zuckerman T, Tadmor T, Harel R, Ganelin-Cohen E, Beyar-Katz O. CCR5 predicts neurotoxicity in CAR-T-cell therapy British journal of haematology 2025. PMID 40572013.
- Schulenburg A, Rüsing LZ, Bumberger A, Mitterbauer M, Rabitsch W. S100 as marker for immune effector cell-associated neurotoxicity syndrome Wiener klinische Wochenschrift 2025. PMID 39365474.
- Schulenburg A, Schulenburg A, Rüsing L, Bumberger A, Mitterbauer M, Cserna J, Petrasch C, Oesterreicher S, Worel N, Rabitsch W. Retrospective Analysis: S100 as Marker for Immune Effector Cell-Associated Neurotoxicity Syndrome Oncology 2025. PMID 39999826.
- Suresh A, Wishart HA, Arslan MN, Lizcano RA, Shah PS, PonnamReddy S, Hayes CA, Jacobson BS, Moncrief G, Martinez-Camblor P, Chan AM, Meehan KR, Hill JM. Novel Neurocognitive Testing Tool for Early Neurotoxicity Detection Following Anti-CD19 and Anti-BCMA Chimeric Antigen Receptor (CAR) T-cell Therapy: A Pilot Study Clinical lymphoma, myeloma & leukemia 2025. PMID 39814673.
- Takahata A, Akita K, Shimada T, Bando K, Toyota S. Utility of CSF IL-6 monitoring in managing ICANS associated with Epcoritamab treatment: a case report and literature review Journal of clinical and experimental hematopathology : JCEH 2025. PMID 40159285.
- Ursu R, Barsan C, Di Blasi R, Belin C, Cuzzubbo S, El Soudany M, Cristinelli C, Sivapragassame R, Ouedrani A, Thieblemont C, Caillat-Zucman S, Carpentier AF. Baseline Cognitive and Neurological Status Does Not Modify the Occurrence of ICANS in CAR T-Cell Therapy for Aggressive B-Cell Lymphoma European journal of neurology 2025. PMID 40542557.
- Ursu R, Cuzzubbo S, Carpentier AF. Neurotoxicity from chimeric antigen receptor T-cells: an update on diagnosis and treatment Current opinion in neurology 2025. PMID 40748015.
- Vichare A, Lee JS, Duong TQ. Neuroimaging Findings of CAR T-Cell-Associated Neurotoxicity: A Review Neurology. Clinical practice 2025. PMID 40687961.
- Vilaseca A, Ariño H, Iacoboni G, Feijóo S, Zabalza A, Fissolo N, Arévalo MJ, Carpio C, Sánchez M, Tintoré M, Comabella M, Montalban X, Barba P, Vidal-Jordana Á. Severity-Dependent Neuroaxonal Damage Assessed by Serum Neurofilaments in ICANS Patients Undergoing CD19-Targeted CAR T-Cell Therapy European journal of neurology 2025. PMID 40765152.
- Vonberg FW, Malik I, O'Reilly M, Hyare H, Carr AS, Roddie C. Neurotoxic complications of chimeric antigen receptor (CAR) T-cell therapy Journal of neurology, neurosurgery, and psychiatry 2025. PMID 40185628.
- Wang X, He X, Zhang T, Liu J, Zhao M. Latest updates on pathogenesis mechanisms and management strategies for cytokine release syndrome, neurotoxicity, and hemophagocytic lymphohistiocytosis related to CAR-T cell therapies Annals of hematology 2025. PMID 40536724.
- Wesley SF, Friedman AR, Gordillo C, Kent PF, Reshef R. Electroencephalography in patients with immune effector cell-associated neurotoxicity syndrome (ICANS) Journal of the neurological sciences 2025. PMID 40743599.
- Wronski M, Wesselingh R, Kazzi C, Seery N, Ko K, Li J, Tan TH, Simpson T, Abbott C, Fleming S, Inam S, Tam CS, Wong SM, O'Brien T, van der Walt A, Spencer A, Butzkueven H, Monif M. Neurotoxicity associated with chimeric antigen receptor T-cell therapy Journal of neuroimmunology 2025. PMID 40812205.

- Zandaki D, Selukar S, Bi Y, Li Y, Zinsky M, Bonifant CL, Epperly R, Keerthi D, Triplett BM, Gottschalk S, Naik S, Talleur AC. EASIX and m-EASIX predict CRS and ICANS in pediatric and AYA patients after CD19-CAR T-cell therapy Blood advances 2025. PMID 39325974.
- [no author listed]. IMMU-13. TUMOR INFLAMMATION-ASSOCIATED NEUROTOXICITY (TIAN): A TOXICITY SYNDROME IN PATIENTS TREATED WITH IMMUNOTHERAPY FOR CENTRAL NERVOUS SYSTEM TUMORS Neuro-oncology 2024. abstract.
- Fenton GA, Dean EA. Exacerbations of Persistent Neurotoxicity following Axicabtagene Ciloleucel in a Patient with Relapsed/Refractory Diffuse Large B-Cell Lymphoma: A Case Report Case reports in oncology 2024. PMID 38196817.
- Ferrer-Lores B, Ortiz-Algarra A, Picó-Peris A, Estepa-Fernández A, Bellvís-Bataller F, Weiss GJ, Fuster-Matanzo A, Fernández JP, Jimenez-Pastor A, Hernani R, Saus-Carreres A, Benzaquen A, Ventura L, Piñana JL, Teruel AB, Serrano-Alcalá A, Dosdá R, Sopena-Novales P, Balaguer-Rosello A, Guerreiro M, Sanz J, Martí-Bonmatí L, Terol MJ, Alberich-Bayarri Á. Predicting survival, neurotoxicity and response in B-cell lymphoma patients treated with CAR-T therapy using an imaging features-based model EJNMMI research 2024. PMID 39567446.
- Finsterer J. Glofitamab-Associated Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS) Presenting as Serial Seizures and Responding Positively to Antiseizure Drugs and Anakinra: A Case Report Cureus 2024. PMID 38910651.
- Frenking JH, Zhou X, Wagner V, Hielscher T, Kauer J, Mai EK, Friedrich MJ, Michel CS, Hajiyianni M, Breitkreutz I, Costello P, Nadeem O, Weinhold N, Goldschmidt H, Schmitt A, Luft T, Schmitt M, Müller-Tidow C, Topp M, Einsele H, Dreger P, Munshi NC, Sperling AS, Rasche L, Sauer S, Raab MS. EASIX-guided risk stratification for complications and outcome after CAR T-cell therapy with ide-cel in relapsed/refractory multiple myeloma Journal for immunotherapy of cancer 2024. PMID 39379098.
- Han MW, Jeong SY, Suh CH, Park H, Guenette JP, Huang RY, Kim KW, Yoon DH. Incidence of immune effector cell-associated neurotoxicity among patients treated with CAR T-cell therapy for hematologic malignancies: systematic review and meta-analysis Frontiers in neurology 2024. PMID 39474369.
- Hernani R, Aiko M, Victorio R, Benzaquén A, Pérez A, Piñana JL, Hernández-Boluda JC, Amat P, Pastor-Galán I, Remigia MJ, Ferrer-Lores B, Micó M, Carbonell N, Ferreres J, Blasco-Cortés ML, Santonja JM, Dosdá R, Estellés R, Campos S, Martínez-Ciarpaglini C, Ferrández-Izquierdo A, Goterris R, Gómez M, Teruel A, Saus A, Ortiz A, Morello D, Martí E, Carretero C, Calabuig M, Tormo M, Terol MJ, Cases P, Solano C. EEG before chimeric antigen receptor T-cell therapy and early after onset of immune effector cell-associated neurotoxicity syndrome Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology 2024. PMID 38733703.
- Jiang P, Yang P, Wang W, Cao J, Chen W, Fu J, Lu L, Lu Y, Zhu X. Management of neurotoxicity syndrome complicated by autologous hematopoietic stem cell transplantation bridge to chimeric antigen receptor T-Cell therapy: A case report Asia-Pacific journal of oncology nursing 2024. PMID 38426045.
- Katsin M, Shman T, Migas A, Lutskovich D, Serada Y, Khalankova Y, Kostina Y, Dubovik S. Case report: Rapid resolution of grade IV ICANS after first line intrathecal chemotherapy with methotrexate, cytarabine and dexamethasone Frontiers in immunology 2024. PMID 38765003.
- Kikuchi T, Kunisada K, Sato K, Tsukada N, Ishida T. Severe Immune Effector Cell-Associated Neurotoxicity Syndrome in a Patient With Multiple Myeloma Treated With Elranatamab Cureus 2024. PMID 39744308.
- Larue M, Bouvier A, Maillard A, Cuffel A, Allain V, Ursu R, Carpentier AF, Azoulay E, Thieblemont C, Di Blasi R, Caillat-Zucman S. Neurofilament light chain levels as an early predictive biomarker of neurotoxicity after CAR T-cell therapy Journal for immunotherapy of cancer 2024. PMID 39317455.
- Ma Y, Zhou H, Zhang J, Zhang Q, Li Y, Xie R, Zhang B, Shen Z, Li P, Liang A, Zhou K, Han L, Hu Y, Xu K, Sang W, Wang X. The influence of CRS and ICANS on the efficacy of anti-CD19 CAR-T treatment for B-cell acute lymphoblastic leukemia Frontiers in immunology 2024. PMID 39399502.
- Mao D, Reiner AS, Chen X, Park J, Pennisi M, Perales MA, Avila EK, Santomasso BD. EEG features and synek scale indicate severity of neurotoxicity in adult patients treated with CD19 CAR T-cell therapy Scientific reports 2024. PMID 39580601.
- Mauget M, Lemercier S, Quelven Q, Maamar A, Lhomme F, De Guibert S, Houot R, Manson G. Impact of diagnostic investigations in the management of CAR T-cell-associated neurotoxicity Blood advances 2024. PMID 38501964.
- Nie EH, Su YJ, Baird JH, Agarwal N, Bharadwaj S, Weng WK, Smith M, Dahiya S, Han MH, Dunn JE, Kipp LB, Miklos DB, Scott BJ, Frank MJ. Clinical features of neurotoxicity after CD19 CAR T-cell therapy in mantle cell lymphoma Blood advances 2024. PMID 38295285.
- Ochenduszko S, Landete L, Martinez DC, Feria AG, Francés C, Torregrosa MD, Maiques IM. Cytokine release syndrome and immune effector cell-associated neurotoxicity syndrome in a melanoma patient treated with adjuvant pembrolizumab Experimental and therapeutic medicine 2024. PMID 39301256.
- Palazzo L, Pieri V, Berzero G, Filippi M. CAR-T Cells for the Treatment of Central Nervous System Tumours: Known and Emerging Neurotoxicities Brain sciences 2024. PMID 39766419.

- Pensato U, Pondrelli F, de Philippis C, Asioli GM, Crespi A, Buizza A, Mannina D, Casadei B, Maffini E, Straffi L, Marcheselli S, Zinzani PL, Bonifazi F, Guarino M, Bramanti S. Primary vs. pre-emptive anti-seizure medication prophylaxis in anti-CD19 CAR T-cell therapy Neurological sciences : official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology 2024. PMID 38512531.
- Rochate D, González-García AM, Santos Marcos C, Pérez-López E, Martín-López AÁ, Alaña M, Martín Martín L, López Parra M, Sánchez Guijo F, López Corral L. Autonomic dysfunction as manifestation of ICANS: A case report Medicine 2024. PMID 39252328.
- Sales C, Anderson MA, Kuznetsova V, Rosenfeld H, Malpas CB, Roos I, Dickinson M, Harrison S, Kalincik T. Patterns of neurotoxicity among patients receiving chimeric antigen receptor T-cell therapy: A single-centre cohort study European journal of neurology 2024. PMID 38085272.
- Shalabi H, Harrison C, Yates B, Calvo KR, Lee DW, Shah NN. Intrathecal hydrocortisone for treatment of children and young adults with CAR T-cell immune-effector cell-associated neurotoxicity syndrome Pediatric blood & cancer 2024. PMID 37897136.
- Storci G, De Felice F, Ricci F, Santi S, Messelodi D, Bertuccio SN, Laprovitera N, Dicataldo M, Rossini L, De Matteis S, Casadei B, Vaglio F, Ursi M, Barbato F, Roberto M, Guarino M, Asioli GM, Arpinati M, Cortelli P, Maffini E, Tomassini E, Tassoni M, Cavallo C, Iannotta F, Naddeo M, Tazzari PL, Dan E, Pellegrini C, Guadagnuolo S, Carella M, Sinigaglia B, Pirazzini C, Severi C, Garagnani P, Kwiatkowska KM, Ferracin M, Zinzani PL, Bonafè M, Bonifazi F. CAR+ extracellular vesicles predict ICANS in patients with B cell lymphomas treated with CD19-directed CAR T cells The Journal of clinical investigation 2024. PMID 38833312.
- Tanaka T, Taoka M, Makimoto G, Ninomiya K, Higo H, Fujii M, Ichihara E, Ohashi K, Hotta K, Tabata M, Maeda Y. Severe Cytokine Release Syndrome and Immune Effector Cell-associated Neurotoxicity Syndrome in a Man Receiving Immune Checkpoint Inhibitors for Lung Cancer Internal medicine (Tokyo, Japan) 2024. PMID 37722894.
- Tang JP, Lafeuille P, Socolov A, Diamond SS, Aptekar J, Moore TB, Nie EH, Hanudel MR, Nowicki TS. Hypophosphatemia Correction Reduces ICANS Incidence and Duration in CAR T-cell Therapy: A Pooled Clinical Trial Analysis Cancer research communications 2024. PMID 39269033.
- Yorichika Y, Neshige S, Edahiro T, Aoki S, Maruyama H. An Early Neurological Indicator of Immune Effector Cell-Associated Neurotoxicity Syndrome Cureus 2024. PMID 38690494.
- Zhao Y, Zhang X, Zhang M, Guo R, Zhang Y, Pu Y, Zhu H, Liu P, Zhang Y, He X, Lyu C, Lyu H, Xiao X, Zhao M. Modified EASIX scores predict severe CRS/ICANS in patients with acute myeloid leukemia following CLL1 CAR-T cell therapy Annals of hematology 2024. PMID 38214708.
- [no author listed]. IMMU-05. TUMOR INFLAMMATION-ASSOCIATED NEUROTOXICITY (TIAN): A TOXICITY SYNDROME IN PATIENTS TREATED WITH IMMUNOTHERAPY FOR CENTRAL NERVOUS SYSTEM TUMORS Neuro-oncology 2023. abstract.
- Barker K, Koza S, Katsanis E, Husnain M. Hypophosphatemia and pre-infusion thrombocytopenia as biomarkers for CRS and ICANS after CAR T therapy Bone marrow transplantation 2023. PMID 37580398.
- Dean EA, Peters KR, Adams CB, Hiemenz JW. Pattern of brexucabtagene autoleucel-related neurotoxicity on magnetic resonance imaging of the brain in a patient with relapsed/refractory B-cell acute lymphoblastic leukemia and prior leptomeningeal disease Radiology case reports 2023. PMID 36660565.
- Eckhardt CA, Sun H, Malik P, Quadri S, Santana Firme M, Jones DK, van Sleuwen M, Jain A, Fan Z, Jing J, Ge W, Danish HH, Jacobson CA, Rubin DB, Kimchi EY, Cash SS, Frigault MJ, Lee JW, Dietrich J, Westover MB. Automated detection of immune effector cell-associated neurotoxicity syndrome via quantitative EEG Annals of clinical and translational neurology 2023. PMID 37545104.
- Fontanelli L, Pizzanelli C, Milano C, Cassano Cassano R, Galimberti S, Rossini MI, Santo I, Turco F, Bonanni E, Siciliano G, Orciuolo E, Baldacci F. Pre-existing frontal lobe dysfunction signs as predictors of subsequent neurotoxicity in CAR T cell therapy: insights from a case series Neurological sciences : official journal of the Italian Neurological Society and of the Italian Society of Clinical Neurophysiology 2023. PMID 37160803.
- Gatto L, Ricciotti I, Tosoni A, Di Nunno V, Bartolini S, Ranieri L, Franceschi E. CAR-T cells neurotoxicity from consolidated practice in hematological malignancies to fledgling experience in CNS tumors: fill the gap Frontiers in oncology 2023. PMID 37397356.
- Genoud V, Migliorini D. Novel pathophysiological insights into CAR-T cell associated neurotoxicity Frontiers in neurology 2023. PMID 36970518.
- Huby S, Gelisse P, Tudesq JJ, Labauge P, Duflos C, Cartron G, Gallerand MA, Platon L, Badiou S, Lamure S, Menjot de Champfleur N, Ayrignac X, Taieb G. Frontal Intermittent Rhythmic Delta Activity Is a Useful Diagnostic Tool of Neurotoxicity After CAR T-Cell Infusion Neurology(R) neuroimmunology & neuroinflammation 2023. PMID 37059470.
- Jacobson CA, Rosenthal AC, Arnason J, Agarwal S, Zhang P, Wu W, Amber V, Yared JA. A phase 2 trial of defibrotide for the prevention of chimeric antigen receptor T-cell-associated neurotoxicity syndrome Blood advances 2023. PMID 37399456.

- Karschnia P, Arrillaga-Romany IC, Eichler A, Forst DA, Gerstner E, Jordan JT, Ly I, Plotkin SR, Wang N, Martinez-Lage M, Winter SF, Tonn JC, Rejeski K, von Baumgarten L, Cahill DP, Nahed BV, Shankar GM, Abramson JS, Barnes JA, El-Jawahri A, Hochberg EP, Johnson PC, Soumerai JD, Takvorian RW, Chen YB, Frigault MJ, Dietrich J. Neurotoxicity and management of primary and secondary central nervous system lymphoma after adoptive immunotherapy with CD19-directed chimeric antigen receptor T-cells Neuro-oncology 2023. PMID 37402650.
- Lutfi F, Abdallah AO, Nashatizadeh M, Ahmed N, Nelson M, Hamideh J, Mahmoudjafari Z, Shune L. Exploring the High-Grade and Refractory Neurotoxicity of Teclistamab: An Underreported Entity Cureus 2023. PMID 38130552.
- Mahdi J, Dietrich J, Straathof K, Roddie C, Scott BJ, Davidson TB, Prolo LM, Batchelor TT, Campen CJ, Davis KL, Gust J, Lim M, Majzner RG, Park JR, Partap S, Ramakrishna S, Richards R, Schultz L, Vitanza NA, Wang LD, Mackall CL, Mackall CL, Monje M. Tumor inflammation-associated neurotoxicity Nature medicine 2023. PMID 37024595.
- Nie EH, Ahmadian SS, Bharadwaj SN, Acosta-Alvarez L, Threlkeld ZD, Frank MJ, Miklos DB, Monje M, Scott BJ, Vogel H. Multifocal demyelinating leukoencephalopathy and oligodendroglial lineage cell loss with immune effector cell-associated neurotoxicity syndrome (ICANS) following CD19 CAR T-cell therapy for mantle cell lymphoma Journal of neuropathology and experimental neurology 2023. PMID 36592076.
- Pensato U, Amore G, Muccioli L, Sammali S, Rondelli F, Rinaldi R, D'Angelo R, Nicodemo M, Mondini S, Sambati L, Asioli GM, Rossi S, Santoro R, Cretella L, Ferrari S, Spinardi L, Faccioli L, Fanti S, Paccagnella A, Pierucci E, Casadei B, Pellegrini C, Zinzani PL, Bonafè M, Cortelli P, Bonifazi F, Guarino M. CAR t-cell therapy in BOlogNa-NEUrotoxicity TReatment and Assessment in Lymphoma (CARBON-NEUTRAL): proposed protocol and results from an Italian study Journal of neurology 2023. PMID 36869888.
- Pensato U, de Philippis C, Pistolese F, Mannina D, Marcheselli S, Politi LS, Santoro A, Bramanti S. Case report: Reversible punctate inflammatory foci in the corpus callosum: A novel radiological finding of CAR T-cell therapy-related neurotoxicity Frontiers in neurology 2023. PMID 36824415.
- Santomasso BD, Gust J, Perna F. How I treat unique and difficult-to-manage cases of CAR T-cell therapy-associated neurotoxicity Blood 2023. PMID 36877916.
- Satyanarayan S, Spiegel J, Hovsepian D, Markert M, Thomas R, Muffly L, Miklos D, Graber K, Scott BJ. Continuous EEG monitoring detects nonconvulsive seizure and Ictal-Interictal Continuum abnormalities in moderate to severe ICANS following systemic CAR-T therapy The Neurohospitalist 2023. PMID 36531846.
- Velasco R, Mussetti A, Villagrán-García M, Sureda A. CAR T-cell-associated neurotoxicity in central nervous system hematologic disease: Is it still a concern? Frontiers in neurology 2023. PMID 37090983.
- Amidi Y, Eckhardt CA, Quadri SA, Malik P, Firme MS, Jones DK, Jain A, Danish HH, Rubin DB, Jacobson CA, Cash SS, Lee JW, Dietrich J, Westover MB. Forecasting immune effector cell-associated neurotoxicity syndrome after chimeric antigen receptor t-cell therapy Journal for immunotherapy of cancer 2022. PMID 36450377.
- Beuchat I, Danish HH, Rubin DB, Jacobson C, Robertson M, Vaitkevicius H, Lee JW. EEG findings in CAR T-cell-associated neurotoxicity: Clinical and radiological correlations Neuro-oncology 2022. PMID 34265061.
- Cohen AD, Parekh S, Santomasso BD, Gállego Pérez-Larraya J, van de Donk NWCJ, Arnulf B, Mateos MV, Lendvai N, Jackson CC, De Braganca KC, Schecter JM, Marquez L, Lee E, Cornax I, Zudaire E, Li C, Olyslager Y, Madduri D, Varsos H, Pacaud L, Akram M, Geng D, Jakubowiak A, Einsele H, Jagannath S. Incidence and management of CAR-T neurotoxicity in patients with multiple myeloma treated with ciltacabtagene autoleucel in CARTITUDE studies Blood cancer journal 2022. PMID 35210399.
- Faulhaber LD, Phuong AQ, Hartsuyker KJ, Cho Y, Mand KK, Harper SD, Olson AK, Garden GA, Shih AY, Gust J. Brain capillary obstruction during neurotoxicity in a mouse model of anti-CD19 chimeric antigen receptor T-cell therapy Brain communications 2022. PMID 35169706.
- Garcia-Robledo JE, Valencia-Sanchez C, Knox MG, Goodman BP, Rosenthal AC, Patel B, Castro JE. It Is in the Eye of the Beholder: Ocular Ultrasound Enhanced Monitoring of Neurotoxicity after CAR-T Cell Therapy Hematology reports 2022. PMID 36648879.
- Grant SJ, Grimshaw AA, Silberstein J, Murdaugh D, Wildes TM, Rosko AE, Giri S. Clinical Presentation, Risk Factors, and Outcomes of Immune Effector Cell-Associated Neurotoxicity Syndrome Following Chimeric Antigen Receptor T Cell Therapy: A Systematic Review Transplantation and cellular therapy 2022. PMID 35288347.
- Gu T, Hu K, Si X, Hu Y, Huang H. Mechanisms of immune effector cell-associated neurotoxicity syndrome after CAR-T treatment WIREs mechanisms of disease 2022. PMID 35871757.
- Jones DK, Eckhardt CA, Sun H, Tesh RA, Malik P, Quadri S, Firme MS, van Sleuwen M, Jain A, Fan Z, Jing J, Ge W, Nascimento FA, Sheikh IS, Jacobson C, Frigault M, Kimchi EY, Cash SS, Lee JW, Dietrich J, Westover MB. EEG-based grading of immune effector cell-associated neurotoxicity syndrome Scientific reports 2022. PMID 36414694.
- Khorasanchi A, Ansari AM, Bottinor W, Simmons G, Abbate A, Toor AA. Transient left ventricular dysfunction following chimeric antigen receptor T-cell-mediated encephalopathy: A form of stress cardiomyopathy EJHaem 2022. PMID 35846197.

- Lapidus AH, Anderson MA, Harrison SJ, Dickinson M, Kalincik T, Lasocki A. Neuroimaging findings in immune effector cell associated neurotoxicity syndrome after chimeric antigen receptor T-cell therapy Leukemia & lymphoma 2022. PMID 35570737.
- Morris EC, Neelapu SS, Giavridis T, Sadelain M. Cytokine release syndrome and associated neurotoxicity in cancer immunotherapy Nature reviews. Immunology 2022. PMID 34002066.
- Pensato U, Amore G, D'Angelo R, Rinaldi R, Nicodemo M, Rondelli F, Mondini S, Santoro R, Sammali S, Farolfi A, Spinardi L, Faccioli L, Casadei B, Dicataldo M, Bonifazi F, Zinzani P, Cortelli P, Stracciari A, Guarino M. Frontal predominant encephalopathy with early paligraphia as a distinctive signature of CAR T-cell therapy-related neurotoxicity Journal of neurology 2022. PMID 34424399.
- Reveron-Thornton R, Scott BJ, Post D, Caulfield AF, Werbaneth K, Hovsepian DA, Spiegel J, Miklos D, Thomas RP, Patel CB. Recurrent Status Epilepticus in the Setting of Chimeric Antigen Receptor (CAR)-T Cell Therapy The Neurohospitalist 2022. PMID 34950390.
- Saw JL, Sidiqi MH, Ruff M, Hocker S, Alkhateeb H, Ansell SM, Bennani NN, Dingli D, Hayman SR, Johnston PB, Kapoor P, Kenderian SJ, Kourelis TV, Kumar SK, Paludo J, Shah MV, Siddiqui MA, Warsame R, Rosenthal A, Grill M, Castro JE, Siegel J, Abdel Rahman ZH, Kharfan-Dabaja MA, So E, Lin Y. Acute seizures and status epilepticus in immune effector cell associated neurotoxicity syndrome (ICANS) Blood cancer journal 2022. PMID 35418113.
- Shalabi H, Martin S, Yates B, Wolters PL, Kaplan C, Smith H, Sesi CR, Jess J, Toledo-Tamula MA, Struemph K, Delbrook CP, Khan OI, Mackall CL, Lee DW, Shah NN. Neurotoxicity following CD19/CD28ζ CAR T-cells in children and young adults with B-cell malignancies Neuro-oncology 2022. PMID 35148417.
- Shalabi H, Nellan A, Shah NN, Gust J. Immunotherapy Associated Neurotoxicity in Pediatric Oncology Frontiers in oncology 2022. PMID 35265526.
- Sterner RC, Sterner RM. Immune effector cell associated neurotoxicity syndrome in chimeric antigen receptor-T cell therapy Frontiers in immunology 2022. PMID 36081506.
- Tang JP, Peters CW, Quiros C, Wang X, Klomhaus AM, Yamada RE, Timmerman JM, Moore TB, Nowicki TS. Hypophosphatemia Due to Increased Effector Cell Metabolic Activity Is Associated with Neurotoxicity Symptoms in CD19-Targeted CAR T-cell Therapy Cancer immunology research 2022. PMID 36259217.
- Wieczorek A, Manzitti C, Garaventa A, Gray J, Papadakis V, Valteau-Couanet D, Zachwieja K, Poetschger U, Pribill I, Fiedler S, Ladenstein R, Lode HN. Clinical Phenotype and Management of Severe Neurotoxicity Observed in Patients with Neuroblastoma Treated with Dinutuximab Beta in Clinical Trials Cancers 2022. PMID 35454826.
- Śliwa-Tytko P, Kaczmarska A, Lejman M, Zawitkowska J. Neurotoxicity Associated with Treatment of Acute Lymphoblastic Leukemia Chemotherapy and Immunotherapy International journal of molecular sciences 2022. PMID 35628334.
- Bhojwani D, Bansal R, Wayne AS. Managing therapy-associated neurotoxicity in children with ALL Hematology. American Society of Hematology. Education Program 2021. PMID 34889354.
- Brown BD, Tambaro FP, Kohorst M, Chi L, Mahadeo KM, Tewari P, Petropoulos D, Slopis JM, Sadighi Z, Khazal S. Immune Effector Cell Associated Neurotoxicity (ICANS) in Pediatric and Young Adult Patients Following Chimeric Antigen Receptor (CAR) T-Cell Therapy: Can We Optimize Early Diagnosis? Frontiers in oncology 2021. PMID 33763368.
- Burton LB, Eskian M, Guidon AC, Reynolds KL. A review of neurotoxicities associated with immunotherapy and a framework for evaluation Neuro-oncology advances 2021. PMID 34859238.
- Castaneda-Puglianini O, Chavez JC. Assessing and Management of Neurotoxicity After CAR-T Therapy in Diffuse Large B-Cell Lymphoma Journal of blood medicine 2021. PMID 34466048.
- Foster MC, Savoldo B, Lau W, Rubinos C, Grover N, Armistead P, Coghill J, Hagan RS, Morrison K, Buchanan FB, Cheng C, Laing S, Ivanova A, West J, Foster A, Serody J, Dotti G. Utility of a safety switch to abrogate CD19.CAR T-cell-associated neurotoxicity Blood 2021. PMID 33624095.
- Gonzalez Castro LN, Dietrich J. Evaluation and management of chimeric antigen receptor (CAR) T-cell-associated neurotoxicity Neuro-oncology practice 2021. PMID 34055373.
- Gust J, Annesley CE, Gardner RA, Bozarth X. EEG Correlates of Delirium in Children and Young Adults With CD19-Directed CAR T Cell Treatment-Related Neurotoxicity Journal of clinical neurophysiology : official publication of the American Electroencephalographic Society 2021. PMID 31851018.
- Holtzman NG, Xie H, Bentzen S, Kesari V, Bukhari A, El Chaer F, Lutfi F, Siglin J, Hutnick E, Gahres N, Ruehle K, Ahmad H, Shanholtz C, Kocoglu MH, Badros AZ, Yared JA, Hardy NM, Rapoport AP, Dahiya S. Immune effector cell-associated neurotoxicity syndrome after chimeric antigen receptor T-cell therapy for lymphoma: predictive biomarkers and clinical outcomes Neuro-oncology 2021. PMID 32750704.
- Pensato U, Muccioli L, Cani I, Janigro D, Zinzani PL, Guarino M, Cortelli P, Bisulli F. Brain dysfunction in COVID-19 and CAR-T therapy: cytokine storm-associated encephalopathy Annals of clinical and translational neurology 2021. PMID 33780166.
- Sheth VS, Gauthier J. Taming the beast: CRS and ICANS after CAR T-cell therapy for ALL Bone marrow transplantation 2021. PMID 33230186.

- Xiao X, Huang S, Chen S, Wang Y, Sun Q, Xu X, Li Y. Mechanisms of cytokine release syndrome and neurotoxicity of CAR T-cell therapy and associated prevention and management strategies Journal of experimental & clinical cancer research : CR 2021. PMID 34794490.
- Yuen C, Rezania K, Kelly T, Bishop M. NCMP-10. DYSGRAPHIA AS THE EARLIEST PRESENTING SYMPTOM OF SEVERE CHIMERIC ANTIGEN RECEPTOR (CAR) T-CELL THERAPY NEUROTOXICITY: A SINGLE CENTER EXPERIENCE Neuro-oncology 2021. abstract.
- Achbergerová M, Hederová S, Mikesková M, Husáková K, Hrašková A, Kolenová A. Implementation of immunotherapy into the treatment of neuroblastoma - single center experience with the administration of dinutuximab and management of its adverse effects Klinicka onkologie : casopis Ceske a Slovenske onkologicke spolecnosti 2020. PMID 33108882.
- Belin C, Devic P, Ayrignac X, Dos Santos A, Paix A, Sirven-Villaros L, Simard C, Lamure S, Gastinne T, Ursu R, Berger C, Platon L, Tessoulin B, Azoulay E, Wallet F, Thieblemont C, Bachy E, Cartron G, Laplaud DA, Carpentier AF. Description of neurotoxicity in a series of patients treated with CAR T-cell therapy Scientific reports 2020. PMID 33149178.
- Brown B, Mahadeo K, Khazal S, Petropoulos D, Tewari P, Slopis J, Tambaro P, Sadighi Z. IMMU-52. IMMUNE EFFECTOR CELL ASSOCIATED NEUROTOXICITY (ICANS) AMONG PEDIATRIC AND AYA PATIENTS: MD ANDERSON CANCER CENTER EXPERIENCE Neuro-oncology 2020. abstract.
- Brown B, Tambaro P, Mahadeo K, Khazal S, Tewari P, Petropoulos D, Slopis J, Sadighi Z. IMMU-07. IMMUNE EFFECTOR CELL ASSOCIATED NEUROTOXICITY (ICANS) AMONG PEDIATRIC AND AYA PATIENTS: MD ANDERSON CANCER CENTER EXPERIENCE Neuro-oncology 2020. abstract.
- Butt O, Zhou A, Schindler S, Fagan A, Dipersio J, Ghobadi A, Ances B. NCMP-24. PRE-INFUSION NEUROFILAMENT LIGHT CHAIN (NFL) LEVELS PREDICT THE DEVELOPMENT OF IMMUNE EFFECTOR CELL-ASSOCIATED NEUROTOXICITY SYNDROME (ICANS) Neuro-oncology 2020. abstract.
- Freyer CW, Porter DL. Cytokine release syndrome and neurotoxicity following CAR T-cell therapy for hematologic malignancies The Journal of allergy and clinical immunology 2020. PMID 32771558.
- Gust J, Ponce R, Liles WC, Garden GA, Turtle CJ. Cytokines in CAR T Cell-Associated Neurotoxicity Frontiers in immunology 2020. PMID 33391257.
- Gust J, Wilson A, Finney O, Hartsuyker K, Narayanaswamy P, Wu V, Garden G, Annesley C, Gardner R. IMMU-02. CHIMERIC ANTIGEN RECEPTOR (CAR) T CELL NEUROTOXICITY CORRELATES WITH PRETREATMENT AND ACUTE CSF NEUROFILAMENT LIGHT CHAIN (NFL) LEVELS Neuro-oncology 2020. abstract.
- Mokhtari S, Asquith J, Bachmeier C, Faramand R, Kim Y, Peguero E, Sahebjam S, Jain M, Vogelbaum M, Davila M, Forsyth P, Locke F, Aleksandr L. NCMP-27. THE USE OF INTRAVENOUS IMMUNOGLOBULIN (IVIG) DURING SEVERE NEUROTOXICITY AMONG THE RECIPIENTS OF CHIMERIC ANTIGEN RECEPTOR T-CELLS (CAR-T) THERAPY Neuro-oncology 2020. abstract.
- Strati P, Nastoupil LJ, Westin J, Fayad LE, Ahmed S, Fowler NH, Hagemeister FB, Lee HJ, Iyer SP, Nair R, Parmar S, Rodriguez MA, Samaniego F, Steiner RE, Wang M, Pinnix CC, Adkins S, Claussen CM, Martinez CS, Hawkins MC, Johnson NA, Singh P, Mistry HE, Horowitz S, George S, Feng L, Kebriaei P, Shpall EJ, Neelapu SS, Tummala S, Chi TL. Clinical and radiologic correlates of neurotoxicity after axicabtagene ciloleucel in large B-cell lymphoma Blood advances 2020. PMID 32822484.
- Wang M, Jain P, Chi TL, Chen SE, Heimberger A, Weathers SP, Zheng L, Rao AV, Rossi JM. Management of a patient with mantle cell lymphoma who developed severe neurotoxicity after chimeric antigen receptor T-cell therapy in ZUMA-2 Journal for immunotherapy of cancer 2020. PMID 33067318.
- Gonzalez Castro LN, Dietrich J, Forst DA. Stuttering as the first sign of CAR-T-cell-related encephalopathy syndrome (CRES) Journal of cancer research and clinical oncology 2019. PMID 30539279.
- Gust J, Finney OC, Li D, Brakke HM, Hicks RM, Futrell RB, Gamble DN, Rawlings-Rhea SD, Khalatbari HK, Ishak GE, Duncan VE, Hevner RF, Jensen MC, Park JR, Gardner RA. Glial injury in neurotoxicity after pediatric CD19-directed chimeric antigen receptor T cell therapy Annals of neurology 2019. PMID 31074527.
- Gofshteyn JS, Shaw PA, Teachey DT, Grupp SA, Maude S, Banwell B, Chen F, Lacey SF, Melenhorst JJ, Edmonson MJ, Panzer J, Barrett DM, McGuire JL. Neurotoxicity after CTL019 in a pediatric and young adult cohort Annals of neurology 2018. PMID 30178481.
- Gust J, Taraseviciute A, Turtle CJ. Neurotoxicity Associated with CD19-Targeted CAR-T Cell Therapies CNS drugs 2018. PMID 30387077.
- Neelapu SS, Tummala S, Kebriaei P, Wierda W, Gutierrez C, Locke FL, Komanduri KV, Lin Y, Jain N, Daver N, Westin J, Gulbis AM, Loghin ME, de Groot JF, Adkins S, Davis SE, Rezvani K, Hwu P, Shpall EJ. Chimeric antigen receptor T-cell therapy - assessment and management of toxicities Nature reviews. Clinical oncology 2018. PMID 28925994.
- Santomasso BD, Park JH, Salloum D, Riviere I, Flynn J, Mead E, Halton E, Wang X, Senechal B, Purdon T, Cross JR, Liu H, Vachha B, Chen X, DeAngelis LM, Li D, Bernal Y, Gonen M, Wendel HG, Sadelain M, Brentjens RJ. Clinical and Biological Correlates of Neurotoxicity Associated with CAR T-cell Therapy in Patients with B-cell Acute Lymphoblastic Leukemia Cancer discovery 2018. PMID 29880584.

- Taraseviciute A, Tkachev V, Ponce R, Turtle CJ, Snyder JM, Liggitt HD, Myerson D, Gonzalez-Cuyar L, Baldessari A, English C, Yu A, Zheng H, Furlan SN, Hunt DJ, Hoglund V, Finney O, Brakke H, Blazar BR, Berger C, Riddell SR, Gardner R, Kean LS, Jensen MC. Chimeric Antigen Receptor T Cell-Mediated Neurotoxicity in Nonhuman Primates Cancer discovery 2018. PMID 29563103.
- Gust J, Hay KA, Hanafi LA, Li D, Myerson D, Gonzalez-Cuyar LF, Yeung C, Liles WC, Wurfel M, Lopez JA, Chen J, Chung D, Harju-Baker S, Özpolat T, Fink KR, Riddell SR, Maloney DG, Turtle CJ. Endothelial Activation and Blood-Brain Barrier Disruption in Neurotoxicity after Adoptive Immunotherapy with CD19 CAR-T Cells Cancer discovery 2017. PMID 29025771.

cytokine release syndrome (185)

- [no author listed]. P-2148. Infection Following Cytokine Release Syndrome in Patients Receiving Chimeric Antigen Receptor T-Cell Therapy Open forum infectious diseases 2026. abstract.
- Alarcon Tomas A, Segura A, Chavez J. Managing cytokine release syndrome in hematological malignancies being treated with CAR-T cell therapy Expert review of hematology 2026. PMID 42233607.
- Alotaibi A, Alajmi N, Alnuhait M. Severe Cytokine Release Syndrome After CAR T Cell Therapy in a Pediatric Patient With Relapsed ALL Case reports in oncological medicine 2026. PMID 41696549.
- Arai Y, Jo T, Sato T, Sakurai M, Kaji D, Kitawaki T, Shimada K, Shimoyama T, Yoshihara S, Makita S, Fujii N, Yamamoto G, Kataoka K, Sakaida E, Goto H, Nakashima Y, Yoshida A, Umezawa Y, Kim H, Kato M, Atsuta Y, Kato K. Decreased renal function predicts severe cytokine release syndrome after CAR-T-cell therapy for large B-cell lymphoma British journal of haematology 2026. PMID 41986280.
- Attia MS, Dyer B, McMillan N, Zunk M. A systematic review and meta-analysis of shRNA-IL-6-engineered CAR-T cells for B-cell acute lymphoblastic leukemia: a stepping stone toward risk-free immunotherapy Bioscience reports 2026. PMID 42012511.
- Bamba S, Jo T, Kitawaki T, Yoshinaga N, Sakamoto T, Kanda J, Nishikori M, Yamashita K, Nagao M, Takaori-Kondo A, Arai Y. Prolonged Cytokine Release Syndrome is Associated with Hypofibrinogenemia after CD19-Directed CAR T-Cell Therapy in Adult Patients with Relapsed/Refractory Lymphoma Transplantation and cellular therapy 2026. PMID 42173255.
- Berna A, Fahrmann J, Irajizad E, Rudsari H, Liu Y, Logan J, Murtada K, Grandy J, Edwards M, Ayers A, Ahmed S, Neelapu S, Saini N, John A, John T. Pre-infusion Exhaled breath volatile organic compounds predict severe CRS and ICANS after CAR T-cell therapy medRxiv 2026. DOI 10.64898/2026.05.28.26354352.
- Besbassi H. A PK-Driven Quantitative Systems Pharmacology Model Predicts Cytokine Release Syndrome Severity Across T Cell-Activating Therapies via a Locked Cytokine Amplification Network bioRxiv 2026. DOI 10.64898/2026.05.05.722920.
- Bland V, Przybylski D. Infectious complications associated with tocilizumab following chimeric antigen receptor T-cell therapy Journal of oncology pharmacy practice : official publication of the International Society of Oncology Pharmacy Practitioners 2026. PMID 42154770.
- Clerbaux LA, Alb M, Fogal B, Hartung T, Hench VK, Morgan H, Gavins FNE, Lindeman B, Margiotta-Casaluci L, Ostaszewski M, Reiche K, Sachana M, Wittwehr C, Sewald K. Filling the mechanistic and methodological gaps for predicting cytokine release syndrome in humans Frontiers in pharmacology 2026. PMID 42382173.
- Easton N, Andreoli M, van Besien H, Gribbin C, Pasciolla M, Alperovich A, Saldarriaga MM, Ma B, Chokr N, Assal A, Fein J, Mayer S, Arteaga AG, Choi D, Shore T, Hackett CS, Barker J, Yamshon S. Real-World Assessment of Prophylactic Anakinra on Neurotoxicity and Cytokine Release Syndrome after CD19 Chimeric Antigen Receptor T-Cell Therapy in Relapsed and Refractory B-Cell Lymphoma: An Inverse Probability of Treatment Weighting Analysis Transplantation and cellular therapy 2026. PMID 42309461.
- Frigault MJ, Maziarz RT, Amoloja T, Burke L, Bhatt V, Moskop A, Pasquini M, DiPersio JF. Cytokine Release Syndrome and Immune Effector Cell-Associated Neurotoxicity Syndrome Following Axicabtagene Ciloleucel or Brexucabtagene Autoleucel Chimeric Antigen Receptor T-Cell Therapy: Center for International Blood and Marrow Transplant Research Data Transplantation and cellular therapy 2026. PMID 42537846.
- Gabriel K, Heinzerling L, von Baumgarten L, Subklewe M, Kobold S. The role of cytokines in cytokine release syndrome (CRS) after CAR T cell therapy Biochimica et biophysica acta. Molecular cell research 2026. PMID 41581566.
- Georgiev V, Anderka I, Bucher D, Preiss L, Zhang JD, Umehara K, Parrott N. Current status of prediction of IL-6 mediated cytochrome P450 activity modulation using in vitro data and PBPK modeling Drug metabolism and pharmacokinetics 2026. PMID 41499926.
- Goala P, Zhang Y, Beatty N, Pavy A, McSain S, Sailer C, Tariq MJ, Hamid S, Cortes Gomez E, Wang J, Massillon D, Ilecki M, Boucher JC, Savid-Frontera C, Lee SB, Kotani H, Stone ML, Jain MD, Davila ML. IFN-γ-driven skewing towards Th1 over Th17 differentiation underlies CRS and neutropenia in CAR-T therapy The Journal of clinical investigation 2026. PMID 41165751.

- Golec S, Fu W, Moshier E, Peleg A, Pileggi B, Rossi A, Sahni G. Relationship between early use of tocilizumab during chimeric antigen receptor T-cell therapy for multiple myeloma and cardiovascular risk and progression-free survival European heart journal open 2026. PMID 42368427.
- Gomez-Llobell M, Escribano Serrat S, Bromberg M, Seshan V, Devlin S, Einarsdottir S, Ben Valid O, Golan-Aceb N, Marcus R, Simoni A, Beyar-Katz O, Perek S, Abed Al Wahad A, Corona M, Rejeski K, Raj SS, Dahi P, Scordo M, Shah GL, Palomba ML, Salles G, Park JH, Shouval R, Avigdor A, Sanz J, Perales MA. Impact of ASTCT Toxicity Grading System on Outcomes After CAR-T for Mature B-Cell Malignancies Transplantation and cellular therapy 2026. PMID 42178058.
- He M, Liu L, Chen Z, Liu Y, Liu C, Ma M, Li J, Dong B, Gong J, Shen L, Tang L, Qi C. A two-step scoring model incorporating visceral-to-subcutaneous fat ratio and systemic immunoinflammatory index for predicting cytokine release syndrome severity in patients with gastric cancer receiving Claudin18.2-targeted CAR-T cell therapy Cancer immunology, immunotherapy : CII 2026. PMID 41801430.
- Hu T, Li J, He L, Zhang X, Xu J, Li T, Zhou L. Case Report: Early introduction of anti-GD2 immunotherapy during induction chemotherapy in high-risk neuroblastoma: two pediatric cases Frontiers in oncology 2026. PMID 42614217.
- Hunter BD, Lunning M, Shadman M, Ahmed S, Abramson JS, Perales MA, Ahmed N, Mirza AS, Isufi I, Frigault MJ, Crombie JL, Miklos DB, Vasconcelos A, Crotta A, Bernasconi D, Roy D, Bleickardt E, Pasquini MC, Kamdar M. CRS or ICANS Are Rare Beyond 2 Weeks After Lisocabtagene Maraleucel Infusion: Data From Clinical Trials and the Real-World Setting Transplantation and cellular therapy 2026. PMID 41235976.
- Irajizad E, Fahrmann JF, Katayama H, Strati P, Nair R, Chihara D, Ahmed S, Iyer SP, Locke FL, Davila M, Flowers CR, Shpall E, Jenq R, Neelapu SS, Hanash S, Westin J, Jain MD, John TM, Saini NY. Pre-infusion plasma proteomics identifies an endothelial–immune priming signature predictive of severe cytokine release syndrome and neurotoxicity following CAR T-cell therapy in relapsed/refractory lymphoma medRxiv 2026. DOI 10.64898/2026.03.29.26349664.
- Khoury A, El Halabi L, Al Bayeh A, Katrib M, Nasr J, Reddy P. Tocilizumab: From Bench to Bedside - A Comprehensive Review Kansas journal of medicine 2026. PMID 42137382.
- Liang J, Hu F, Li J, Zhai Y, Guo X, Xu J, Ye X, He J. Spectrum and prognosis of CAR-T-related endocrine toxicity: glucose/calcium dysregulation, pituitary axis disorders, and overlap with CRS Frontiers in medicine 2026. PMID 42058415.
- Lu ER, Hoo CPL, Cheung HM, Wong IWC, Fung KFK, Chang SLY, Lee ACW, Fu ECH, Ku DTL, Yau JPW, Shing MMK, Mak CYK, Liu APY, Chan GCF. End-of-Induction Response and Tolerability of High-Risk Neuroblastoma Treated with Chemoimmunotherapy-Modified N7 Regimen with Dinutuximab Beta Cancers 2026. PMID 41899628.
- Luo Q, Zhang X, Qiu X, Jiang H, Xu Y. Teclistamab-Induced Localized Pleural Cavity Cytokine Release Syndrome in a Multiple Myeloma Patient Managed with Intrapleural Dexamethasone Administration Journal of blood medicine 2026. PMID 42051974.
- Maggi R, Ricciuti EF, Galli E, Sorà F, Fiorita A, Sica S, Chiusolo P. Organ-specific local cytokine release syndrome after anti-CD19 CAR-T therapy with salivary gland involvement: a case report, literature review, and a diagnostic alert for tocilizumab-associated cervical swelling Annals of hematology 2026. PMID 42014477.
- Miyashita A, Castellanos MSJ, Rodgers T. Efficacy and Safety of Bendamustine Versus Fludarabine and Cyclophosphamide as Lymphodepleting Regimens for CAR-T Therapy in Hematological Malignancies: A Systematic Review and Meta-Analysis European journal of haematology 2026. PMID 42063324.
- Mochizuki T, Shibata K, Naito T, Shimoyama K, Ogawa N, Kawakami J. Relationships of serum levels of tocilizumab to interleukin-6 and endogenous markers of CYP3A activity in patients with rheumatoid arthritis Annals of clinical biochemistry 2026. PMID 40562694.
- Monaca F, Woodford R, Joshi K, Thistlethwaite F, Lim KHJ, Gomez-Randulfe I. Cytokine release syndrome in solid tumours: mechanism-based therapeutic strategies for prevention and management Therapeutic advances in medical oncology 2026. PMID 42306495.
- Moon S, Kang DW, Choi YJ, Shin S. Signal Detection and Machine Learning-Based Prediction of Cytokine Release Syndrome in B-Cell Maturation Antigen-Targeting Immunotherapies Using FAERS Data Pharmaceuticals (Basel, Switzerland) 2026. PMID 42198343.
- Rajeeve S, Wilkes M, Zahradka N, Tomalin L, Quidwai M, Pan D, Calafat NJ, Cusack M, Aleman A, Serebryakova K, Kappes K, Jackson H, Agte S, Thibaud S, Sanchez L, Richard S, Richter J, Rodriguez C, Cho HJ, Chari A, Jagannath S, Laganà A, Rossi AC, Parekh S. Detection of cytokine release syndrome using wearables and cytokine profiling following CAR-T therapy for myeloma JCI insight 2026. PMID 42084945.
- Ren Y, Lin Z, Li S, Zhang R, Lai Z, Jiang H, Qu Z, Huo G, Zhang D, Yang Y, Li B, Geng X. An in vitro co-culture model with CAR-T cells, antigen-presenting cells, and tumor cells to evaluate CAR-T cell-induced cytokine release syndrome Frontiers in cellular and infection microbiology 2026. PMID 41669611.

- Rossi JF, Cailleux T, Wisnewski E, Coussirou J, de Crozals F. Case Report: Targeted interleukin-6 blockade by siltuximab for cytokine release syndrome control and infection limitation in thirteen patients treated with bi-specific T-cell engagers Frontiers in immunology 2026. PMID 41766898.
- Sainatham C, Goloubeva O, Margiotta P, Alharthy H, Patel I, Gaddipati G, Bukhari A, Law J, Lee ST, Kocoglu M, Yared J, Hardy N, Nelson M, Ahmed N, Mushtaq MU, McGuirk J, Rapoport AP, Dahiya S, Lutfi F. Real-World Experience With a ZUMA-1 Cohort 4 Adopted Approach to CRS and ICANS in CAR-T Recipients European journal of haematology 2026. PMID 42119568.
- Suzuki K, Uehara J, Takeda K, Ishizuki S, Aoki S, Sekiyama R, Akiya M, Yoshino K. Cytokine Release Syndrome Induced by Combination Therapy With Nivolumab and Ipilimumab in Lower Palpebral Conjunctival Melanoma: A Case Report The Journal of dermatology 2026. PMID 42493823.
- Tashiro Y, Jo T, Kitawaki T, Yoshinaga N, Sakamoto T, Shirakawa K, Kanda J, Nishikori M, Yamashita K, Nagao M, Takaori-Kondo A, Arai Y. Low reticulocyte count at infusion is a risk factor for high-grade cytokine release syndrome in chimeric antigen receptor T cell therapy International journal of hematology 2026. PMID 41247641.
- Thukral J, Sindhwani N, Khan K, Moudgil P, Singla R, Garg K, Shah RK, Kaur H, Thukral N, Agrawal SP, Frishman WH, Aronow WS. Pre-emptive and Prophylactic Strategies to Prevent Cardiovascular and Neurotoxicity in Chimeric Antigen Receptor T-Cell Therapy: The Role of Tocilizumab, Anakinra, and Cytokine-Targeted Interventions Cardiology in review 2026. PMID 42190079.
- Wang J, Arora K, de Lima M. Cytokine release syndrome after allogeneic hematopoietic stem cell transplantation using posttransplant cyclophosphamide: current understanding and management Frontiers in immunology 2026. PMID 42158862.
- Wang WL, Lee D, Cheung E, Jensen A, Rana MS, Goyal A, Hamilton M, Ghalehsari N, Latchford T, Scott B, Mallampet J, Heng E, Khan H, Zarak MS, Lohman C, Sahaf B, Sidana S, Weng WK, Smith M, Bharadwaj S, Frank MJ, Miklos DB, Dahiya S. Early steroid and anakinra use to manage axicabtagene ciloleucel toxicity reduces the total duration of CRS and ICANS Blood advances 2026. PMID 41855506.
- Wu M, Guo W, Wang B, Gu M, Wang C. Serial monitoring of cytokine profile, liver function, and hematological biomarkers in severe CRS and CLS following CAR-T cell therapy for relapsed hepatocellular carcinoma: a case report Clinica chimica acta; international journal of clinical chemistry 2026. PMID 42097197.
- Yi M, Chen S, Xu H, Liu C, Lang H, Yuan X. Dinutuximab beta-based first-line maintenance therapy for newly diagnosed high-risk neuroblastoma following a COG-based pathway: a single-center real-world study in China Research Square 2026. DOI 10.21203/rs.3.rs-10192839/v1.
- Yomogida D, Hasegawa S, Mizuta S, Horikawa S, Koshida Y. Successful use of Remcyte®-based continuous hemodiafiltration in a case of refractory CRS and ICANS complicated by tumor lysis syndrome-associated AKI CEN case reports 2026. PMID 41632354.
- Zhang J, Shi W, Yang J, Qian J, Su M, An K, Wang T, Jia R, Fei C, Tang Y, Li B. Emapalumab in pediatric patients with high-grade cytokine release syndrome associated with CAR T-cell therapy Frontiers in immunology 2026. PMID 41859101.
- Zuo R, Han M. Beyond the infusion: nursing at the vanguard of cytokine release syndrome rescue in CAR-T cell therapy Clinical & translational oncology : official publication of the Federation of Spanish Oncology Societies and of the National Cancer Institute of Mexico 2026. PMID 41166025.
- Arvanitis P, Tziotis A, Papadimatos S, Farmakiotis D. Pathogenesis, Diagnosis, and Management of Cytokine Release Syndrome in Patients with Cancer: Focus on Infectious Disease Considerations Current oncology (Toronto, Ont.) 2025. PMID 40277755.
- Bajwa A, Zhao Q, Geer M, Lin C, Westholder J, Maakaron J, Ghosh M, Frame D, Galal A, Tossey J, Ahmed N, Bezerra E, Denlinger N, de Lima M, Epperla N, Caimi P, Voorhees T. Siltuximab for chimeric antigen receptor T-cell therapy-related CRS and ICANS: a multicenter retrospective analysis Blood advances 2025. PMID 39437770.
- Barker K, Marco T, Husnain M, Katsanis E. Addition of Phosphorous and IL6 to m-EASIX Score Improves Detection of ICANS and CRS, as Well as CRS Progression Cancers 2025. PMID 40149255.
- Cai L, Wen X, Qiu Z, Fu A, Guo D, Zhu M. Characteristics, risk factors and a risk prediction model of tocilizumab-induced hypofibrinogenemia: a retrospective real-world study of inpatients BMC pharmacology & toxicology 2025. PMID 39789607.
- Cai W, Lu Y, He H, Li J, Liu S, Geng H, Yang Q, Zeng L, Wu D, Li C. Efficacy of emapalumab in the management of anti-CD19 chimeric antigen receptor T-cell therapy-associated cytokine release syndrome: A report of two cases Oncology letters 2025. PMID 39628826.
- Cheng J, Song Y. Serum ferritin as a prognostic biomarker in CAR-T therapy for multiple myeloma: A meta-analysis Biomolecules & biomedicine 2025. PMID 40198890.
- Driscoll DF, Bistrian BR. The physiological and pharmaceutical roles of MCTs and other ingredients in intravenous emulsions containing omega-3 enriched fish oil designed to mitigate cytokine release syndrome Frontiers in pharmacology 2025. PMID 41059195.

- Du M, Mayombo RTM, Liu J, Zhang Y, Liao D, Hu Y, Mei H. The impact of obesity and its related underlying diseases on cytokine release syndrome and the efficacy of CAR-T therapy in treating B-cell malignancies Annals of hematology 2025. PMID 40195173.
- Frigault MJ, Yao N, Berger TR, Wehrli M, Gallagher KME, Horick N, Graham CE, Jacobson CA, Chen YB, Leick MB, DeFilipp Z, El-Jawahri AR, Johnson PC, Dolaher M, Katsis K, Kim AI, Crombie J, Merryman RW, Cook D, Trailor M, Cho H, Jeffrey R, Shen R, Filosto S, Nater J, Getz G, Haradhvala NJ, Maus MV. Single-cell dynamics of breakthrough toxicities after anakinra prophylaxis for axicabtagene ciloleucel in lymphoma Blood advances 2025. PMID 39928957.
- Galán-Gómez V, González-Martínez B, Alonso-Saladrigues A, Rives S, Herrero B, Kwon M, Sánchez-Pina J, Minguillón J, Martínez-Romera I, Mirones Aguilar I, Mestre-Durán C, Casado G, Sánchez-Martín M, Echecopar C, González-Pérez C, León-Triana O, Aguirre-Portolés C, Molinos-Quintana Á, Barba P, Balsalobre P, Pérez-Martínez A. Siltuximab for the treatment of early complications after chimeric antigen receptor T-cell therapy for acute lymphoblastic leukemia in children, adolescents, and young adults Experimental hematology & oncology 2025. PMID 40176077.
- Gligorevic S, Brezic N, Jagodzinski J, Radulovic A, Peranovic A, Dumic I. Multiorgan Failure Resembling Grade 5 (Fatal) Cytokine Release Syndrome in Patient with Multiple Myeloma Following Carfilzomib Infusion: A Case Report Journal of clinical medicine 2025. PMID 40649097.
- Hayashino K, Kitamura W, Fujii N, Terao T, Kobayashi H, Kamoi C, Kondo T, Seike K, Fujiwara H, Asada N, Ennishi D, Fujii K, Maeda Y. Severe hypophosphatemia following idecabtagene vicleucel regardless of the severity of cytokine release syndrome Cytotherapy 2025. PMID 39846935.
- Hu J, Feng C, He L, Wang Y. Chimeric antigen receptor T cell immunotherapy-associated hemophagocytic lymphohistiocytosis: Pathogenesis, clinical manifestation, diagnosis and management compared with cytokine release syndrome (Review) Molecular medicine reports 2025. PMID 40417883.
- Ibraheem A, Dalby M. Proposed observational study protocol for early differentiation of cytokine release syndrome and sepsis in CAR-T recipients with haematological malignancies using the IL-6/PCT ratio: the DRACARYS study Frontiers in oncology 2025. PMID 41347097.
- Islam J, Rashid Z, Munster AB, Hughes P, Badzek S, Huda I. Recurrent Cytokine Release Syndrome in Metastatic Gastric Cancer on Nivolumab-FOLFOX: A Case Report and Mechanistic Review Journal of gastrointestinal cancer 2025. PMID 41186785.
- Jamois C, Turner DC, Gibiansky L, Li F, Menuet JF, Pereira LR, Lundberg L, Guerini E, Carlile DJ, Relf J, Wei MC, Kwan A, Frey N, Grange S. Tocilizumab Dosing for Management of T Cell-Engaging Bispecific Antibody-Related CRS in Patients With R/R B-Cell NHL Clinical pharmacology and therapeutics 2025. PMID 40576759.
- Jiang YH, Zhou M, Cheng MD, Chen S, Guo YQ. CAR-engineered cytolytic Tregs reverse pulmonary fibrosis and remodel the fibrotic niche with limited CRS JCI insight 2025. PMID 40627450.
- Ke S, Zhang TY, Wu ZL, Xie W, Liu L, Du MY. Novel Prognostic Scoring Systems for Severe CRS after Anti-CD19 CAR-T-Cells in Acute B-Lymphoblastic Leukemia Current medical science 2025. PMID 40928684.
- Khalafallah SA, Mohammed Elzein A, Faris IM, Attaelmanan GA, Alamin MF, Almleeh A, Bader M, Alfaki M. Identification of Hub Genes and Pathways in Preinfusion Chimeric Antigen Receptor (CAR) T-cell Products Associated With Cytokine Release Syndrome Cureus 2025. PMID 40370902.
- Khan MA, Khan FH, Khan A, Majeed A, Gorsline C, Ahmed N. Predicators and Outcomes of Cytomegalovirus Reactivation in Chimeric Antigen Receptor T-Cell Therapy: A Systematic Review Clinical lymphoma, myeloma & leukemia 2025. PMID 40940256.
- Kowalski A, Lykon J, Diamond B, Coffey D, Kaddoura M, Maura F, Hoffman J, Pandey A, Kazandjian D, Landgren O. Tocilizumab prophylaxis for patients with multiple myeloma treated with bispecific antibodies Blood advances 2025. PMID 40590849.
- Long JP, Prakash R, Edelkamp P, Knafl M, Lionel AC, Nair R, Ahmed S, Strati P, Castillo LEM, Al-Zaki A, Chien K, Chihara D, Westin J, Khawaja F, Nastoupil LJ, Mulanovich V, Futreal A, Woodman SE, Daver NG, Flowers CR, Neelapu S, Manzano JG, Iyer SP, Data-Driven Determinants for COVID-19 Discovery Effort (D3CODE) Team. Cytokine Storms in COVID-19, Hemophagocytic Lymphohistiocytosis, and CAR-T Therapy JAMA network open 2025. PMID 40193078.
- Martel E, Li S, Hoteit M, Bham Z. Cytokine Release Syndrome Induced by Pembrolizumab for Metastatic Anal Melanoma Case reports in hematology 2025. PMID 40630742.
- Mazein A, Lopata O, Reiche K, Sewald K, Alb M, Sakellariou C, Gogesch P, Morgan H, Neuhaus V, Pham NN, Sommer C, Perkins E, Fogal B, Shoaib M, Schneider R, Satagopam V, Ostaszewski M. An explorable model of an adverse outcome pathway of cytokine release syndrome related to the administration of immunomodulatory biotherapeutics and cellular therapies Frontiers in immunology 2025. PMID 40861455.
- Menardi G, Castellino A, Castellino C, Bersia ME, Grande E, Fraternali G, Massaia M, Fruttero C. Management of chimeric antigen receptor T-cell-related toxicity of a patient affected by cytokine release syndrome and immune effector cell-associated neurotoxicity syndrome, followed by an intestinal perforation: a case report Journal of medical case reports 2025. PMID 39833958.

- Moreno-Castaño AB, Fernández S, Brillembourg H, Iraola G, Moreno DF, Ventosa-Capell H, Martinez-Sanchez J, De Moner B, Molina P, Ramos A, Palomo M, Gómez-Ramírez P, Ortiz-Maldonado V, Martínez-Cibrián N, Delgado J, Oliver-Caldés A, Fernández de Larrea C, Salas MQ, Urbano Á, Téllez A, Nicolás JM, Juan M, González-Navarro A, Penack O, Escolar G, Carreras E, Fernández-Avilés F, Castro P, Diaz-Ricart M. m-EASIX (better than EASIX) predicts severe CAR T-cell toxicities, worse overall survival, and discriminates cytokine release syndrome from sepsis Frontiers in immunology 2025. PMID 41573561.
- Nie J, Zhou L, Tian W, Liu X, Yang L, Yang X, Zhang Y, Wei S, Wang DW, Wei J. Deep insight into cytokine storm: from pathogenesis to treatment Signal transduction and targeted therapy 2025. PMID 40234407.
- Robert PA, Denck J, Do CT, Ozkirimli E, Jamois C, Corso C, Wang K, Berger CT. Guideline-based strategies to identify severe cytokine release syndrome in COVID-19 and cancer immunotherapy using large-scale electronic health records Frontiers in digital health 2025. PMID 41782841.
- Ruemmele T, Macedo R, Stein MN, Chan HT, Mapara MY, Jacquemont CF, Reshef R. Emapalumab for severe cytokine release syndrome in solid tumor CAR-T: a case report Frontiers in oncology 2025. PMID 40236652.
- Shouval R, Strouse C, Kim S, Oloyede T, Ahmed S, Awan FT, Luan D, Bachanova V, Badar T, Bar M, Barba P, Beitinjaneh AM, Cashen A, Dholaria B, Elsawy M, Ganguly S, Geethakumari PR, Greenbaum U, Hashmi H, Hill LC, Jain MD, Jain T, Kebriaei P, Kittai AS, Locke FL, Lulla PD, Mead E, McGuirk JP, Mussetti A, Nishihori T, Olson AL, Pennisi M, Perales MA, Riedell PA, Saber W, Mirza AS, Magalhaes-Silverman M, Shpall EJ, Sorror M, Wudhikarn K, Turtle CJ, Moskop A, Pasquini MC. Cytokine Release Syndrome and Neurotoxicity Following CD19 CAR-T in B-Cell Lymphoma Transplantation and cellular therapy 2025. PMID 40288610.
- Speth K, Xie J, Song Q, Mattie M, Kim JJ, Barrett DM, Andrade J, Shen R, Bedognetti D, Adhikary S. Classification of patients with relapsed/refractory large B-cell lymphoma who do not develop early CRS/NE toxicity using ZUMA clinical trial data Journal for immunotherapy of cancer 2025. PMID 40759437.
- Stolz SM, Musa A, Bachofner A, Bankova AK, Gourri E, Manz MG, Rieger MJ, Schneidawind D, Wolfensberger N, Zenz T, Rösler W. Prophylactic Tocilizumab prior to infusion of CD19 CAR T-cells reduces therapy-related complications in older lymphoma patients Annals of hematology 2025. PMID 40711524.
- Synnott D, O'Reilly D, De Freitas D, Naidoo J. Cytokine release syndrome in solid tumors Cancer 2025. PMID 40847830.
- Takagi K, Saito G, Inazaki T, Shojima H, Miyakoshi J, Naito A, Sato S, Shimazui T, Anzai H, Imai C, Suzuki T. Rapid-onset Severe Cytokine Release Syndrome With Marked Interleukin-6 Increase and Acute Liver Injury After the First Tarlatamab Dose in SCLC: Case Report JTO clinical and research reports 2025. PMID 41362703.
- Wang Y, Ding L, Wang Y, Li H, Wu M, Li S, Xu Q, Chen Y, Wang X. Tetrandrine alleviates macrophage activation syndrome after CAR-T cell therapy Phytomedicine : international journal of phytotherapy and phytopharmacology 2025. PMID 39947004.
- Yamamoto K, Hirano K, Shimada T, Hata A. Tocilizumab for Steroid Pulse-Refractory Cytokine Release Syndrome in Chemotherapy With Durvalumab Plus Tremelimumab for NSCLC: A Case Report JTO clinical and research reports 2025. PMID 40822628.
- Yu X, Wang Q, Halimulati T, Lv J, Zhou K, Chen G, Yin L, Liu Y, Bi J, Xiang Z, Wang Q. Machine learning-based predictive model for high- grade cytokine release syndrome in chimeric antigen receptor T-cell therapy Frontiers in immunology 2025. PMID 41357188.
- Zhang H, Wang Y, Ji Q, Zhang Q, Qiu C, Huang S, Dong X, Pan J, Lu J, Bai Z, Hu S, Wu S. Low-dose glucocorticoid improves progression-free survival of children with B cell acute lymphoblastic leukaemia following chimeric antigen receptor T-cell therapy Frontiers in immunology 2025. PMID 41235219.
- Zhang J, Xie D, Tang M, Gong Q, Huang X, Li J, Xu S, Luo N, Sun H, Tan F. Utilizing plasma exchange for severe cytokine release syndrome after CAR-T cell therapy: clinical experience and literature insights Frontiers in immunology 2025. PMID 40630950.
- Zhou Q, An Y, Zhang X, Xiao X, Bai X, Liu P, Pu Y, Meng J, Zhu H, Lyu C, Zhang H, Zhang Y, Xie T, Meng H, Lyu H. Efficacy and safety of tocilizumab in managing cytokine release syndrome after CD19 CAR-T therapy for relapsed or refractory B-cell acute lymphoblastic leukemia Frontiers in immunology 2025. PMID 40160812.
- [no author listed]. Anticytokine Therapy and Corticosteroids for Cytokine Release Syndrome and for Neurotoxicity Following T-Cell Engager or CAR T-Cell Therapy: Rapid Review Canadian Agency for Drugs and Technologies in Health, Ottawa (ON) 2024. PMID 38985918.
- Barker K, Marco T, Husnain M, Katsanis E. Further modification of the EASIX score through the addition of phosphorous and IL-6 improves prediction of CRS and ICANS after CAR-T therapy Research Square 2024. DOI 10.21203/rs.3.rs-4650639/v1.
- Brem S. Vagus nerve stimulation: Novel concept for the treatment of glioblastoma and solid cancers by cytokine (interleukin-6) reduction, attenuating the SASP, enhancing tumor immunity Brain, behavior, & immunity - health 2024. PMID 39512605.
- Daryanani AE, Abbasi MA, Gomez Ardila MF, Tellez-Garcia E, Garzon-Dangond JM, Lin Y, Paludo J, Herrmann J, Ansell SM, Rosenthal AC, Villarraga HR. Baseline echocardiographic variables as predictors of hemodynamically

significant cytokine release syndrome in adults treated with CD19 CAR T-cell therapy for hematological malignancies Cardio-oncology (London, England) 2024. PMID 39709512.

- Driscoll DF, Bistrian BR. Cytokine Release Syndrome Associated With Immune-Modulating Chemotherapy: Potential Mitigating Role of Intravenous Omega-3 Fatty Acid Triglycerides Cancer reports (Hoboken, N.J.) 2024. PMID 39376028.
- Furqan F, Bhatlapenumarthi V, Dhakal B, Fenske TS, Farrukh F, Longo W, Akhtar O, D'Souza A, Pasquini M, Guru Murthy GS, Runaas L, Abedin S, Mohan M, Shah NN, Hamadani M. Outpatient administration of CAR T-cell therapies using a strategy of no remote monitoring and early CRS intervention Blood advances 2024. PMID 38889435.
- Harb R, Coverdell TC, Shalabi H, Shah NN. Diagnostic pitfalls in assessment of ferritin following CAR-T-cell therapy: Understanding the hook effect Pediatric blood & cancer 2024. PMID 38961599.
- Haroun G, Gordon EM. DeltaRex-G, tumor targeted retrovector encoding a CCNG1 inhibitor, for CAR-T cell therapy induced cytokine release syndrome Frontiers in molecular medicine 2024. PMID 39359418.
- Helms KA. Cytokine Release Syndrome After CAR T-Cell Therapy in a 35-Year-Old Patient With Pneumocystis jiroveci Pneumonia and Cytomegalovirus Viremia Case reports in medicine 2024. PMID 39717043.
- Hoyt R, Ye Z, Dasgupta A. Very unusual extremely high ferritin with cytokine release syndrome in a patient with hematological malignancy after experimental chimeric antigen receptor (CAR)-T-Cell therapy Clinica chimica acta; international journal of clinical chemistry 2024. PMID 38697457.
- Huntington SF, Lin D, Lafeuille MH, Thompson-Leduc P, Shah A, Kim N, Hester L, Tardif-Samson A, Moore B, Fowler J, Marshall A, Zhang X, Gifkins D, Wu B. Identification of cytokine release syndrome and indicators of severity in retrospective databases among patients receiving immunotherapy Pharmacology research & perspectives 2024. PMID 39487576.
- Kawase S, Sakurai M, Masuda K, Kubota Y, Shindo T, Inokuchi A, Hayashi H, Nakayasu M, Shiozawa Y, Hirai T, Inoue T, Fujii T, Kim H, Koda Y, Kato J, Kataoka K. Salivary gland swelling as a characteristic manifestation of local cytokine release syndrome after anti-CD19 chimeric antigen receptor T cell therapy: A case series Journal of clinical and experimental hematopathology : JCEH 2024. PMID 39343612.
- Kikuchi T, Tsukada N, Kunisada K, Nomura-Yogo M, Ishida T. Local Cytokine Release Syndrome After Idecabtagene Vicleucel Therapy in Patients With Multiple Myeloma: Two Case Reports Cureus 2024. PMID 39588407.
- Kitamura W, Asada N, Ikegawa S, Fujiwara H, Kamoi C, Ennishi D, Nishimori H, Fujii K, Fujii N, Matsuoka KI, Maeda Y. Activated CD4+ T Cell Proportion in the Peripheral Blood Correlates with the Duration of Cytokine Release Syndrome and Predicts Clinical Outcome after Chimeric Antigen Receptor T Cell Therapy Internal medicine (Tokyo, Japan) 2024. PMID 38945932.
- Locke FL, Neelapu SS, Bartlett NL, Lekakis LJ, Jacobson CA, Braunschweig I, Oluwole OO, Siddiqi T, Lin Y, Timmerman JM, Kersten MJ, Zheng Y, Zhang T, Nater J, Shen R, Miao H, Kim JJ, Miklos DB. Tocilizumab Prophylaxis Following Axicabtagene Ciloleucel in Relapsed or Refractory Large B-Cell Lymphoma Transplantation and cellular therapy 2024. PMID 39187161.
- Ma JF, Yan JW, Liu MJ, Yan CL, Tang XW, Qiu HY, Miao M, Han Y, Li LM, Kang LQ, Xu N, Yu Z, Tan JW, Zhu HJ, Jia X, Zhang ZZ, Wang M, Dai HP, Yu L, Xue SL, Wu DP, Gong WJ. Safe and potent anti-CD19 CAR T-cells with shRNA-IL-6 gene silencing element in patients with refractory or relapsed B-cell acute lymphoblastic leukemia HemaSphere 2024. PMID 39380843.
- Maheswaran J, Rafiee MJ, Cordes S, Lefebvre B, Chetrit M. Multimodality Imaging of Acute Myocarditis in Cytokine Release Syndrome (CRS) Following CAR-T Therapy CJC open 2024. PMID 39211749.
- Nakamura N, Jo T, Arai Y, Kitawaki T, Nishikori M, Mizumoto C, Kanda J, Yamashita K, Nagao M, Takaori-Kondo A. Clinical Impact of Cytokine Release Syndrome on Prolonged Hematotoxicity after Chimeric Antigen Receptor T Cell Therapy: KyoTox A-Score, a Novel Prediction Model Transplantation and cellular therapy 2024. PMID 38281589.
- Ntwali F, Gilliaux Q, Honoré PM. Nivolumab-Induced Cytokine Release Syndrome: A Case Report and Literature Review The American journal of case reports 2024. PMID 38625840.
- Nwozichi C, Ogunmuyiwa AO, Ojewale MO. Nurses' roles in CAR-T therapy for B-cell malignancies and managing associated cytokine release syndrome Asia-Pacific journal of oncology nursing 2024. PMID 38304228.
- Perl M, Herfeld K, Harrer DC, Höpting M, Schweiger M, Sterz U, Knödler L, Heimerl S, Hansmann L, Herr W, Poeck H, Wolff D, Edinger M, Hart C, Fante MA. Tocilizumab administration in cytokine release syndrome is associated with hypofibrinogenemia after chimeric antigen receptor T-cell therapy for hematologic malignancies Haematologica 2024. PMID 38546698.
- Sekimata M, Kinjo Y, Tohyama A, Murakami M, Hashiwaki S, Saito Y, Higami S, Hagimoto M, Taketomi R, Hoshino K, Harada H, Ueda T, Kurita T, Matsuura Y, Yoshino K. Cytokine release syndrome induced by immune checkpoint inhibitor treatment for uterine cervical cancer recurrence: A case report Oncology letters 2024. PMID 38807673.
- Sesques P, Kirkwood AA, Kwon M, Rejeski K, Jain MD, Di Blasi R, Brisou G, Gros FX, le Bras F, Bories P, Choquet S, Rubio MT, Iacoboni G, O'Reilly M, Casasnovas RO, Bay JO, Mohty M, Joris M, Abraham J, Castilla Llorente C,

Loschi M, Carras S, Chauchet A, La Rochelle LD, Hermine O, Guidez S, Cony-Makhoul P, Fogarty P, Le Gouill S, Morschhauser F, Gastinne T, Cartron G, Subklewe M, Locke FL, Sanderson R, Barba P, Houot R, Bachy E. Novel prognostic scoring systems for severe CRS and ICANS after anti-CD19 CAR T cells in large B-cell lymphoma Journal of hematology & oncology 2024. PMID 39107847.

- Su M, Chen L, Xie L, Fleurie A, Jonquieres R, Cao Q, Li B, Liang J, Tang Y. Identification of early predictive biomarkers for severe cytokine release syndrome in pediatric patients with chimeric antigen receptor T-cell therapy Frontiers in immunology 2024. PMID 39328408.
- Tu S, Luo X, Mei H, Hu Y, Liu Y, Li P, Zou D, Niu T, Xu K, Zhang X, Qiu L, Gao L, Gao G, Zhang L, Feng Y, Wang Y, Zhao M, Mi J, Hou M, Yang J, Huang H, Wang J, Hu Y, Zhao W, Wu D, Ma J, Li Y, Qian W, Huang X, Han W, Liang A. Recommendations for the timing, dosage, and usage of corticosteroids during cytokine release syndrome (CRS) caused by chimeric antigen receptor (CAR)-T cell therapy for hematologic malignancies Chinese medical journal 2024. PMID 39497436.
- Walton ZE, Frigault MJ, Maus MV. Current and emerging pharmacotherapies for cytokine release syndrome, neurotoxicity, and hemophagocytic lymphohistiocytosis-like syndrome due to CAR T cell therapy Expert opinion on pharmacotherapy 2024. PMID 38588525.
- Willemin ME, Gong J, Hilder BW, Masterson T, Tolbert J, Renaud T, Heuck C, Kane C, De Zwart L, Girgis S, Ma X, Ouellet D. Evaluation of Drug-Drug Interaction Potential of Talquetamab, a T-Cell-Redirecting GPRC5D × CD3 Bispecific Antibody, as a Result of Cytokine Release Syndrome in Patients with Relapsed/Refractory Multiple Myeloma in MonumenTAL-1, Using a Physiologically Based Pharmacokinetic Model Targeted oncology 2024. PMID 39285155.
- Willemin ME, Wang Lin SX, De Zwart L, Wu LS, Miao X, Verona R, Banerjee A, Liu B, Kobos R, Qi M, Ouellet D, Goldberg JD, Girgis S. Evaluating drug interaction potential from cytokine release syndrome using a physiologically based pharmacokinetic model: A case study of teclistamab CPT: pharmacometrics & systems pharmacology 2024. PMID 38831634.
- Winkelmann M, Blumenberg V, Rejeski K, Quell C, Bücklein VL, Ingenerf M, Unterrainer M, Schmidt C, Dekorsy FJ, Bartenstein P, Ricke J, von Bergwelt-Baildon M, Subklewe M, Kunz WG. Predictive value of pre-infusion tumor growth rate for the occurrence and severity of CRS and ICANS in lymphoma under CAR T-cell therapy Annals of hematology 2024. PMID 37861736.
- Yankelevich M, Thakur A, Modak S, Chu R, Taub J, Martin A, Schalk D, Schienshang A, Whitaker S, Rea K, Lee DW, Liu Q, Shields AF, Cheung NV, Lum LG. Targeting refractory/recurrent neuroblastoma and osteosarcoma with anti-CD3×anti-GD2 bispecific antibody armed T cells Journal for immunotherapy of cancer 2024. PMID 38519053.
- Yoshikawa T, Ito Y, Wu Z, Kasuya H, Nakashima T, Okamoto S, Amaishi Y, Zhang H, Li Y, Matsukawa T, Inoue S, Kagoya Y. Development of a chimeric cytokine receptor that captures IL-6 and enhances the antitumor response of CAR-T cells Cell reports. Medicine 2024. PMID 38670095.
- Zheng X, Zhang S, Wu H, Xia J, Zheng K, Wang Y, Qin Y. Plasma exchange as an effective treatment for cytokine release syndrome following T cell receptor-engineered T cell immunotherapy: A case report Oncology letters 2024. PMID 39483965.
- Zia K, Nur-E-Alam M, Ahmad A, Ul-Haq Z. Taming the cytokine storm: small molecule inhibitors targeting IL-6/IL-6α receptor Molecular diversity 2024. PMID 38366102.
- Caballero-Bellón M, Alonso-Saladrigues A, Bobillo-Perez S, Faura A, Arqués L, Rivera C, Català A, Dapena JL, Rives S, Jordan I. Risk factors and outcome of Chimeric Antigen Receptor T-Cell patients admitted to Pediatric Intensive Care Unit: CART-PICU study Frontiers in immunology 2023. PMID 37600775.
- Dou B, Ren S, Qiu L, Zhang X, Zhang N, Cai J, Chen D, Zhang Q, Yao H, Fan F. Prophylactic use of interleukin 6 monoclonal antibody can reduce CRS response of CAR-T cell therapy Frontiers in medicine 2023. PMID 38264058.
- Gazeau N, Liang EC, Wu QV, Voutsinas JM, Barba P, Iacoboni G, Kwon M, Ortega JLR, López-Corral L, Hernani R, Ortiz-Maldonado V, Martínez-Cibrian N, Martinez AP, Maziarz RT, Williamson S, Nemecek ER, Shadman M, Cowan AJ, Green DJ, Kimble E, Hirayama AV, Maloney DG, Turtle CJ, Gauthier J. Anakinra for Refractory Cytokine Release Syndrome or Immune Effector Cell-Associated Neurotoxicity Syndrome after Chimeric Antigen Receptor T Cell Therapy Transplantation and cellular therapy 2023. PMID 37031746.
- Jain MD, Smith M, Shah NN. How I treat refractory CRS and ICANS after CAR T-cell therapy Blood 2023. PMID 36989488.
- Liu C, Cojutti PG, Giannella M, Roberto M, Casadei B, Cristiano G, Papayannidis C, Vianelli N, Zinzani PL, Viale P, Bonifazi F, Pea F. Does Cytokine-Release Syndrome Induced by CAR T-Cell Treatment Have an Impact on the Pharmacokinetics of Meropenem and Piperacillin/Tazobactam in Patients with Hematological Malignancies? Findings from an Observational Case-Control Study Pharmaceutics 2023. PMID 36986882.
- Park JH, Nath K, Devlin SM, Sauter CS, Palomba ML, Shah G, Dahi P, Lin RJ, Scordo M, Perales MA, Shouval R, Tomas AA, Cathcart E, Mead E, Santomasso B, Holodny A, Brentjens RJ, Riviere I, Sadelain M. CD19 CAR T-cell

therapy and prophylactic anakinra in relapsed or refractory lymphoma: phase 2 trial interim results Nature medicine 2023. PMID 37400640.

- Patel R, Patel M, Laxmidhar F, Lakhatariya K, Patel D, Patel Z, Shaikh S. Cytokine Release Syndrome in Patients Treated With Chimeric Antigen Receptor T-cell Therapy: A Retrospective Study Analyzing Risks, Outcomes, and Healthcare Burden Cureus 2023. PMID 38152777.
- Shah D, Soper B, Shopland L. Cytokine release syndrome and cancer immunotherapies - historical challenges and promising futures Frontiers in immunology 2023. PMID 37304291.
- Tsutsui T, Hata K, Kawaguchi M, Kobayashi H, Kakizaki Y, Miyashita Y. Cytokine release syndrome complicated with severe rashes induced by nivolumab plus ipilimumab therapy in a patient with non-small cell lung cancer: A case report Thoracic cancer 2023. PMID 37381088.
- Wei Z, Xu J, Zhao C, Zhang M, Xu N, Kang L, Lou X, Yu L, Feng W. Prediction of severe CRS and determination of biomarkers in B cell-acute lymphoblastic leukemia treated with CAR-T cells Frontiers in immunology 2023. PMID 37854590.
- Yankelevich M, Thakur A, Modak S, Chu R, Taub J, Martin A, Schalk DL, Schienshang A, Whitaker S, Rea K, Lee DW, Liu Q, Shields A, Cheung N, Lum LG. Targeting GD2-positive Refractory/Resistant Neuroblastoma and Osteosarcoma with Anti- CD3 x Anti-GD2 Bispecific Antibody Armed T cells Research Square 2023. DOI 10.21203/rs.3.rs-3570311/v1.
- Yoshida M, Matsuoka Y, Mitsuyuki S, Yonetani N, Kawai J, Kondo T, Ishikawa T. Early prediction of cytokine release syndrome by measuring phosphate and magnesium levels following chimeric antigen receptor T cell therapy Blood cell therapy 2023. PMID 38149020.
- Zhang Q, Zu C, Jing R, Feng Y, Zhang Y, Zhang M, Lv Y, Cui J, Zhou L, Meng Y, Wang L, Cen Z, Chang AH, Hu Y, Huang H. Incidence, clinical characteristics and prognosis of tumor lysis syndrome following B-cell maturation antigen-targeted chimeric antigen receptor-T cell therapy in relapsed/refractory multiple myeloma Frontiers in immunology 2023. PMID 37215107.
- Balagopal S, Sasaki K, Kaur P, Nikolaidi M, Ishihara J. Emerging approaches for preventing cytokine release syndrome in CAR-T cell therapy Journal of materials chemistry. B 2022. PMID 35912720.
- Baur K, Heim D, Beerlage A, Poerings AS, Kopp B, Medinger M, Dirks JC, Passweg JR, Holbro A. Dasatinib for treatment of CAR T-cell therapy-related complications Journal for immunotherapy of cancer 2022. PMID 36455991.
- Diorio C, Vatsayan A, Talleur AC, Annesley C, Jaroscak JJ, Shalabi H, Ombrello AK, Hudspeth M, Maude SL, Gardner RA, Shah NN. Anakinra utilization in refractory pediatric CAR T-cell associated toxicities Blood advances 2022. PMID 35395068.
- Dos Santos DMC, Rejeski K, Winkelmann M, Liu L, Trinkner P, Günther S, Bücklein VL, Blumenberg V, Schmidt C, Kunz WG, Von Bergwelt-Baildon M, Theurich S, Subklewe M. Increased visceral fat distribution and body composition impact cytokine release syndrome onset and severity after CD19 chimeric antigen receptor T-cell therapy in advanced B-cell malignancies Haematologica 2022. PMID 35172565.
- Ferreros P, Trapero I. Interleukin Inhibitors in Cytokine Release Syndrome and Neurotoxicity Secondary to CAR-T Therapy Diseases (Basel, Switzerland) 2022. PMID 35892735.
- Gong WJ, Qiu Y, Li MH, Chen LY, Li YY, Yu JQ, Kang LQ, Sun AN, Wu DP, Yu L, Xue SL. Investigation of the risk factors to predict cytokine release syndrome in relapsed or refractory B-cell acute lymphoblastic leukemia patients receiving IL-6 knocking down anti-CD19 chimeric antigen receptor T-cell therapy Frontiers in immunology 2022. PMID 36105799.
- Logan GE, Miller K, Kohler ME, Loi M, Maddux AB. Outcomes of Critically Ill Children With Acute Lymphoblastic Leukemia and Cytokine Release Syndrome Due to Chimeric Antigen Receptor T Cell Therapy: US, Multicenter PICU, Cohort Database Study Pediatric critical care medicine : a journal of the Society of Critical Care Medicine and the World Federation of Pediatric Intensive and Critical Care Societies 2022. PMID 36194016.
- McNerney KO, DiNofia AM, Teachey DT, Grupp SA, Maude SL. Potential Role of IFNγ Inhibition in Refractory Cytokine Release Syndrome Associated with CAR T-cell Therapy Blood cancer discovery 2022. PMID 35015687.
- Moriyama S, Fukata M, Yokoyama T, Ueno S, Nunomura T, Mori Y, Kato K, Miyamoto T, Akashi K. Case Report: Cardiac Tamponade in Association With Cytokine Release Syndrome Following CAR-T Cell Therapy Frontiers in cardiovascular medicine 2022. PMID 35387436.
- Narayan V, Barber-Rotenberg JS, Jung IY, Lacey SF, Rech AJ, Davis MM, Hwang WT, Lal P, Carpenter EL, Maude SL, Plesa G, Vapiwala N, Chew A, Moniak M, Sebro RA, Farwell MD, Marshall A, Gilmore J, Lledo L, Dengel K, Church SE, Hether TD, Xu J, Gohil M, Buckingham TH, Yee SS, Gonzalez VE, Kulikovskaya I, Chen F, Tian L, Tien K, Gladney W, Nobles CL, Raymond HE, Prostate Cancer Cellular Therapy Program Investigators, Hexner EO, Siegel DL, Bushman FD, June CH, Fraietta JA, Haas NB. PSMA-targeting TGFβ-insensitive armored CAR T cells in metastatic castration-resistant prostate cancer: a phase 1 trial Nature medicine 2022. PMID 35314843.
- Ojemolon PE, Kalidindi S, Ahlborn TA, Aihie OP, Awoyomi MI. Cytokine Release Syndrome Following Blinatumomab Therapy Cureus 2022. PMID 35228941.

- Totzeck M, Michel L, Lin Y, Herrmann J, Rassaf T. Cardiotoxicity from chimeric antigen receptor-T cell therapy for advanced malignancies European heart journal 2022. PMID 35257157.
- Wang L, Chen Y, Zhou W, Miao X, Zhou H. Utilization of physiologically-based pharmacokinetic model to assess disease-mediated therapeutic protein-disease-drug interaction in immune-mediated inflammatory diseases Clinical and translational science 2022. PMID 34581012.
- Wehrli M, Gallagher K, Chen YB, Leick MB, McAfee SL, El-Jawahri AR, DeFilipp Z, Horick N, O'Donnell P, Spitzer T, Dey B, Cook D, Trailor M, Lindell K, Maus MV, Frigault MJ. Single-center experience using anakinra for steroid-refractory immune effector cell-associated neurotoxicity syndrome (ICANS) Journal for immunotherapy of cancer 2022. PMID 34996813.
- Xu Z, Tian F, Chen B, Kong X, Dai X, Cao J, Jiang P, Lu L, Lv Q, Yang A, Ma L, Hu L, Zhu X. CAR-T cell-induced cytokine release syndrome is rapidly alleviated by tripterygium glycosides Research Square 2022. DOI 10.21203/rs.3.rs-2331958/v1.
- Yoshihara K, Orihara Y, Hoshiyama T, Tamaki H, Sunayama I, Matsuda I, Nishikawa A, Kumamoto T, Samori M, Utsunomiya N, Min KD, Asakura M, Hirota S, Ishihara M, Higasa S, Yoshihara S. Severe acute heart failure during or following cytokine release syndrome after CAR T-cell therapy Leukemia research reports 2022. PMID 35898695.
- Zhang X, Guo Y, Ji Y, Gao Y, Zhang M, Liu Y, Zhu W, Lu P. Cytokine Release Syndrome After Modified CAR-NK Therapy in an Advanced Non-small Cell Lung Cancer Patient: A Case Report Cell transplantation 2022. PMID 35506155.
- Zhou L, Yu N, Li T, Ji H, Jiang L, Wang D, Xu B, Zhou X. Clinical characteristics and prognosis of 16 relapsed/refractory B-cell malignancy patients with CAR T-cell-related hyperferritinaemia Frontiers in oncology 2022. PMID 36313658.
- Bates PD, Rakhmilevich AL, Cho MM, Bouchlaka MN, Rao SL, Hales JM, Orentas RJ, Fry TJ, Gilles SD, Sondel PM, Capitini CM. Combining Immunocytokine and Ex Vivo Activated NK Cells as a Platform for Enhancing Graft-Versus-Tumor Effects Against GD2+ Murine Neuroblastoma Frontiers in immunology 2021. PMID 34489927.
- Browne EK, Daut E, Hente M, Turner K, Waters K, Duffy EA. Evidence-Based Recommendations for Nurse Monitoring and Management of Immunotherapy-Induced Cytokine Release Syndrome: A Systematic Review from the Children's Oncology Group Journal of pediatric oncology nursing : official journal of the Association of Pediatric Oncology Nurses 2021. PMID 34460332.
- Caimi PF, Pacheco Sanchez G, Sharma A, Otegbeye F, Ahmed N, Rojas P, Patel S, Kleinsorge Block S, Schiavone J, Zamborsky K, Boughan K, Hillian A, Reese-Koc J, Maschan M, Dropulic B, Sekaly RP, de Lima M. Prophylactic Tocilizumab Prior to Anti-CD19 CAR-T Cell Therapy for Non-Hodgkin Lymphoma Frontiers in immunology 2021. PMID 34712233.
- Cosenza M, Sacchi S, Pozzi S. Cytokine Release Syndrome Associated with T-Cell-Based Therapies for Hematological Malignancies: Pathophysiology, Clinical Presentation, and Treatment International journal of molecular sciences 2021. PMID 34299273.
- Dave P, Pallares Vela E, Cancarevic I. Is Prophylaxis the Only Way Out for Cytokine Release Syndrome Associated With Chimeric Antigen T-cell Therapy? Cureus 2021. PMID 34650883.
- Guo H, Qian L, Cui J. Focused evaluation of the roles of macrophages in chimeric antigen receptor (CAR) T cell therapy associated cytokine release syndrome Cancer biology & medicine 2021. PMID 34570442.
- Hong R, Zhao H, Wang Y, Chen Y, Cai H, Hu Y, Wei G, Huang H. Clinical characterization and risk factors associated with cytokine release syndrome induced by COVID-19 and chimeric antigen receptor T-cell therapy Bone marrow transplantation 2021. PMID 32943758.
- Mühlbacher J, Schörgenhofer C, Doberer K, Dürr M, Budde K, Eskandary F, Mayer KA, Schranz S, Ely S, Reiter B, Chong E, Adler SH, Jilma B, Böhmig GA. Anti-interleukin-6 antibody clazakizumab in late antibody-mediated kidney transplant rejection: effect on cytochrome P450 drug metabolism Transplant international : official journal of the European Society for Organ Transplantation 2021. PMID 34153143.
- Oluwole OO, Bouabdallah K, Muñoz J, De Guibert S, Vose JM, Bartlett NL, Lin Y, Deol A, McSweeney PA, Goy AH, Kersten MJ, Jacobson CA, Farooq U, Minnema MC, Thieblemont C, Timmerman JM, Stiff P, Avivi I, Tzachanis D, Kim JJ, Bashir Z, McLeroy J, Zheng Y, Rossi JM, Johnson L, Goyal L, van Meerten T. Prophylactic corticosteroid use in patients receiving axicabtagene ciloleucel for large B-cell lymphoma British journal of haematology 2021. PMID 34296427.
- Pennisi M, Sanchez-Escamilla M, Flynn JR, Shouval R, Alarcon Tomas A, Silverberg ML, Batlevi C, Brentjens RJ, Dahi PB, Devlin SM, Diamonte C, Giralt S, Halton EF, Jain T, Maloy M, Mead E, Palomba ML, Ruiz J, Santomasso B, Sauter CS, Scordo M, Shah GL, Park JH, Yanez San Segundo L, Perales MA. Modified EASIX predicts severe cytokine release syndrome and neurotoxicity after chimeric antigen receptor T cells Blood advances 2021. PMID 34432870.
- Perrin P, Collongues N, Baloglu S, Bedo D, Bassand X, Lavaux T, Gautier-Vargas G, Keller N, Kremer S, Fafi-Kremer S, Moulin B, Benotmane I, Caillard S. Cytokine release syndrome-associated encephalopathy in patients with COVID-19 European journal of neurology 2021. PMID 32853434.

- Sackstein P, Zaemes J, Kim C. Pembrolizumab-induced cytokine release syndrome in a patient with metastatic lung adenocarcinoma: a case report Journal for immunotherapy of cancer 2021. PMID 34330765.
- Seitz CM, Flaadt T, Mezger M, Lang AM, Michaelis S, Katz M, Syring D, Joechner A, Rabsteyn A, Siebert N, Troschke-Meurer S, Zumpe M, Lode HN, Yang SF, Atar D, Mast AS, Scheuermann S, Heubach F, Handgretinger R, Lang P, Schlegel P. Immunomonitoring of Stage IV Relapsed Neuroblastoma Patients Undergoing Haploidentical Hematopoietic Stem Cell Transplantation and Subsequent GD2 (ch14.18/CHO) Antibody Treatment Frontiers in immunology 2021. PMID 34367149.
- Singbartl K, Rosenthal A, Leis J, Patel B, Sen A. Novel Use of Extracorporeal Blood Purification for Treatment of Severe, Refractory Neurotoxicity After Chimeric Antigen Receptor T-Cell Therapy-A Case Report Critical care explorations 2021. PMID 34235458.
- Topp MS, van Meerten T, Houot R, Minnema MC, Bouabdallah K, Lugtenburg PJ, Thieblemont C, Wermke M, Song KW, Avivi I, Kuruvilla J, Dührsen U, Zheng Y, Vardhanabhuti S, Dong J, Bot A, Rossi JM, Plaks V, Sherman M, Kim JJ, Kerber A, Kersten MJ. Earlier corticosteroid use for adverse event management in patients receiving axicabtagene ciloleucel for large B-cell lymphoma British journal of haematology 2021. PMID 34590303.
- Tvedt THA, Vo AK, Bruserud Ø, Reikvam H. Cytokine Release Syndrome in the Immunotherapy of Hematological Malignancies: The Biology behind and Possible Clinical Consequences Journal of clinical medicine 2021. PMID 34768710.
- Yomota M, Mirokuji K, Sakaguchi M, Kitahara Y, Chin F, Setoguchi K, Hosomi Y. Cytokine Release Syndrome Induced by Immune-checkpoint Inhibitor Therapy for Non-small-cell Lung Cancer Internal medicine (Tokyo, Japan) 2021. PMID 33775995.
- Chen LY, Kang LQ, Zhou HX, Gao HQ, Zhu XF, Xu N, Yu L, Wu DP, Xue SL, Sun AN. Successful application of anti-CD19 CAR-T therapy with IL-6 knocking down to patients with central nervous system B-cell acute lymphocytic leukemia Translational oncology 2020. PMID 32745961.
- Chou CK, Turtle CJ. Assessment and management of cytokine release syndrome and neurotoxicity following CD19 CAR-T cell therapy Expert opinion on biological therapy 2020. PMID 32067497.
- Cicek F, Troschke-Meurer S, Ceylan K, Jahns LJ, Zumpe M, Siebert N, Ehlert K, Lode HN. Impact of IL-2 on Treatment Tolerance in Patients With High-Risk Neuroblastoma Treated With Dinutuximab Beta-Based Immunotherapy Frontiers in pediatrics 2020. PMID 33392114.
- Dixon BN, Daley RJ, Buie LW, Hsu M, Park JH, Brentjens RJ, Purdon TJ, Latcha S. Correlation of IL-6 secretion and hyponatremia with the use of CD19+ chimeric antigen receptor T-cells Clinical nephrology 2020. PMID 31670650.
- Frigault MJ, Nikiforow S, Mansour MK, Hu ZH, Horowitz MM, Riches ML, Hematti P, Turtle CJ, Zhang MJ, Perales MA, Pasquini MC. Tocilizumab not associated with increased infection risk after CAR T-cell therapy: implications for COVID-19? Blood 2020. PMID 32457999.
- Güneş S, Gök H, Ataman Ş, Kutlay Ş. Spontaneous Spinal Epidural Hematoma During Simultaneous Tocilizumab and Warfarin Use in a Patient With Rheumatoid Arthritis: Is There a Drug Interaction Between Tocilizumab and Oral Anticoagulants? Archives of rheumatology 2020. PMID 33758819.
- Hao Z, Li R, Li R, Meng L, Han Z, Hong Z. Macrophage, the potential key mediator in CAR-T related CRS Experimental hematology & oncology 2020. PMID 32665874.
- Jatiani SS, Aleman A, Madduri D, Chari A, Cho HJ, Richard S, Richter J, Brody J, Jagannath S, Parekh S. Myeloma CAR-T CRS Management With IL-1R Antagonist Anakinra Clinical lymphoma, myeloma & leukemia 2020. PMID 32553791.
- Shalabi H, Sachdev V, Kulshreshtha A, Cohen JW, Yates B, Rosing DR, Sidenko S, Delbrook C, Mackall C, Wiley B, Lee DW, Shah NN. Impact of cytokine release syndrome on cardiac function following CD19 CAR-T cell therapy in children and young adults with hematological malignancies Journal for immunotherapy of cancer 2020. PMID 32883871.
- Si S, Teachey DT, Teachey DT. Spotlight on Tocilizumab in the Treatment of CAR-T-Cell-Induced Cytokine Release Syndrome: Clinical Evidence to Date Therapeutics and clinical risk management 2020. PMID 32801727.
- Xu Z, Tian F, Chen B, Kong X, Dai X, Cao J, Jiang P, Tan J, Lu L, Wang X, Lv Q, Kang D, Xu M, Hu Y, Yang A, Wang Q, Yang Z, Sun X, Ma L, Hu L, Zhu X. Tripterygium glycosides as a potential treatment for CAR-T induced cytokine release syndrome: implication of monocyte depletion medRxiv 2020. DOI 10.1101/2020.11.22.20232801.
- Kotch C, Barrett D, Teachey DT. Tocilizumab for the treatment of chimeric antigen receptor T cell-induced cytokine release syndrome Expert review of clinical immunology 2019. PMID 31219357.
- Giavridis T, van der Stegen SJC, Eyquem J, Hamieh M, Piersigilli A, Sadelain M. CAR T cell-induced cytokine release syndrome is mediated by macrophages and abated by IL-1 blockade Nature medicine 2018. PMID 29808005.
- Le RQ, Li L, Yuan W, Shord SS, Nie L, Habtemariam BA, Przepiorka D, Farrell AT, Pazdur R. FDA Approval Summary: Tocilizumab for Treatment of Chimeric Antigen Receptor T Cell-Induced Severe or Life-Threatening Cytokine Release Syndrome The oncologist 2018. PMID 29622697.

- Norelli M, Camisa B, Barbiera G, Falcone L, Purevdorj A, Genua M, Sanvito F, Ponzoni M, Doglioni C, Cristofori P, Traversari C, Bordignon C, Ciceri F, Ostuni R, Bonini C, Casucci M, Bondanza A. Monocyte-derived IL-1 and IL-6 are differentially required for cytokine-release syndrome and neurotoxicity due to CAR T cells Nature medicine 2018. PMID 29808007.
- Wang Z, Han W. Biomarkers of cytokine release syndrome and neurotoxicity related to CAR-T cell therapy Biomarker research 2018. PMID 29387417.
- Ploessl C, Pan A, Maples KT, Lowe DK. Dinutuximab: An Anti-GD2 Monoclonal Antibody for High-Risk Neuroblastoma The Annals of pharmacotherapy 2016. PMID 26917818.
- Davila ML, Riviere I, Wang X, Bartido S, Park J, Curran K, Chung SS, Stefanski J, Borquez-Ojeda O, Olszewska M, Qu J, Wasielewska T, He Q, Fink M, Shinglot H, Youssif M, Satter M, Wang Y, Hosey J, Quintanilla H, Halton E, Bernal Y, Bouhassira DC, Arcila ME, Gonen M, Roboz GJ, Maslak P, Douer D, Frattini MG, Giralt S, Sadelain M, Brentjens R. Efficacy and toxicity management of 19-28z CAR T cell therapy in B cell acute lymphoblastic leukemia Science translational medicine 2014. PMID 24553386.
- Albertini MR, Hank JA, Gadbaw B, Kostlevy J, Haldeman J, Schalch H, Gan J, Kim K, Eickhoff J, Gillies SD, Sondel PM. Phase II trial of hu14.18-IL2 for patients with metastatic melanoma Cancer immunology, immunotherapy : CII 2012. PMID 22678096.
- Gilman AL, Ozkaynak MF, Matthay KK, Krailo M, Yu AL, Gan J, Sternberg A, Hank JA, Seeger R, Reaman GH, Sondel PM. Phase I study of ch14.18 with granulocyte-macrophage colony-stimulating factor and interleukin-2 in children with neuroblastoma after autologous bone marrow transplantation or stem-cell rescue: a report from the Children's Oncology Group Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2009. PMID 19047298.

infection immune (92)

- Al-Bazaz M, Pischke S, Alsdorf W, Sonnemann P, Cichutek S, Wiesch JSZ, Schaefers C, Artzenroth J, Leypoldt L, Kamili A, Bokemeyer C, Weisel K, Kosch R. Hepatitis E virus infections in people with multiple myeloma: an emerging challenge in the era of immunotherapeutic approaches Haematologica 2026. PMID 40994343.
- Alameer RM, Alamro B, Alanazi K, Alahmari A, Almousa G, Almohaizeie A, Samarkandi H, Almaghrabi RS. Use of Cytomegalovirus Immunoglobulin with Antiviral Therapy for Cytomegalovirus Infection in Transplant Recipients: A Tertiary Care Single-Center Experience Viruses 2026. PMID 42357610.
- Bouvier A, Durand A, Dupont V, Gower N, Lorente CC, Campidelli A, Dulery R, Gangneux JP, Joris M, Eloit M, Latiere C, di Blasi R, Denis B, Gabellier L, Morschhauser F, Casasnovas O, Houot R, Angebault C, Bachy E, Le Bras F, Gastinne T, Tudesq JJ, Lemonnier F, Melica G. Invasive fungal infections after CD19 chimeric antigen receptor T-cell therapy for B-cell lymphoma: a Lymphoma study association study from the DESCAR-T (Dispositif d'Enregistrement et Suivi des patients traités par CAR-T cells) registry Clinical microbiology and infection : the official publication of the European Society of Clinical Microbiology and Infectious Diseases 2026. PMID 41109429.
- Buvelot H, Baleydier F, Pittet L, Blanchard-Rohner G. Factors Affecting Immune Reconstitution Post-Allogeneic HSCT in Children: The Case for an Individualized Approach to Vaccination European journal of haematology 2026. PMID 41408960.
- Cao W, Meng F, Feng S, Zhao M, Song J, Sun Y, Cao W, Tian W, Hu Y, Fan F, Tang X, Qian W, Zhang Y, Wei J. Expert Consensus on Cytomegalovirus Management in Recipients of CAR-T Cell and Bispecific Antibody Therapies Journal of evidence-based medicine 2026. PMID 41531379.
- Chakraborty S, Pramanik J, Alviter-Raymundo G, Ward CJ, Shaji SK, Yamashita-Kanemaru Y, Ali FAZ, Banik D, Zhang Z, Veiga-Villauriz C, Homer NZM, Simpson J, Laforest S, Tong S, Zhao Q, Roy J, Iqbal M, Morris AC, Chapman MA, Roychoudhuri R, Hussein H, Klenerman D, Saeb-Parsy K, Mahata B. Cortisol-resistant CAR-NK cells overcome steroid-induced immunosuppression in lung cancer Signal transduction and targeted therapy 2026. PMID 41956993.
- Chalmin Katz A, Egan KP, Syeda Z, Son S, Watson B, Gopalakrishnan M, Bromberg V, Radaelli E, Assenmacher CA, Awasthi S, Cohen GH, Friedman HM. Immunogenicity and Efficacy of a Trivalent HSV-2 gC2, gD2, gE2 Nucleoside-Modified mRNA-LNP Vaccine Against HSV-1 Eye Infection and Neuroinvasion in Mice Vaccines 2026. PMID 41893790.
- Choo HS, Yu T, Pham T, Singh N, Joshi S. Myeloid derived suppressor cells in neuroblastoma: mechanisms of immune evasion and therapeutic opportunities Frontiers in immunology 2026. PMID 41972140.
- Cohen I, Lebel E, Grisariu S, Avni B, Kfir-Erenfeld S, Asherie N, Zimran E, Vainstein V, Assayag M, Sharon TD, Alexander-Shani R, Bessig N, Shehadeh A, Ishtay A, Schlossberg M, Pick M, Gatt ME, Bdolah-Abram T, Stepensky P, Elias S. Viral Reactivation in Multiple Myeloma Patients Receiving Anti-BCMA Chimeric Antigen Receptor T-Cell Therapy International journal of molecular sciences 2026. PMID 41977300.

- Dello Strologo A, De Liso C, Scarsella G, Trotta A, Strazza C, Donatone L, Pesce F, Grandaliano G. The immunosuppressive tumor microenvironment in post-transplant lymphoproliferative disorder: pathogenesis and novel therapeutic frontiers Frontiers in immunology 2026. PMID 42023228.
- Fan JB, Li QY, Feng XF, Huang SY, Wang R, Liao FY, Liu D, Liu WY, Sun JH, Zhang HC, Zhou HT, Jiang JX, Wang Z, Zeng L. The "cytokine storm" in infection and sepsis: win the battle but lose the war Military Medical Research 2026. PMID 41521316.
- Gallardo-Pizarro A, Ariño M, Lopera C, Martinez-Urrea A, Terrones-Campos C, Castiella-Aranzasti A, González-Navarro EA, Albiol N, Teijon-Lumbreras C, Pitart C, Marcos MÁ, Ortíz-Maldonado V, Mensa J, Delgado J, Urbano-Ispizua Á, Fernández de Larrea C, Soriano A, Martinez-Cibrian N, Castro P, Garcia-Vidal C. Infections in Patients With Cytokine Release Syndrome/Immune Effector Cell-Associated Neurotoxicity Syndrome Following Chimeric Antigen Receptor T-Cell Therapy Open forum infectious diseases 2026. PMID 42281893.
- Han Q, Zhang X, Chen L, Zhang Y. Chimeric antigen receptor T-cell therapy in chronic viral infections: a review Journal of translational medicine 2026. PMID 41535864.
- Hirao T, Kaji D, Kuno M, Watanabe O, Yamaguchi K, Kageyama K, Taya Y, Nishida A, Takagi S, Yamamoto H, Asano-Mori Y, Uchida N, Wake A, Yamamoto G. Hypogammaglobulinemia at Leukapheresis Predicts Delayed Humoral Immune Reconstitution and Increased Risk of Infections After CD19 CAR-T for B-Cell Lymphoma Transplantation and cellular therapy 2026. PMID 41698482.
- Imataki O, Takagi H, Ishida T, Uemura M. Triple Infections With Campylobacter coli, Cytomegalovirus, and SARS-CoV-2 in a Lymphoma Patient Treated With Epcoritamab Cureus 2026. PMID 41835636.
- Kaplan D, Blyth E, Sutrave G, Yong MK, Gottlieb DJ, Abendroth A, Slobedman B, Stern L. Human Cytomegalovirus Infection in Haematopoietic Stem Cell Transplant Recipients and CAR T Cell Recipients - PART 1: Risk Factors, Clinical Impact and Immune Response Reviews in medical virology 2026. PMID 41982051.
- Khan MA, Shafiq M, Khan MA, Khan FH, McGuirk JP, Okut H, Ellerbeck E, Ahmed N. Cytomegalovirus Infection and Clinical Outcomes in Hospitalized Chimeric Antigen Receptor T-Cell Therapy Recipients Cureus 2026. PMID 41664759.
- Lang MV, Hou A, Kornas M, Merilan V, Skelly R, Buckshaw R. Disseminated Angioinvasive Fusarium Infection following Chimeric Antigen Receptor T-Cell Therapy in Refractory Diffuse Large B-Cell Lymphoma: A Case Report Case reports in oncology 2026. PMID 42325639.
- Liu AJ, Zhu VZ, Teh BW, Yong MK. Antiviral prophylaxis against herpesviruses (herpes simplex virus, varicella-zoster virus and cytomegalovirus) among patients with cancer Current opinion in infectious diseases 2026. PMID 42068587.
- Liu KJ, Zhang L, Zhou B, Hong JS, Pak A, Yee AJ, Raje NS, Frigault MJ, Maus MV, Barmettler S. Characterization of Hypogammaglobulinemia, Infection Incidence, and Mortality in Patients Receiving B-cell Maturation Antigen Chimeric Antigen Receptor (CAR) T-cell Therapy Transplantation and cellular therapy 2026. PMID 41825801.
- Lou Y, He X, Li G, Sikhondze M, Ma J, Du H, Ouyang X, Dotti G, Sajadi M, Su L. CD4-Based Chimeric Antigen Receptor (CAR)-T Cells With Resistance to HIV-1 Infection and Enhanced Anti-HIV Efficacy: Covalent Interaction Between CD4-CAR and HIV-1 Envelope Glycoprotein Journal of medical virology 2026. PMID 41736532.
- Min KI, Min GJ, Eom KS, Cho SG. Rapidly progressive varicella zoster virus vasculopathy in a chemotherapy- and steroid-immunosuppressed patient with refractory diffuse large B-cell lymphoma: diagnostic and therapeutic challenges Blood research 2026. PMID 41790173.
- Othman T, Baird JH, Wang Y, Pak S, Zhong H, Wen YP, Shouse G, Herrera A, Popplewell L, Aribi A, Htut M, Palmer J, Budde LE, Dadwal SS. High Incidence of Cytomegalovirus Infection in Patients Receiving Standard of Care CD19- or BCMA-Targeted CAR-T Cell Therapies for Hematologic Malignancies Transplantation and cellular therapy 2026. PMID 42162873.
- Paramythiotou E, Pappa E, Iliadi V, Vernikos P. Fatal invasive Trichosporon asahii infection with exophytic bronchial lesions in a patient with lymphoma undergoing CAR T-cell therapy BMJ case reports 2026. PMID 41932713.
- Reynolds GK, Anderson MA, Thursky K, Teh BW, Slavin MA. Longitudinal real-world surveillance of infection outcomes in CAR-T and bispecific therapy recipients: the CLARITY study protocol BMJ open 2026. PMID 42097661.
- Reynolds GK, Dowling MR, Valencia-Klug J, Teh BW, Anderson MA, Vanguru V, Harrison SJ, Ho PJ, Slavin MA, Thursky K. Infections in the First 30 Days after Chimeric Antigen Receptor T-Cell Therapy in Patients Not Receiving Fluoroquinolone Prophylaxis Transplantation and cellular therapy 2026. PMID 42114808.
- Secreto C, Fasano F, Stella D, Novo M, Botto B, Cerrano M, Freilone R, Busca A. Infections and CAR-T cells for the treatment of lymphoid malignancies: a narrative review Frontiers in medicine 2026. PMID 42383061.
- Singh P, Hafeez AS, Zaman A, Faisal AR, Faizan M, Qureshi AJ, Saleem H, Khan MA, Khan A. Clinical outcomes of Epstein-Barr virus infection/reactivation following CAR-T cell therapy: A systematic review Transplant immunology 2026. PMID 42442615.

- Spataro F, Desantis V, Dicuonzo G, Molica S, Einsele H, Vacca A, Ria R, Solimando AG. Real-world evidence on infection risk in multiple myeloma treated with BiTEs and CAR-T cells: a meta-analysis Experimental hematology & oncology 2026. PMID 42323640.
- Synnott D, Bowden A, Sexton DJ. Systematic Review of Immunosuppression After Chimeric Antigen Receptor T-Cell Therapy for Posttransplant Lymphoproliferative Disorder Kidney international reports 2026. PMID 41907816.
- Tao L, He M, Zhang X, Sun J, Shen N, Shen B. Construction of a predictive model for lower respiratory tract infection in children with leukemia after chimeric antigen receptor T cell therapy Translational pediatrics 2026. PMID 41657452.
- Weiss JJ, Messina J, Saullo J, Li Y, Andermann TM, Smith M, Schrum D, Eberwein E, Kennedy E, Rowe-Nichols K, Kelsey C, Choi T, McKinney M, Kang Y, Huggins J. Respiratory Viral Infections Following CD19 CAR T-Cell Therapy Journal of medical virology 2026. PMID 42047272.
- Zhang X, Wang L, Kuang N, Yang J, Wang H, Qiu L, Wang D, Sun J, Long J, Lu P. Immune reconstitution after CD7 CAR-T cell therapy for refractory/relapsed acute T-lymphoblastic leukaemia/lymphoblastic lymphoma (R/R T-ALL/LBL) British journal of haematology 2026. PMID 41111252.
- Zuber M, Elshafie S, Chastain DB, Tan CJ, Villa Zapata L. Incidence, Recurrence, Timing, and Economic Impact of Infections After CAR T-Cell Therapy: A Real-World Analysis From MarketScan Database Transplantation and cellular therapy 2026. PMID 41713650.
- infante ms, pagano l, einarsdottir s, cornely o, salmanton-garcia j. Practice patterns, gaps, and priorities in infection management for patients receiving bispecific antibodies and CAR-T cell therapy – Results from the E-Impact survey of the EHA-SWG Infections in Hematology Research Square 2026. DOI 10.21203/rs.3.rs-10195291/v1.
- [no author listed]. P-1013. Incidence and Risk Factors for Fungal Infection after CD19-targeted Chimeric Antigen Receptor T cell Therapy for Non-Hodgkin Lymphoma Open forum infectious diseases 2025. abstract.
- Cheng L, Wang L, You X, Xiong L, Dai Q, Wang Q. Impact of Fungal Co-Infection on Teicoplanin Plasma Trough Concentration in Critically Ill Adults: A Novel Consideration for Dose Adjustment Drug design, development and therapy 2025. PMID 40521008.
- Cho Y, Lee C, Park SI, Kim Y, Lee YS, Lee S, Yoon S, Roh G, Ha D, Oh A, Cho K, Lee J, Park HJ, Lee HR, Nam JH. Design and immunogenicity of a quadrivalent mRNA vaccine targeting HSV-2 with comparative evaluation of co-formulated and admixed formulations Frontiers in immunology 2025. PMID 41425559.
- Dhamdhere MR, Gowda CP, Imamura Y, Wang HG, Schell TS, Spiegelman VS. IGF2BP1 fosters an immunosuppressive tumor microenvironment in high-risk neuroblastoma, contributing to their resistance to immunotherapy Oncoimmunology 2025. PMID 41328513.
- Gentilini M, Casadei B, Pellegrini C, Argnani L, Gugliotta G, Carella M, Stefoni V, Fabbri N, Gabrielli G, Salvatore F, Desiderio A, Mazzoni C, Maglio P, Bagnato G, Broccoli A, Bonifazi F, Zinzani PL. SARS-CoV-2 infection does not affect the outcome of patients with relapsed/refractory large B-cell lymphoma treated with CAR T-cell therapy Annals of hematology 2025. PMID 40464897.
- Gerber JM, Dehdashtian E, Hu G, Gregoire C, Borie D, Bindal P, Cerny J, Geara A, Schett G, Caricchio R. Successful autologous CD19 CAR T cell therapy following severe lupus flare during immunosuppressive washout in refractory lupus nephritis Lupus science & medicine 2025. PMID 41266084.
- Gwadera J, Grajewski M, Chowaniec H, Gucia K, Michoń J, Mikulicz Z, Knast M, Pujanek P, Tołkacz A, Murawa A, Dobosz P. Can We Use CAR-T Cells to Overcome Immunosuppression in Solid Tumours? Biology 2025. PMID 40906384.
- Han J, Lim SY, Hyung J, Cho H, Yoon DH, Kim SH. Cytomegalovirus Infection After Chimeric Antigen Receptor T-Cell Therapy or Bispecific Antibody Treatment for Hematologic Malignancies Transplant infectious disease : an official journal of the Transplantation Society 2025. PMID 41313664.
- Hayashino K, Seike K, Masunari T, Hashida R, Oka S, Fujiwara Y, Terao T, Kitamura W, Kobayashi H, Kamoi C, Kondo T, Fujiwara H, Asada N, Ennishi D, Fujii K, Fujii N, Maeda Y. Cytomegalovirus reactivation in patients with large B-cell lymphoma treated with chimeric antigen receptor T-cell therapy International journal of hematology 2025. PMID 40526215.
- Henden AS, Lineburg KE, Panikkar A, Mahajan A, Nelles R, Wright E, Crooks P, Raju J, Le Texier L, Swaminathan S, Beagley L, Best S, Solomon M, Reddiex H, Kennedy G, Tey SK, Neller MA, Khanna R, Smith C. Allogeneic 'off-the-shelf' SARS-CoV-2-specific adoptive T-cell therapy for refractory viral infection and end organ disease Clinical & translational immunology 2025. PMID 40497132.
- Luan D, DeWolf S, Fei T, Raj S, Shah GL, Lareau CA, Alhomoud M, Salles G, Rivas-Delgado A, Rejeski K, Park JH, Luttwak E, Luna de Abia A, Corona M, Ntrivalas E, Cassanello G, Gomez-Llobell M, Parascondola A, Scordo M, Hsu KC, Palomba ML, Perales MA, Shouval R. Dynamics of Immune Reconstitution and Impact on Outcomes across CAR-T Cell Products in Large B-cell Lymphoma Blood cancer discovery 2025. PMID 39666878.

- Mannavola CM, De Maio F, Marra J, Fiori B, Santarelli G, Posteraro B, Sica S, D'Inzeo T, Sanguinetti M. Bloodstream infection by Lactobacillus rhamnosus in a haematology patient: why metagenomics can make the difference Gut pathogens 2025. PMID 40544256.
- Melchio M, Hill JA, Shah M, Neofytos D, Gambella M, Raiola AM, Delfino E, Balletto E, Angelucci E, Bassetti M, Mikulska M. Old Pathogens-New Patient Types: Infections in a CAR T-Cell Recipient. Could It Get Any More Complicated? Transplant infectious disease : an official journal of the Transplantation Society 2025. PMID 40838528.
- Mikulska M, Robin C, Neofytos D, Oltolini C, Piekarska A, Reigadas E, Gil L, Chemaly RF, Groll AH, Muñoz P, Teh BW. Management of Clostridioides difficile infection in patients with haematological malignancies and after cellular therapy: guidelines from 10th European Conference on Infections in Leukaemia (ECIL-10) EClinicalMedicine 2025. PMID 40823499.
- Nishiyama R, Makita S, Shiotsuka M, Kobayashi O, Watanabe A, Maeshima AM, Ochi T, Iwaki N, Fukuhara S, Munakata W, Izutsu K. Schizophyllum commune infection following chimeric antigen receptor T-cell therapy in a patient with lymphoma Journal of clinical and experimental hematopathology : JCEH 2025. PMID 41016810.
- Rangarajan HG, Satwani P, Herr MM, Chen M, Martens MJ, Wudhikarn K, John S, Fabrizio VA, Hsieh EM, Kelkar AH, Doherty E, Marks DI, Ringden O, Friend B, Kelly MS, Farhadfar N, Prestidge T, Hossain NM, Liu H, Hashmi S, Modi D, Winestone LE, El Boghdadly Z, Murthy HS, Perales MA, Chemaly RF, Dandoy CE, Hill JA, Huppler A, Riches M, Auletta JJ. Real-world outcomes of infections following tisagenlecleucel in patients with B-cell ALL: a CIBMTR analysis Blood advances 2025. PMID 40737539.
- Riedel A, Phely L, Hug S, Faustmann P, Schroeder JC, Besemer B, Paczulla Stanger AM, Faul C, Lengerke C, Weller JF, Bethge W. Immunological consequences of CAR T-cell therapy: an analysis of infectious complications and immune reconstitution Blood advances 2025. PMID 40188456.
- Rosas D, Cox A, Martinez D, Fares A, Ouansafi I, Melendez D, Sandoval-Sus J. Cytomegalovirus Lymphadenitis After Anti-CD19 Chimeric Antigen Receptor T-cell (CAR-T): An Underappreciated Etiology of Hypermetabolic Lymphadenopathy in the Post-CAR-T Setting Cureus 2025. PMID 41583270.
- Sakalihasan S, Lifrange F, Czajkowski M, Goncette V, Duysinx B, Lovinfosse P, Can D, Schils R, Hayette MP, De Voeght A. Severe dual fungal infection after bispecific antibody therapy: A case of invasive aspergillosis and mucormycosis in immunocompromised patient IDCases 2025. PMID 39810812.
- Sassine J, Agudelo Higuita NI, Siegrist EA, Saeedi A, Corbisiero MF, Connelly P, Bastias AG, Dib RW, Henao-Cordero J, Chastain DB, Chiu CY, Henao-Martínez AF. Timeline and outcomes of viral and fungal infections after chimeric antigen receptor T-cell therapy: a large database analysis Clinical microbiology and infection : the official publication of the European Society of Clinical Microbiology and Infectious Diseases 2025. PMID 39528086.
- Sassine J, Siegrist EA, Chemaly RF. Herpesvirus Infections After Chimeric Antigen Receptor T-Cell Therapy and Bispecific Antibodies: A Review Viruses 2025. PMID 39861922.
- Stock S, Bücklein VL, Blumenberg V, Magno G, Emhardt AJ, Holzem AME, Cordas Dos Santos DM, Schmidt C, Grießhammer S, Frölich L, Kobold S, von Bergwelt-Baildon M, Rejeski K, Subklewe M. Prognostic significance of immune reconstitution following CD19 CAR T-cell therapy for relapsed/refractory B-cell lymphoma HemaSphere 2025. PMID 39807276.
- Sutherland NM, Zhou B, Zhang L, Ong MS, Hong JS, Pak A, Liu KJ, Frigault MJ, Maus MV, Hill JA, Reynolds K, Walter JE, Camargo CA, Barmettler S. Association of CD19+-targeted chimeric antigen receptor (CAR) T-cell therapy with hypogammaglobulinemia, infection, and mortality The Journal of allergy and clinical immunology 2025. PMID 39505278.
- Wonnaparhown A, Hilal T, Chong A, Fonseca R. Exploring the depths of hypogammaglobulinemia in lymphoid malignancies: Pathophysiology, clinical implications, management options, and future directions Blood reviews 2025. PMID 40533299.
- Wudhikarn K, Herr MM, Chen M, Martens MJ, Baird JH, Gowda L, Rangarajan HG, Abid MB, Kharfan-Dabaja MA, Williams KM, Ganguly S, Young JH, Sharma A, Fatobene G, Jain T, Kanakry CG, Modi D, Grover NS, Salem B, Batista MV, Vergidis P, Yin DE, Beitinjaneh AM, Kelkar AH, Nishihori T, Holter Chakrabarty J, Gergis U, Smith M, El Boghdadly Z, Dandoy CE, Murthy HS, Huppler AR, Perales MA, Chemaly RF, Hill JA, Riches M, Auletta JJ. Infection after CD19 chimeric antigen receptor T-cell therapy for large B-cell lymphoma: real-world analysis from CIBMTR Blood advances 2025. PMID 40435511.
- Zheng F, Zhang S, Chang AE, Moon JJ, Wicha MS, Wang SX, Chen J, Liu J, Cheng F, Li Q. Breaking Immunosuppression to Enhance Cancer Stem Cell-Targeted Immunotherapy International journal of biological sciences 2025. PMID 39990669.
- van Besien H, Easwar N, Demetres M, Pasciolla M, Shore T, Leonard J, Barker J, Martin P, Yamshon S. Comparative infection risk in CAR T vs bispecific antibodies in B-cell lymphoma: a systematic review and meta-analysis Blood advances 2025. PMID 40590871.
- Ahmed N, Oluwole O, Mahmoudjafari Z, Suleman N, McGuirk JP. Managing Infection Complications in the Setting of Chimeric Antigen Receptor T cell (CAR-T) Therapy Clinical hematology international 2024. PMID 38817309.

- An J, Zhao J, Zou P, Zhang Y, Wei J, Tian W, Wei J. Infections associated with CAR-T cell therapy in patients with relapsed refractory multiple myeloma: Risks and prevention strategies Cancer medicine 2024. PMID 38923216.
- Hou Z, Meng D, Ruan M, Liang X, Ni J, Jifei D, Wu Z, Xia R, Ge J, Long Z. A lesson from fatal invasive fungal infections after CAR-T cell therapy: a case report and literature review Immunotherapy 2024. PMID 39373599.
- Kampouri E, Ibrahimi SS, Xie H, Wong ER, Hecht JB, Sekhon MK, Vo A, Stevens-Ayers TL, Green DJ, Gauthier J, Maloney DG, Perez A, Jerome KR, Leisenring WM, Boeckh MJ, Hill JA. Cytomegalovirus (CMV) Reactivation and CMV-Specific Cell-Mediated Immunity After Chimeric Antigen Receptor T-Cell Therapy Clinical infectious diseases : an official publication of the Infectious Diseases Society of America 2024. PMID 37975819.
- Kampouri E, Reynolds G, Teh BW, Hill JA. Chimeric antigen receptor-T-cell therapies going viral: latent and incidental viral infections Current opinion in infectious diseases 2024. PMID 39361275.
- Khawaja F, Ahmed S, Iyer SP, Sassine J, Handley G, Prakash R, VanWierren T, Jackson J, Zubovskaia A, Ramdial J, Rondon G, Patel KK, Spallone A, Ariza-Heredia EJ, Mulanovich V, Angelidakis G, Jiang Y, Chemaly RF. Cytomegaloviral Infections in Recipients of Chimeric Antigen Receptor T-Cell Therapy: An Observational Study With Focus on Oncologic Outcomes Open forum infectious diseases 2024. PMID 39086466.
- Lin RY, Anderson AD, Natori Y, Raja M, Morris MI, Jimenez AJ, Beitinjaneh A, Wang T, Goodman M, Lekakis L, Spiegel J, Holtzman NG, Pereira D, Benjamin C, Natori A, Komanduri KV, Camargo JF. Incidence and outcomes of cytomegalovirus reactivation after chimeric antigen receptor T-cell therapy Blood advances 2024. PMID 38838226.
- Liu Y, Cai J, Wang T, Wang J, Tang Y, Wan X, Li W, Li B, Cao Q. Glycoprotein 350-targeted chimeric antigen receptor T-cell therapy for nonneoplastic chronic active Epstein-Barr virus infection: a case report Translational pediatrics 2024. PMID 39822998.
- Matsuo T, Wurster S, Ivan D, Hicklen R, McConn K, Bagwell KA, Khawaja F, Chemaly RF, Kontoyiannis DP. Lethal Disseminated Mucorales Infection With Positive Blood Cultures With Purpura Fulminans Complicating Hemophagocytic Lymphohistiocytosis After Chimeric Antigen Receptor T-Cell Therapy Open forum infectious diseases 2024. PMID 39507882.
- O'Reilly MA, Neill L, Collin SM, Stone N, Springell D, Mensah J, Cheok KPL, Jalowiec K, Benjamin R, Kuhnl A, Roddie C, Sanderson R. High pretreatment disease burden as a risk factor for infectious complications following CD19 chimeric antigen receptor T-cell therapy for large B-cell lymphoma HemaSphere 2024. PMID 38434533.
- Pu LF, Zheng HM, Feng XJ, Charwudzi A, Liang X, Hu LH, Ding YY, Liu ZL, Liao Y, Xiong SD. Rapid identification of early infections in febrile patients after CD19 target CAR-T cell therapy for B-cell malignancies Journal of translational medicine 2024. PMID 38956649.
- Shahid Z, Jain T, Dioverti V, Pennisi M, Mikkilineni L, Thiruvengadam SK, Shah NN, Dadwal S, Papanicolaou G, Hamadani M, Carpenter PA, Alfaro GM, Seo SK, Hill JA. Best Practice Considerations by The American Society of Transplant and Cellular Therapy: Infection Prevention and Management After Chimeric Antigen Receptor T Cell Therapy for Hematological Malignancies Transplantation and cellular therapy 2024. PMID 39084261.
- Tabbara N, Dioverti-Prono MV, Jain T. Mitigating and managing infection risk in adults treated with CAR T-cell therapy Hematology. American Society of Hematology. Education Program 2024. PMID 39644015.
- Ye H, Zou X, Fang X. Advancing cell-based therapy in sepsis: An anesthesia outlook Chinese medical journal 2024. PMID 38708689.
- Liles WC. MICROVASCULAR ENDOTHELIAL ACTIVATION/DYSFUNCTION AND DYSREGULATION OF THE ANGIOPOIETIN-TIE2 SYSTEM IN THE PATHOGENESIS OF LIFE-THREATENING INFECTIONS Transactions of the American Clinical and Climatological Association 2023. PMID 37701588.
- Little JS, Tandon M, Hong JS, Nadeem O, Sperling AS, Raje N, Munshi N, Frigault M, Barmettler S, Hammond SP. Respiratory infections predominate after day 100 following B-cell maturation antigen-directed CAR T-cell therapy Blood advances 2023. PMID 37486599.
- Mohan M, Chakraborty R, Bal S, Nellore A, Baljevic M, D'Souza A, Pappas PG, Berdeja JG, Callander N, Costa LJ. Recommendations on prevention of infections during chimeric antigen receptor T-cell and bispecific antibody therapy in multiple myeloma British journal of haematology 2023. PMID 37287117.
- Raje N, Anderson K, Einsele H, Efebera Y, Gay F, Hammond SP, Lesokhin AM, Lonial S, Ludwig H, Moreau P, Patel K, Ramasamy K, Mateos MV. Monitoring, prophylaxis, and treatment of infections in patients with MM receiving bispecific antibody therapy: consensus recommendations from an expert panel Blood cancer journal 2023. PMID 37528088.
- Sanders H, Callas C, St Amant H, Chung J, Dimitriades VR, Nakra NA. Case report: Clinical course and treatment of SARS-CoV-2 in a pediatric CAR-T cell recipient with persistent hypogammaglobulinemia Frontiers in pediatrics 2023. PMID 36969291.
- Stip MC, Teeuwen L, Dierselhuis MP, Leusen JHW, Krijgsman D. Targeting the myeloid microenvironment in neuroblastoma Journal of experimental & clinical cancer research : CR 2023. PMID 38087370.
- Liu X, Wills CA, Chen L, Zhang J, Zhao Y, Zhou M, Sundstrom JM, Schell T, Spiegelman VS, Young MM, Wang HG. Small extracellular vesicles induce resistance to anti-GD2 immunotherapy unveiling tipifarnib as an adjunct to neuroblastoma immunotherapy Journal for immunotherapy of cancer 2022. PMID 35483745.

- Liu Z, Zhou Z, Dang Q, Xu H, Lv J, Li H, Han X. Immunosuppression in tumor immune microenvironment and its optimization from CAR-T cell therapy Theranostics 2022. PMID 36168626.
- Oh BLZ, Chan LWY, Chai LYA. Manipulating NK cellular therapy from cancer to invasive fungal infection: promises and challenges Frontiers in immunology 2022. PMID 36969979.
- Garner W, Samanta P, Haidar G. Invasive Fungal Infections after Anti-CD19 Chimeric Antigen Receptor-Modified T-Cell Therapy: State of the Evidence and Future Directions Journal of fungi (Basel, Switzerland) 2021. PMID 33672208.
- Korell F, Schubert ML, Sauer T, Schmitt A, Derigs P, Weber TF, Schnitzler P, Müller-Tidow C, Dreger P, Schmitt M. Infection Complications after Lymphodepletion and Dosing of Chimeric Antigen Receptor T (CAR-T) Cell Therapy in Patients with Relapsed/Refractory Acute Lymphoblastic Leukemia or B Cell Non-Hodgkin Lymphoma Cancers 2021. PMID 33918340.
- Los-Arcos I, Iacoboni G, Aguilar-Guisado M, Alsina-Manrique L, Díaz de Heredia C, Fortuny-Guasch C, García-Cadenas I, García-Vidal C, González-Vicent M, Hernani R, Kwon M, Machado M, Martínez-Gómez X, Maldonado VO, Pla CP, Piñana JL, Pomar V, Reguera-Ortega JL, Salavert M, Soler-Palacín P, Vázquez-López L, Barba P, Ruiz-Camps I. Recommendations for screening, monitoring, prevention, and prophylaxis of infections in adult and pediatric patients receiving CAR T-cell therapy: a position paper Infection 2021. PMID 32979154.
- Meir J, Abid MA, Abid MB. State of the CAR-T: Risk of Infections with Chimeric Antigen Receptor T-Cell Therapy and Determinants of SARS-CoV-2 Vaccine Responses Transplantation and cellular therapy 2021. PMID 34587552.
- Blom T, Lurvink R, Aleven L, Mensink M, Wolfs T, Dierselhuis M, van Eijkelenburg N, Kraal K, van Noesel M, van Grotel M, Tytgat G. Treatment-Related Toxicities During Anti-GD2 Immunotherapy in High-Risk Neuroblastoma Patients Frontiers in oncology 2020. PMID 33680926.
- Hill JA, Seo SK. How I prevent infections in patients receiving CD19-targeted chimeric antigen receptor T cells for B-cell malignancies Blood 2020. PMID 32582924.
- Stoner A, Miller JO, Flatt T, Wallisch JS. Extracorporeal Membrane Oxygenation as a Bridge to Chimeric Antigen Receptor T-cell Therapy for Severe Refractory Sepsis in the Setting of Relapsed Refractory Pediatric Acute Lymphoblastic Leukemia: A Case Report Critical care explorations 2020. PMID 32426735.

hematologic coagulopathy (67)

- Ansari S, Bono K, Madendere B, Shah A, Kra J. Factors Associated With Mortality in Diffuse Large B-Cell Lymphoma (DLBCL) With Disseminated Intravascular Coagulation (DIC): A National Inpatient Sample Analysis Advances in hematology 2026. PMID 42609468.
- Bailey SR, Maus MV. Fanning the flames: IFN-γ fuels CAR-T inflammation and cytopenia The Journal of clinical investigation 2026. PMID 41480762.
- Beyar-Katz O, Stern A. After CAR-T therapy for myeloma: challenge of persistent cytopenias and infections Haematologica 2026. PMID 40836786.
- Dima D, Logue JM, Waqar SHB, Peres LC, Colin-Leitzinger CM, De Avila G, Smith EC, Skelson L, Matte KL, Blue B, Hovanky VN, Gaballa M, Pasvolsky O, Oswald LB, Fortuna GGM, Wagner CB, DeJarnette S, Dillard C, Perna F, Mikkilineni L, Hosoya H, Freeman CL, Shain KH, Baz RC, Grajales-Cruz A, Puglianini OC, Alsina M, Locke FL, Shune LO, Sborov DW, Patel KK, Sidana S, Hansen DK. Cytopenias and infections following ciltacabtagene autoleucel in heavily pretreated relapsed or refractory multiple myeloma Haematologica 2026. PMID 40637727.
- Gebremedhen AI, Mohammad A, Gundakaram S, Tesfamariam S, Bodiwala R, Souleymane M, Jamil M. Disseminated intravascular coagulation in chimeric antigen receptor T-cell therapy: a 6-year nationwide analysis of clinical outcomes and health care resource utilization in patients with hematologic malignancies Research and practice in thrombosis and haemostasis 2026. PMID 42633171.
- Kaur G, Logue J, Brazauskas R, Oloyede T, Abid MB, Ahmed S, Aljurf M, Banerjee R, Bashey A, Beitinjaneh AM, Callander NS, Castillo P, Dholaria B, Goldsmith SR, Hashmi S, Kharfan-Dabaja MA, Liu H, Mian H, Nishihori T, Rejeski K, Saad A, Slade MJ, Turtle CJ, Vogl DT, Williams KM, Wirk B, Pasquini MC, Moskop A, Akhtar OS, Hansen DK, Janakiram M. Prolonged Cytopenia After Idecabtagene Vicleucel for Multiple Myeloma is Associated with Poor Overall Survival Transplantation and cellular therapy 2026. PMID 42302965.
- Ko AK, Mutsaers PGNJ, van Kleij LM, Vledder A, Pennings ERA, Spanjaart AM, de Boer JW, Keijzer K, Demandt AMP, van Doesum JA, van Dorp S, Jak M, Kuipers MT, van Meerten T, van der Poel MWM, Vermaat JSP, Leebeek FWG, Kersten MJ, Lauw MN. Risk of thrombosis and bleeding following CAR T-cell therapy: Insights from the Dutch "Follow that CAR!" registry HemaSphere 2026. PMID 42453534.
- Korte IK, Kharboutli S, Völkl S, Heberling L, Scholz JK, Pfeiffer H, Strobel J, Harrer T, Kretschmann S, Vasova I, Gerbitz A, Achenbach S, Spriewald B, Krause S, Gutsche K, von Bonin M, Röllig C, Aigner M, Mackensen A, Müller F. CD19 CAR-T therapy induces remission in refractory autoimmune hemolytic anemia with ITP and antiphospholipid syndrome Med (New York, N.Y.) 2026. PMID 41962541.

- Kuter DJ, Berentsen S, Cooper N, Zhang L, Shi J, Chen Y, Zhang L, Zhou C, Wang Y, Dudani JS, Mozaffarian N. Towards achieving immune reset in autoimmune cytopenias: Insights from clinical experiences with B cell depleting therapies Blood reviews 2026. PMID 41850974.
- Li R, Pan H, Zhang L, Ma J, Li W, Gao Z, Fang L, Tian L, Shen Y, Yang F, Zhao J, Nie N, Li J, Wang W, Pan X, Lian Y, Li X, Peng G, Li L, Yu X, Xu C, Liu Y, Kuang Z, Huang J, Zhao X, Ge M, Liu L, Chen S, Feng Y, Chang AH, Deng B, Dai M, Huang L, Lv L, Zheng Y, Luo Y, Xiong H, Shi J. CD19 CAR T-Cell Therapy for Autoimmune Hemolytic Anemia The New England journal of medicine 2026. PMID 41534043.
- Su B, Wang X. CAR-T cell therapy-associated coagulopathy: From immunothrombotic storm mechanisms to precision management strategies Critical reviews in oncology/hematology 2026. PMID 42624179.
- Wong CH, Hui MY, Chan WYK, Liu APY, Ku DTL, So JCC, Lee PPW, Cheuk DKL, Chan GCF, Leung W. Quadruple immunotherapy with allogeneic natural killer cell infusion for recurrent neuroblastoma Cytotherapy 2026. PMID 42202757.
- Wudhikarn K, Bromberg M, Brower J, Seshan VE, Bachanova V, Ahmed S, McGuirk JP, Manu G, Maziarz RT, Oluwole OO, Schuster SJ, Porter DL, Bishop MR, Riedell PA, Perales MA, Dahi PB. Late cytopenia after CD19 chimeric antigen receptor T cell in large B cell lymphoma: A Cell Therapy Consortium Analysis Blood advances 2026. PMID 42341321.
- Akiyama H, Kakiuchi S, Sakamoto A, Tomigaki N, Fujioka S, Harima I, Niwa R, Takagi I, Kozuki Y, Miyata Y, Yoshihara K, Yoshihara S, Iwata N. Case Report: autologous stem cell boost enables hematopoietic recovery after severe cytopenia induced by BCMA-targeted bispecific antibody therapy in multiple myeloma Frontiers in oncology 2025. PMID 41561742.
- Antherieu G, Thibaud V, Sesques P, Jacot W, Menia K, Balducci L, Falandry C. Neutropenic events associated with new anticancer treatments, from pathophysiology to practice: a scoping review-NEW Group (Neutropenic Events Working Group) Supportive care in cancer : official journal of the Multinational Association of Supportive Care in Cancer 2025. PMID 41331540.
- Arad A, Katz M, Lebel E, Kalish Y, Assayag M, Avni B, Elias S, Grisariu S, Shai E, Kfir-Erenfeld S, Asherie N, Gatt ME, Stepensky P, Zimran E. High Rate of Cytokine Release Syndrome-Related Coagulopathy with Low Incidence of Bleeding and Thrombosis in Patients Treated with B-Cell Maturation Antigen (BCMA)-Targeted Chimeric Antigen Receptor T-Cells (CAR-T) Cancers 2025. PMID 41228344.
- Beyar-Katz O, Rejeski K, Shouval R. Immune effector cell-associated hematotoxicity: mechanisms, clinical manifestations, and management strategies Haematologica 2025. PMID 39973361.
- Capes A, Morin A, Banet A, Suner L, Ricard L, Corre E, Brissot E, Stocker N, Marjanovic Z, Sarkozy C, Mohty M, Malard F. Severe cytopenia after chimeric antigen receptor-T cell driven by large granular lymphocytes and responsive to steroids British journal of haematology 2025. PMID 39425562.
- Cheng H, Shao L, Wang D, Chen Y, Sun Y, Chen Z, Liu J, Wang X, Chen W, Sang W, Qi K, Li Z, Sun C, Shi M, Qiao J, Wu Q, Zeng L, Zheng J, Li L, Xu K, Cao J. Comprehensive characterization of cytopenia after chimeric antigen receptor-T cell infusion in patients with relapsed or refractory multiple myeloma Cytotherapy 2025. PMID 39283287.
- Dahunsi D, Eleanya C, Akintunde A, Oluwole O. Prolonged Cytopenia with CAR-T Cell Therapy and Management Recommendations Clinical hematology international 2025. PMID 40161263.
- Dong R, Liu R, Yang R, Yang Y, Xu Y, Lei B, Zhao W, Zhang W, Gao G, He A, Bai J. The management of multiple myeloma patients accompanied with disseminated intravascular coagulation before receiving LCAR-B38M therapy Discover oncology 2025. PMID 40295321.
- Frenking JH, Zhou X, Rejeski K, Wagner V, Costello P, Hielscher T, Gatti L, Kauer J, Nadeem O, Mai EK, Michel CS, Friedrich MJ, Sedloev D, Weinhold N, Goldschmidt H, Herfarth K, Schmitt A, Hundemer M, Schmitt M, Müller-Tidow C, Topp M, Einsele H, Dreger P, Munshi NC, Sperling AS, Rasche L, Sauer S, Raab MS. Bridging intensity is associated with impaired hematopoietic recovery after BCMA CAR-T therapy for multiple myeloma Blood advances 2025. PMID 40267180.
- Li Y, Jiang H, Luo L, Mei H. Immunotherapy-associated hemostatic abnormalities: bleeding and thrombotic complications Annals of hematology 2025. PMID 40590914.
- Li Y, Liu J, Luo L, Zhou L, Wu Z, Sang W, Jiang H, Wang J, Wang X, Li P, Chen Z, Shu J, Luo W, Hu Y, Li Y, Mei H. Bone marrow blasts- and modified EASIX-guided risk stratification for coagulopathy and outcomes after CAR-T therapy in relapsed/refractory B-ALL Thrombosis research 2025. PMID 41218266.
- Naik S, Selukar S, Talleur AC, Deshpande S, Llaurador Caraballo G, Fabrizio VA, Rouce RH, Zeng XL, Vatsayan A, Rossoff J, Pacenta HL, John S, Phillips CL, Talano JA, Moskop A, Verneris MR, Myers GD, Hall E, Karras N, Bonifant CL, Qayed M, Bakinowski E, Keating AK, Baumeister SHC, Tomilson E, Hermiston ML, Satwani P, Krupski C, Chinnabhandar V, Stefanski HE, Egeler E, Curran KJ, Laetsch TW, Mackall CL, Prabhu S, Nguyen KP, Baggott C, Schultz LM, McNerney KO. Characterization and prediction of prolonged severe neutropenia in pediatric patients receiving tisagenlecleucel Blood advances 2025. PMID 40690779.

- Pellegrino C, Galli E, Chiusolo P, Ladiana R, Valentini CG, Viscovo M, Sorà F, Sica S, Teofili L. Deciphering the Complex Intertwining Between Cytopenia and Transfusion Needs After CAR-T-Cell Therapy for B-Cell Malignancies Life (Basel, Switzerland) 2025. PMID 41010361.
- Rejeski K, Sanz J, Fei T, Nair MS, Hashmi H, Avigdor A, Beyar-Katz O, Bücklein VL, Curran KJ, Einarsdottir S, Esensten JH, Glaubach N, Golan-Accav N, Gomez-Llobell M, Halamis I, Itzhaki O, Locke FL, Mailankody S, Marcus R, Maus MV, Palomba ML, Park JH, Pasquini M, Raj S, Rajeeve S, Salles G, Scordo M, Shah GL, Shimoni A, Subklewe M, Tix T, Usmani SZ, Valid O, Valtis YK, Zuckerman T, Shah NN, Perales MA, Shouval R. T-ICAHT: grading and prognostic impact of thrombocytopenia after CAR T-cell therapy Blood 2025. PMID 40258181.
- Shamehdi CJ, Gordillo CA, Sawas A, Macedo R, Mapara MY, Reshef R. Life-Threatening Cytopenias in a Jehovah's Witness Following CD19-Directed Chimeric Antigen Receptor T Cell Therapy EJHaem 2025. PMID 40809951.
- Uchibori R, Ohmine K, Teruya T, Mineno J, Ozawa K. BCMA-CAR Therapy for Multiple Myeloma in NOG Mice Prevents the Progression of Anemia and Bone Lesions Human gene therapy 2025. PMID 40372999.
- Wang Y, Zhang L, Ma N, Wang Y. Platelet Transfusion for Patients with Cancer: An Update Transfusion medicine and hemotherapy : offizielles Organ der Deutschen Gesellschaft fur Transfusionsmedizin und Immunhamatologie 2025. PMID 40808881.
- Zhang A, Zheng H, Shang Q, Yin X, Yang Y, Luo Y, Su T, Dou X, Zhao T, Pei X, Liu Z, Lu J, Zhang X, Huang X, Mo X, Lv M, Zhao X. Can one scoring system fit all? Comparative validation of CAR-HEMATOTOX, ALL-HEMATOTOX, and eIPM for predicting immune effector cell-associated hematotoxicity following CAR-T therapy in hematologic malignancies Chinese journal of cancer research = Chung-kuo yen cheng yen chiu 2025. PMID 41229978.
- de Boer JW, Keijzer K, van Dorp S, Mutsaers PGNJ, Niezink AGH, van Doesum JA, Serroukh YIM, Muntendam LW, Pulles AE, Kret EJ, Sijs-Szabo A, Oomen J, Demandt AMP, Stevens WBC, Kuipers MT, Pennings ERA, Spanjaart AM, Kersten MJ, Jak M, van Dijk LV, van der Poel MWM, Vermaat JSP, van Meerten T. Population-based validation of the CAR-HEMATOTOX for hematotoxicity, infections, and survival after CART in R/R LBCL Blood advances 2025. PMID 40668622.
- Bindal P, Patell R, Chiasakul T, Lauw MN, Ko A, Wang TF, Zwicker JI. A meta-analysis to assess the risk of bleeding and thrombosis following chimeric antigen receptor T-cell therapy: Communication from the ISTH SSC Subcommittee on Hemostasis and Malignancy Journal of thrombosis and haemostasis : JTH 2024. PMID 38574863.
- Halloun J, Maza I, Stern A, Henig I, Akria L, Zohar Y, Zuckerman T, Beyar-Katz O. Cytomegalovirus colitis presenting with lower gastrointestinal bleeding following chimeric antigen receptor-T cell therapy Journal of cellular and molecular medicine 2024. PMID 39031465.
- Koike T, Toyama D, Shibata M, Otsuka K, Sugishita Y, Kaneko R, Kawabata N, Fujita S, Akiyama K, Yamamoto S. Prolonged severe cytopenia followed by fatal CAR T-cell-mediated encephalitis in a patient with acute lymphoblastic leukemia Blood cell therapy 2024. PMID 39651063.
- Li J, Mu J, Wang J, Li X, Li Q, Jiang Y, Cui R, Deng Q. Persistent Cytopenia After CD19 CAR T Therapy in Relapsed/Refractory DLBCL Patients Could Be a Predictor of Efficacy and Side Effects Cell transplantation 2024. PMID 38651796.
- Li PR, Li YY, Du MY, Jiang HW, Mei H. [BCMA/CS1 bispecific CAR-T cell therapy for multiple myeloma complicated with thrombocytopenia: a case report and literature review] Zhonghua xue ye xue za zhi = Zhonghua xueyexue zazhi 2024. PMID 40032341.
- Li YY, Li PR, Jiang HW, Mei H. [CD19 CAR-T treatment for B-lymphoblastic lymphoma complicated with disseminated intravascular coagulation: a case report and literature review] Zhonghua xue ye xue za zhi = Zhonghua xueyexue zazhi 2024. PMID 40032330.
- Miller A, Daum R, Wang T, Wu M, Tat C, Pfeiffer T, Navai S, Heczey A, Hegde M, Ahmed N, Whittle SB, Hill L, Martinez C, Krance R, Ramos CA, Rouce RH, Lulla P, Heslop HE, Omer B, Shekar M. Prolonged cytopenias after immune effector cell therapy and lymphodepletion in patients with leukemia, lymphoma and solid tumors Cytotherapy 2024. PMID 38819365.
- Mingot-Castellano M, Reguera-Ortega JL, Zafra Torres D, Hernani R, Lopez-Godino O, Guerreiro M, Herrero B, Lopez-Corral L, Luna A, González-Pinedo L, Chinea-Rodriguez A, Martin-lópez AA, Bailen R, Martinez-Cebrian N, Balsalobre P, Filaferro S, Alonso-Saladrigues A, Barba P, Pérez-Martínez A, Calbacho M, Perez Simon JA, Sanchez-Pina JM. Use of Eltrombopag to Improve Thrombocytopenia and Transfusion Requirement in Anti-CD19 Car-T Cell-Treated Patients Preprints.org 2024. DOI 10.20944/preprints202407.2382.v1.
- Mingot-Castellano ME, Reguera-Ortega JL, Zafra Torres D, Hernani R, Lopez-Godino O, Guerreiro M, Herrero B, López-Corral L, Luna A, Gonzalez-Pinedo L, Chinea-Rodriguez A, Africa-Martín A, Bailen R, Martinez-Cibrian N, Balsalobre P, Filaferro S, Alonso-Saladrigues A, Barba P, Perez-Martinez A, Calbacho M, Perez-Simón JA, Sánchez-Pina JM, On Behalf Of The Spanish Group Of Hematopoietic Transplant And Cell Therapy Geth-Tc. Use of Eltrombopag to Improve Thrombocytopenia and Tranfusion Requirement in Anti-CD19 CAR-T Cell-Treated Patients Journal of clinical medicine 2024. PMID 39274330.
- Palacios-Berraquero ML, Rodriguez-Marquez P, Calleja-Cervantes ME, Berastegui N, Zabaleta A, Burgos L, Alignani D, San Martin-Uriz P, Vilas-Zornoza A, Rodriguez-Diaz S, Inoges S, Lopez-Diaz de Cerio A, Huerga S,

Tamariz E, Rifon J, Alfonso-Pierola A, Lasarte JJ, Paiva B, Hernaez M, Rodriguez-Otero P, San-Miguel J, Ezponda T, Rodriguez-Madoz JR, Prosper F. Molecular mechanisms promoting long-term cytopenia after BCMA CAR-T therapy in multiple myeloma Blood advances 2024. PMID 39058976.

- Rejeski K, Wang Y, Hansen DK, Iacoboni G, Bachy E, Bansal R, Penack O, Müller F, Bethge W, Munoz J, Mohty R, Bücklein VL, Barba P, Locke FL, Lin Y, Jain MD, Subklewe M. Applying the EHA/EBMT grading for ICAHT after CAR-T: comparative incidence and association with infections and mortality Blood advances 2024. PMID 38181508.
- Sahai I, Borgman P, Yates B, Rosenzweig S, Rampertaap S, Rankin AW, Shah NN. Incidence of preexisting B-cell aplasia in B-ALL: implications for post-CAR T-cell monitoring Blood advances 2024. PMID 39454282.
- Vic S, Thibert JB, Bachy E, Cartron G, Gastinne T, Morschhauser F, Le Bras F, Bouabdallah K, Despas F, Bay JO, Rubio MT, Mohty M, Casasnovas O, Choquet S, Castilla-Llorente C, Guidez S, Loschi M, Guffroy B, Carras S, Drieu La Rochelle L, Guillet M, Houot R. Transfusion needs after CAR T-cell therapy for large B-cell lymphoma: predictive factors and outcome (a DESCAR-T study) Blood advances 2024. PMID 38181767.
- Xia Y, Wang L, Shen X, Xu Y, Xu W, Li J, Fan L, Chen L. Mesenchymal stem cell infusion for enhancing hematopoietic recovery and addressing cytopenias in CAR-T cell therapy Stem cell research & therapy 2024. PMID 39334276.
- Zhu HY, Zhao DQ, Zhuang Z, Ruan J, Chen C, Zhang W, Zhou DB, Zhang Y. [Clinical characteristics and prognostic analysis of prolonged cytopenia after CAR-T cell therapy in LBCL patients] Zhonghua nei ke za zhi 2024. PMID 39622728.
- Galli E, Sorà F, Hohaus S, Fresa A, Pansini I, Autore F, Metafuni E, Innocenti I, Limongiello MA, Giammarco S, Laurenti L, Bacigalupo A, Chiusolo P, De Stefano V, Sica S. Endothelial activation predicts disseminated intravascular coagulopathy, cytokine release syndrome and prognosis in patients treated with anti-CD19 CAR-T cells British journal of haematology 2023. PMID 36503182.
- Jain T, Olson TS, Locke FL. How I treat cytopenias after CAR T-cell therapy Blood 2023. PMID 36800563.
- Jess J, Yates B, Dulau-Florea A, Parker K, Inglefield J, Lichtenstein D, Schischlik F, Ongkeko M, Wang Y, Shahani S, Cullinane A, Smith H, Kane E, Little L, Chen D, Fry TJ, Shalabi H, Wang HW, Satpathy A, Lozier J, Shah NN. CD22 CAR T-cell associated hematologic toxicities, endothelial activation and relationship to neurotoxicity Journal for immunotherapy of cancer 2023. PMID 37295816.
- Kenkel TJ, Sridhar N, Hammons LR, Hintzke M, Shah NN. Bone Marrow Aplasia after CAR-T-Cell Therapy for Relapsed/Refractory Burkitt's Lymphoma Medical sciences (Basel, Switzerland) 2023. PMID 37873752.
- Liu R, Lv Y, Hong F, Zhao W, Lei B, Liu J, Zhang W, He A, Wang F. A comprehensive analysis of coagulopathy during anti-B cell maturation antigen chimeric antigen receptor-T therapy in multiple myeloma, a retrospective study based on LEGEND-2 Hematological oncology 2023. PMID 37186314.
- Qiu T, Hu L, Zhang Y, Wang Y, Ma S, Li D, Li Z, Xu K. Cytopenia after CAR-T cell therapy: Analysis of 63 patients with relapsed and refractory B-cell non-Hodgkin lymphoma Oncology letters 2023. PMID 37427346.
- Rejeski K, Greco R, Onida F, Sánchez-Ortega I, Bonini C, Sureda A, Gribben JG, Yakoub-Agha I, Subklewe M. An International Survey on Grading, Diagnosis, and Management of Immune Effector Cell-Associated Hematotoxicity (ICAHT) Following CAR T-cell Therapy on Behalf of the EBMT and EHA HemaSphere 2023. PMID 37125259.
- Song Z, Tu D, Tang G, Liu N, Tai Z, Yang J, Wang Y. Hemophagocytic lymphohistiocytosis and disseminated intravascular coagulation are underestimated, but fatal adverse events in chimeric antigen receptor T-cell therapy Haematologica 2023. PMID 36794498.
- Strati P, Li X, Deng Q, Marques-Piubelli ML, Henderson J, Watson G, Deaton L, Cain T, Yang H, Ravanmehr V, Fayad LE, Iyer SP, Nastoupil LJ, Hagemeister FB, Parra ER, Saini N, Takahashi K, Fowler NH, Westin JR, Steiner RE, Nair R, Flowers CR, Wang L, Ahmed S, Al-Atrash G, Vega F, Neelapu SS, Green MR. Prolonged cytopenia following CD19 CAR T cell therapy is linked with bone marrow infiltration of clonally expanded IFNγ-expressing CD8 T cells Cell reports. Medicine 2023. PMID 37586321.
- Wang X, Li C, Luo W, Zhang Y, Huang Z, Xu J, Mei H, Hu Y. IL-10 plus the EASIX score predict bleeding events after anti-CD19 CAR T-cell therapy Annals of hematology 2023. PMID 37814134.
- Taneja A, Jain T. CAR-T-OPENIA: Chimeric antigen receptor T-cell therapy-associated cytopenias EJHaem 2022. PMID 35844301.
- Johnsrud A, Craig J, Baird J, Spiegel J, Muffly L, Zehnder J, Tamaresis J, Negrin R, Johnston L, Arai S, Shizuru J, Lowsky R, Meyer E, Weng WK, Shiraz P, Rezvani A, Latchford T, Mackall C, Miklos D, Frank M, Sidana S. Incidence and risk factors associated with bleeding and thrombosis following chimeric antigen receptor T-cell therapy Blood advances 2021. PMID 34521106.
- Liu Y, Liang B, Liu Y, Wei G, Wu W, Yang L, Yang L, Huang H, Xie J, Hu Y. Cytokine Release Syndrome Is an Independent Risk Factor Associated With Platelet Transfusion Refractoriness After CAR-T Therapy for Relapsed/Refractory Acute Lymphoblastic Leukemia Frontiers in pharmacology 2021. PMID 34366854.
- Rejeski K, Kunz WG, Rudelius M, Bücklein V, Blumenberg V, Schmidt C, Karschnia P, Schöberl F, Dimitriadis K, von Baumgarten L, Stemmler J, Weigert O, Dreyling M, von Bergwelt-Baildon M, Subklewe M. Severe Candida

glabrata pancolitis and fatal Aspergillus fumigatus pulmonary infection in the setting of bone marrow aplasia after CD19-directed CAR T-cell therapy - a case report BMC infectious diseases 2021. PMID 33509115.

- Rejeski K, Perez A, Sesques P, Hoster E, Berger C, Jentzsch L, Mougiakakos D, Frölich L, Ackermann J, Bücklein V, Blumenberg V, Schmidt C, Jallades L, Fehse B, Faul C, Karschnia P, Weigert O, Dreyling M, Locke FL, von Bergwelt-Baildon M, Mackensen A, Bethge W, Ayuk F, Bachy E, Salles G, Jain MD, Subklewe M. CAR-HEMATOTOX: a model for CAR T-cell-related hematologic toxicity in relapsed/refractory large B-cell lymphoma Blood 2021. PMID 34166502.
- Wang J, Doran J. The Many Faces of Cytokine Release Syndrome-Related Coagulopathy Clinical hematology international 2021. PMID 34595461.
- Xing L, Wang Y, Liu H, Gao S, Shao Q, Yue L, Liu Z, Wang H, Shao Z, Fu R. Case Report: Sirolimus Alleviates Persistent Cytopenia After CD19 CAR-T-Cell Therapy Frontiers in oncology 2021. PMID 35004324.
- Mody R, Yu AL, Naranjo A, Zhang FF, London WB, Shulkin BL, Parisi MT, Servaes SE, Diccianni MB, Hank JA, Felder M, Birstler J, Sondel PM, Asgharzadeh S, Glade-Bender J, Katzenstein H, Maris JM, Park JR, Bagatell R. Irinotecan, Temozolomide, and Dinutuximab With GM-CSF in Children With Refractory or Relapsed Neuroblastoma: A Report From the Children's Oncology Group Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2020. PMID 32343642.
- Mody R, Naranjo A, Van Ryn C, Yu AL, London WB, Shulkin BL, Parisi MT, Servaes SE, Diccianni MB, Sondel PM, Bender JG, Maris JM, Park JR, Bagatell R. Irinotecan-temozolomide with temsirolimus or dinutuximab in children with refractory or relapsed neuroblastoma (COG ANBL1221): an open-label, randomised, phase 2 trial The Lancet. Oncology 2017. PMID 28549783.
- Osenga KL, Hank JA, Albertini MR, Gan J, Sternberg AG, Eickhoff J, Seeger RC, Matthay KK, Reynolds CP, Twist C, Krailo M, Adamson PC, Reisfeld RA, Gillies SD, Sondel PM, Children's Oncology Group. A phase I clinical trial of the hu14.18-IL2 (EMD 273063) as a treatment for children with refractory or recurrent neuroblastoma and melanoma: a study of the Children's Oncology Group Clinical cancer research : an official journal of the American Association for Cancer Research 2006. PMID 16551859.

drug interaction pk (61)

- Drevin G, Picard N, Briet M, Abbara C. Unexpected Voriconazole Overexposure in a CYP2C19 Rapid Metabolizer Due to Inflammation-Induced Phenoconversion: A Short Communication Therapeutic drug monitoring 2026. PMID 42300127.
- Kleinbloesem CH, Braal CL. The Third Dimension of Pharmacokinetic–Pharmacodynamic Theory: Adaptive Rate Capacity as a Conserved Constraint on Biological Tolerability medRxiv 2026. DOI 10.64898/2026.06.02.26354717.
- Klomp SD, Manson ML, Guchelaar HJ, Böhringer S, van Egmond EHM, von dem Borne PA, Swen JJ. Inflammation-driven variability in drug metabolism: Insights from voriconazole treatment of HSCT recipients British journal of clinical pharmacology 2026. PMID 41766069.
- Martins E, Chindalore VL, Sheng XR, Schroeder AV, Chen V, Mody H, Arjomandi A, Mao HA, Hassan I, Dankiewicz-Fares I, Agachi S, Omachi TA, Garg JP, Pendergraft WF. Phase Ib, multicentre, open-label, dose-escalation study to evaluate the safety, tolerability, pharmacokinetics and pharmacodynamics of subcutaneously administered mosunetuzumab in participants with systemic lupus erythematosus Lupus science & medicine 2026. PMID 42309556.
- Steinbronn C, Stanley SE, Bueters T, Ait-Oudhia S. Drug-Drug Interaction Risk Assessment Strategies for Biologics in Inflammatory Bowel Disease: A Literature-Based Evidence Mini-Review Clinical and translational science 2026. PMID 42568100.
- Wang J, Jiang S, Zhang Y, Gao H, Zhao X, Liu L, Li Z. Computational toxicology prioritization of CYP3A4 and DPP7 as candidate triclosan-relevant molecular targets in ulcerative colitis Toxicology mechanisms and methods 2026. PMID 42414864.
- Ye Q, Xing W, Hu X, Liu W, Wu J, Huang H, Xiao Y, Hong Y. Preclinical Pharmacokinetics, ADME, and Drug-Drug Interaction Evaluation of S024, a Novel p38/MK2 Inhibitor for Rheumatoid Arthritis Drug design, development and therapy 2026. PMID 42016385.
- Alrubia S, Achour B, Al-Majdoub ZM, Rostami-Hodjegan A, Barber J. Variability in intestinal drug metabolizing enzymes and transporters in Crohn's disease and potential impact on oral drug absorption British journal of clinical pharmacology 2025. PMID 40040248.
- Boeing LB, Fogaça NS, Kahlow BS, Skare T, Nisihara R. Polypharmacy and drug interactions in the management of rheumatoid arthritis Revista da Associacao Medica Brasileira (1992) 2025. PMID 40802417.
- Coste A, Aubron C, Ansart S, Roberts J, Morio F, Grégoire M. Factors affecting voriconazole pharmacokinetic variability in critically ill patients: a systematic review Critical care (London, England) 2025. PMID 41291870.

- Coutant DE, Rehmel J, Edwards DM, Hall SD. A novel physiologically-based pharmacokinetic model to estimate reduced CYP3A4 activity in cancer patients utilizing the neutrophil-to-lymphocyte ratio as an inflammatory marker European journal of clinical pharmacology 2025. PMID 40186675.
- D'Cunha R, Azam T, Kalabic J, Anschutz T, Lahat A, Pang Y. Evaluation of the Effect of Risankizumab on the Pharmacokinetics of Cytochrome P450 Substrates in Patients with Moderately to Severely Active Ulcerative Colitis or Crohn's Disease Clinical pharmacokinetics 2025. PMID 39707077.
- Lin Y, Li X, Fu B, Xie C, Ye L. Assessment of potential disease-drug interactions when prescribed Fuzi for rheumatoid arthritis treatment, with focus on the pharmacokinetics of aconitine, mesaconitine, and hypaconitine Journal of ethnopharmacology 2025. PMID 40769444.
- Pan X, Liu C, Stader F, Derbalah A, Jamei M, Gardner I. Mechanistic Insights into Cytokine Antagonist-Drug Interactions: A Physiologically Based Pharmacokinetic Modelling Approach with Tocilizumab as a Case Study Pharmaceutics 2025. PMID 40733104.
- Tanaka R. Pharmacokinetic variability and significance of therapeutic drug monitoring for broad-spectrum antimicrobials in critically ill patients Journal of pharmaceutical health care and sciences 2025. PMID 40098009.
- Varela-González-Aller J, Sánchez-Salinas MA, Troconiz I, Iacoboni G, Alonso-Martínez C, Carreras-Soler MJ, Carpio C, Farriols-Danes A, Guerra-González M, Rivas-Delgado A, Rivera Sanchez L, Feijoo S, Valdivia C, Barba P, Miarons M. Towards Personalized Lymphodepletion: A Population Pharmacokinetic Fludarabine Model in Patients Receiving CAR T-Cell Therapy Pharmaceutics 2025. PMID 41471106.
- Agachi S, Beloukhova M, Mould D, Lemak M, Grishin S, Samsonov M. Inflammation-mediated drug interactions of olokizumab and cytochrome P450 activities in patients with rheumatoid arthritis British journal of clinical pharmacology 2024. PMID 38984761.
- Chen Y, Ma F, Jones N, Deng R, Li C, Li CC. Assessment of CYP3A-mediated drug interaction via cytokine (IL-6) elevation for mosunetuzumab using physiologically-based pharmacokinetic modeling CPT: pharmacometrics & systems pharmacology 2024. PMID 38050329.
- Djebli N, Parrott N, Jaminion F, O'Jeanson A, Guerini E, Carlile D. Evaluation of the potential impact on pharmacokinetics of various cytochrome P450 substrates of increasing IL-6 levels following administration of the T-cell bispecific engager glofitamab CPT: pharmacometrics & systems pharmacology 2024. PMID 38044486.
- Klomp SD, Veringa A, Alffenaar JC, de Boer MGJ, Span LFR, Guchelaar HJ, Swen JJ. Inflammation altered correlation between CYP2C19 genotype and CYP2C19 activity in patients receiving voriconazole Clinical and translational science 2024. PMID 39010708.
- Lee KR, Gulnaz A, Chae YJ. Drug Interaction-Informed Approaches to Inflammatory Bowel Disease Management Pharmaceutics 2024. PMID 39598554.
- Morgan ET. Shining a Light on Inflammation as a Critical Modulator of Drug Metabolism Drug metabolism and disposition: the biological fate of chemicals 2024. PMID 39038953.
- Sawant-Basak A, Olabode D, Dai D, Vishwanathan K, Phipps A. Assessing Trends in Cytokine-CYP Drug Interactions and Relevance to Drug Dosing Drug metabolism and disposition: the biological fate of chemicals 2024. PMID 38383116.
- de Jong LM, Harpal C, Berg DVD, Hoekstra M, Peter NJ, Rissmann R, Swen JJ, Manson ML. CYP P450 and non-CYP P450 Drug Metabolizing Enzyme Families Exhibit Differential Sensitivities towards Proinflammatory Cytokine Modulation Drug metabolism and disposition: the biological fate of chemicals 2024. PMID 39349298.
- Dunvald AD, Søltoft K, Sheetal E, Just SA, Frederiksen IEB, Nielsen F, Olsen DA, Madsen JS, Hendricks O, Stage TB. Cytochrome P450 activity in rheumatoid arthritis patients during continuous IL-6 receptor antagonist therapy European journal of clinical pharmacology 2023. PMID 37831074.
- Liu F, Aulin LBS, Manson ML, Krekels EHJ, van Hasselt JGC. Unraveling the Effects of Acute Inflammation on Pharmacokinetics: A Model-Based Analysis Focusing on Renal Glomerular Filtration Rate and Cytochrome P450 3A4-Mediated Metabolism European journal of drug metabolism and pharmacokinetics 2023. PMID 37715056.
- Morales Castro D, Dresser L, Granton J, Fan E. Pharmacokinetic Alterations Associated with Critical Illness Clinical pharmacokinetics 2023. PMID 36732476.
- Yu Y, Henrich C, Wang D. Assessment of the drug-drug interaction potential for therapeutic proteins with pro-inflammatory activities Clinical and translational science 2023. PMID 36890677.
- Zaarur L, Patel A, Pasternak B. Drug Interaction Between Tacrolimus and Paxlovid (Nirmatrelvir/Ritonavir) in an Adolescent with Inflammatory Bowel Disease JPGN reports 2023. PMID 38034448.
- Gatti M, Pea F. The Cytokine Release Syndrome and/or the Proinflammatory Cytokines as Underlying Mechanisms of Downregulation of Drug Metabolism and Drug Transport: A Systematic Review of the Clinical Pharmacokinetics of Victim Drugs of this Drug-Disease Interaction Under Different Clinical Conditions Clinical pharmacokinetics 2022. PMID 36059001.
- Hingorani P, Krailo M, Buxton A, Hutson P, Sondel PM, Diccianni M, Yu A, Morris CD, Womer RB, Crompton B, Randall RL, Teot LA, DuBois SG, Janeway KA, Gorlick RG, Isakoff MS. Phase 2 study of anti-disialoganglioside

antibody, dinutuximab, in combination with GM-CSF in patients with recurrent osteosarcoma: A report from the Children's Oncology Group European journal of cancer (Oxford, England : 1990) 2022. PMID 35809374.

- Huang J, Zhang Z, Hao C, Qiu Y, Tan R, Liu J, Wang X, Yang W, Qu H. Identifying Drug-Induced Liver Injury Associated With Inflammation-Drug and Drug-Drug Interactions in Pharmacologic Treatments for COVID-19 by Bioinformatics and System Biology Analyses: The Role of Pregnane X Receptor Frontiers in pharmacology 2022. PMID 35979235.
- Lenoir C, Niederer A, Rollason V, Desmeules JA, Daali Y, Samer CF. Prediction of cytochromes P450 3A and 2C19 modulation by both inflammation and drug interactions using physiologically based pharmacokinetics CPT: pharmacometrics & systems pharmacology 2022. PMID 34791831.
- Stader F, Battegay M, Sendi P, Marzolini C. Physiologically Based Pharmacokinetic Modelling to Investigate the Impact of the Cytokine Storm on CYP3A Drug Pharmacokinetics in COVID-19 Patients Clinical pharmacology and therapeutics 2022. PMID 34496043.
- Sullivan PM, Kumar R, Li W, Hoglund V, Wang L, Zhang Y, Shi M, Beak D, Cheuk A, Jensen MC, Khan J, Dimitrov DS, Orentas RJ. FGFR4-Targeted Chimeric Antigen Receptors Combined with Anti-Myeloid Polypharmacy Effectively Treat Orthotopic Rhabdomyosarcoma Molecular cancer therapeutics 2022. PMID 35877472.
- Wang X, Rao J, Tan Z, Xun T, Zhao J, Yang X. Inflammatory signaling on cytochrome P450-mediated drug metabolism in hepatocytes Frontiers in pharmacology 2022. PMID 36353494.
- White CM, Sicignano DJ, Smith K. Impact of Interferons and Biological Drug Inhibitors of IL-2 and IL-6 on Small-Molecule Drug Metabolism Through the Cytochrome P450 System The Annals of pharmacotherapy 2022. PMID 34078115.
- Lenoir C, Rodieux F, Desmeules JA, Rollason V, Samer CF. Impact of Inflammation on Cytochromes P450 Activity in Pediatrics: A Systematic Review Clinical pharmacokinetics 2021. PMID 34462878.
- Sun W, Lirio RA, Schneider J, Aubrecht J, Kadali H, Baratta M, Gulati P, Suri A, Lin T, Vasudevan R, Rosario M. Assessment of Vedolizumab Disease-Drug-Drug Interaction Potential in Patients With Inflammatory Bowel Diseases Clinical pharmacology in drug development 2021. PMID 33331142.
- Bishop MW, Hutson PR, Hank JA, Sondel PM, Furman WL, Meagher MM, Navid F, Santana VM. A Phase 1 and pharmacokinetic study evaluating daily or weekly schedules of the humanized anti-GD2 antibody hu14.18K322A in recurrent/refractory solid tumors mAbs 2020. PMID 32643524.
- Boidin C, Moshiri P, Dahyot-Fizelier C, Goutelle S, Lefeuvre S. Pharmacokinetic variability of beta-lactams in critically ill patients: A narrative review Anaesthesia, critical care & pain medicine 2020. PMID 31513935.
- Cojutti PG, Londero A, Della Siega P, Givone F, Fabris M, Biasizzo J, Tascini C, Pea F. Comparative Population Pharmacokinetics of Darunavir in SARS-CoV-2 Patients vs. HIV Patients: The Role of Interleukin-6 Clinical pharmacokinetics 2020. PMID 32856282.
- V Kholodenko I, V Kalinovsky D, V Svirshchevskaya E, I Doronin I, V Konovalova M, V Kibardin A, V Shamanskaya T, S Larin S, M Deyev S, V Kholodenko R. Multimerization through Pegylation Improves Pharmacokinetic Properties of scFv Fragments of GD2-Specific Antibodies Molecules (Basel, Switzerland) 2019. PMID 31653037.
- Khalilieh S, Hussain A, Montgomery D, Levine V, Shaw PM, Bodrug I, Mekokishvili L, Bailey-Smith C, Glasgow XS, Cheng A, Martinho M, Iwamoto M. Effect of tildrakizumab (MK-3222), a high affinity, selective anti-IL23p19 monoclonal antibody, on cytochrome P450 metabolism in subjects with moderate to severe psoriasis British journal of clinical pharmacology 2018. PMID 29926968.
- Cheung IY, Kushner BH, Modak S, Basu EM, Roberts SS, Cheung NV. Phase I trial of anti-GD2 monoclonal antibody hu3F8 plus GM-CSF: Impact of body weight, immunogenicity and anti-GD2 response on pharmacokinetics and survival Oncoimmunology 2017. PMID 29147617.
- Federico SM, McCarville MB, Shulkin BL, Sondel PM, Hank JA, Hutson P, Meagher M, Shafer A, Ng CY, Leung W, Janssen WE, Wu J, Mao S, Brennan RC, Santana VM, Pappo AS, Furman WL. A Pilot Trial of Humanized Anti-GD2 Monoclonal Antibody (hu14.18K322A) with Chemotherapy and Natural Killer Cells in Children with Recurrent/Refractory Neuroblastoma Clinical cancer research : an official journal of the American Association for Cancer Research 2017. PMID 28939747.
- Lee EB, Daskalakis N, Xu C, Paccaly A, Miller B, Fleischmann R, Bodrug I, Kivitz A. Disease-Drug Interaction of Sarilumab and Simvastatin in Patients with Rheumatoid Arthritis Clinical pharmacokinetics 2017. PMID 27722854.
- Mallick P, Taneja G, Moorthy B, Ghose R. Regulation of drug-metabolizing enzymes in infectious and inflammatory disease: implications for biologics-small molecule drug interactions Expert opinion on drug metabolism & toxicology 2017. PMID 28537216.
- Jiang X, Zhuang Y, Xu Z, Wang W, Zhou H. Development of a Physiologically Based Pharmacokinetic Model to Predict Disease-Mediated Therapeutic Protein-Drug Interactions: Modulation of Multiple Cytochrome P450 Enzymes by Interleukin-6 The AAPS journal 2016. PMID 26961818.
- Marachelian A, Desai A, Balis F, Katzenstein H, Qayed M, Armstrong M, Neville KA, Cohn SL, Bush M, Gunawan R, Lim AP, Smith MA, Smith LM. Comparative pharmacokinetics, safety, and tolerability of two sources of ch14.18

in pediatric patients with high-risk neuroblastoma following myeloablative therapy Cancer chemotherapy and pharmacology 2016. PMID 26791869.
- Tran JQ, Othman AA, Wolstencroft P, Elkins J. Therapeutic protein-drug interaction assessment for daclizumab high-yield process in patients with multiple sclerosis using a cocktail approach British journal of clinical pharmacology 2016. PMID 26991517.
- Wei X, Kenny JR, Dickmann L, Maciuca R, Looney C, Tang MT. Assessment of Disease-Related Therapeutic Protein Drug-Drug Interaction for Etrolizumab in Patients With Moderately to Severely Active Ulcerative Colitis Journal of clinical pharmacology 2016. PMID 26412221.
- Zhou H, Sharma A. Therapeutic protein-drug interactions: plausible mechanisms and assessment strategies Expert opinion on drug metabolism & toxicology 2016. PMID 27391296.
- Navid F, Sondel PM, Barfield R, Shulkin BL, Kaufman RA, Allay JA, Gan J, Hutson P, Seo S, Kim K, Goldberg J, Hank JA, Billups CA, Wu J, Furman WL, McGregor LM, Otto M, Gillies SD, Handgretinger R, Santana VM. Phase I trial of a novel anti-GD2 monoclonal antibody, Hu14.18K322A, designed to decrease toxicity in children with refractory or recurrent neuroblastoma Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2014. PMID 24711551.
- Velissaris D, Karamouzos V, Marangos M, Pierrakos C, Karanikolas M. Pharmacokinetic changes and dosing modification of aminoglycosides in critically ill obese patients: a literature review Journal of clinical medicine research 2014. PMID 24883145.
- Wang J, Wang YM, Ahn HY. Biological products for the treatment of psoriasis: therapeutic targets, pharmacodynamics and disease-drug-drug interaction implications The AAPS journal 2014. PMID 24993574.
- Ladenstein R, Weixler S, Baykan B, Bleeke M, Kunert R, Katinger D, Pribill I, Glander P, Bauer S, Pistoia V, Michon J, Garaventa A, Lode HN. Ch14.18 antibody produced in CHO cells in relapsed or refractory Stage 4 neuroblastoma patients: a SIOPEN Phase 1 study mAbs 2013. PMID 23924804.
- Sime FB, Roberts MS, Peake SL, Lipman J, Roberts JA. Does Beta-lactam Pharmacokinetic Variability in Critically Ill Patients Justify Therapeutic Drug Monitoring? A Systematic Review Annals of intensive care 2012. PMID 22839761.
- Clements JD, Jamali F. Pravastatin reverses the down-regulating effect of inflammation on beta-adrenergic receptors: a disease-drug interaction between inflammation, pravastatin, and propranolol Vascular pharmacology 2007. PMID 16949884.
- Aitken AE, Richardson TA, Morgan ET. Regulation of drug-metabolizing enzymes and transporters in inflammation Annual review of pharmacology and toxicology 2006. PMID 16402901.
- Guirguis MS, Jamali F. Disease-drug interaction: Reduced response to propranolol despite increased concentration in the rat with inflammation Journal of pharmaceutical sciences 2003. PMID 12712428.

cardiopulmonary (53)

- Böll B, Garcia Borrega J, Bach F, Grans-Siebel J, Kaul C. [Immunotherapy-Induced Adverse Events: A Focus of Intensive Care Medicine] Deutsche medizinische Wochenschrift (1946) 2026. PMID 42508415.
- Chatrathi AR, Patel K, Saini N, Palaskas NL. Cardiotoxicity induced by multiple myeloma therapies: mechanistic convergence across proteasome inhibitors, CAR-T, and bispecific antibodies Blood cancer journal 2026. PMID 42215456.
- Delafoy M, Vincenti M, Roussel A, Prigent S, Rialland F, Doulet ST, Hauet Q, Broccia MV, Rabian F, Poirée M, Tabone MD, El-Louali F, Saultier P. [Cardiotoxicity in children and adolescents with acute leukemia: Recommendations from the Leukemia Committee of the French Society of Childhood Cancer (SFCE)] Bulletin du cancer 2026. PMID 42315465.
- Gao Y, Li J, Zhang H, Zhang Y, Qian S, Zhu X, Hong L, Xie L, Fu Q, Bao X, Tong Z, Li Y, Li B, Wang L, Shen J, Zhang Q, He X, Zheng Y, Yao C, Liu L, Zhao P, Campos PV, Yang Z, Qian C, Fang W. Hypoxia-responsive CEA-targeted CAR T cells in CEA-positive solid tumors through intraperitoneal or intravenous infusion: a phase 1 trial Nature cancer 2026. PMID 41760800.
- Hidalgo L, Garcia-Rodriguez P, Cubillo I, Zubizarreta M, Perez-Martinez A, García-Castro J. Unaltered NKG2D-CAR T cell function under hypoxia in osteosarcoma in vitro Cancer immunology, immunotherapy : CII 2026. PMID 41677938.
- Hua X, Sun Z, Liang Z, Huang Y, Mo H, Dong F, Mo S, Yang X, Zhang N, Chen X, Liao S, Qi Z, Zhang R, Guan S, Li L, Xu Y, Song J. Engineered T cell therapy for the treatment of cardiac fibrosis during chronic phase of myocarditis Theranostics 2026. PMID 41356193.
- Huang N, Chen Q, Zhang X, Farooq MA, He W, Cui K, Zhu S, Ajmal I, Jiang D, Xu G. Overexpression of HIF-1α via small activating RNA enhances CAR-T cell function in hypoxic microenvironments Pharmacological research 2026. PMID 42105819.

- Li S, Shiekh PA, Li Y, Nan H, Yang L, Mitchell K, yang Y, Zhu Y. Oxygen Delivery Reprograms Hypoxic Tumor Niche to Rejuvenate CAR-NKT Cells for Improved Cancer Immunotherapy Research Square 2026. DOI 10.21203/rs.3.rs-8296891/v1.
- Pedram Rad P, Reyhani A, Dashti M. From barrier to trigger: leveraging tumor hypoxia for precise control of CAR-T cell activity Molecular biology reports 2026. PMID 42455334.
- Raddatz MA, Stein-Merlob AF, Mahmood SS, Ganatra S, Addison D, Lin TL, Larson SM, Yang EH. Cardiotoxicity of T cell immunotherapies Nature reviews. Cardiology 2026. PMID 41730987.
- Sinigiani G, Scapinello G, Berno T, Piazza F, Cipriani A. Chimeric antigen receptor T cell therapy cardiotoxicity: a narrative review Cardio-oncology (London, England) 2026. PMID 42415149.
- Tana M, Piccinini R, Pinterpe G, Porreca E, Berardi R, Tana C. Cardiotoxicity of Antitumor Agents: Therapeutic Challenges in Heart Failure with Reduced and Preserved Ejection Fraction International journal of molecular sciences 2026. PMID 41977162.
- Tu C, Van der Vreken A, Meeus F, van den Broecke L, Brons J, De Veirman K, Vanderkerken K, De Bruyne E, Breckpot K, Menu E. Hypoxia promotes BCMA loss and a suppressive secretome thereby hindering CAR T cell therapy in multiple myeloma Experimental hematology & oncology 2026. PMID 41508031.
- Voß F, Goerke P, Kelm M, Polzin A. Heart failure medication to prevent cancer therapy-related cardiac dysfunction: A narrative review British journal of pharmacology 2026. PMID 42476814.
- Xu L, Wu H, Wu C, Xu Y. Chimeric antigen receptor-T cell therapy-induced cardiotoxicity: Pathophysiological mechanisms and pharmacological intervention strategies Pharmacological research 2026. PMID 42331267.
- Codru IR, Vecerzan L. When and for Whom Does Intensive Care Unit Admission Change the Prognosis in Oncology?-A Scoping Review Cancers 2025. PMID 41301003.
- Grans-Siebel J, Garcia Borrega J, Böll B, Bach F. [Modern immunotherapy and emergency situations : Toxicities of CAR T-cell therapies and BiTEs, their management, and their relevance in intensive care medicine] Medizinische Klinik, Intensivmedizin und Notfallmedizin 2025. PMID 40892214.
- Grunblatt E, Meng Z, Baldridge AS, Patel NP, Stanisic A, Feinstein MJ, Rao A, Gordon LI, Winter JN, Ma S, Mehta J, Singhal S, Karmali R, Akhter N. Variance in development of early and late cardiotoxicities in patients with lymphoma and myeloma receiving CAR T-cell therapies Leukemia & lymphoma 2025. PMID 39772871.
- Hernádfői M, Szabados M, Brückner E, Varga Á, Hauser P, Ottóffy G, Vojcek Á, Csanádi K, Kertész G, Jakab Z, Agócs G, Garami M. Dinutuximab Beta for the Treatment of High-Risk Neuroblastoma: Data from the Hungarian Pediatric Oncology Network Journal of clinical medicine 2025. PMID 41010845.
- Joseph T, Sanchez J, Abbasi A, Zhang L, Sica RA, Duong TQ. Cardiotoxic Effects Following CAR-T Cell Therapy: A Literature Review Current oncology reports 2025. PMID 39836349.
- Keramida K, Kyriakou TC, Antoniades A, Tocchetti CG. Immunotherapy-associated Cardiotoxicity: a Comprehensive Review of Checkpoint inhibitors, monoclonal, cellular, and cytokine-based Treatments Current treatment options in oncology 2025. PMID 41143919.
- Li W, Chen J, Li J, Wang S, Chen Z, Zhao L, Zhao Y, Gu L, Liu J, Zhang Y, Yang X, Chen T, Guo Z. Envirotune-CAR-T: a hypoxia-responsive and glutamine-enhanced CAR-T cell therapy for overcoming tumor microenvironment-mediated suppression Journal for immunotherapy of cancer 2025. PMID 41083281.
- Liu Y, Murphy KM, Hung YF, Won T. Pathogenic drivers of lupus myocarditis and potential therapeutic targets Experimental & molecular medicine 2025. PMID 41203999.
- Micallef Nilsson I, Poiret T, Ryu J, Mohammadpour M, Henriksson J, Österborg A, Mattsson J, Schurich A, Magalhaes I. Production of functional CD19 CAR T cells under hypoxic manufacturing conditions Frontiers in immunology 2025. PMID 41132651.
- Stefanou E, Tsitsinakis G, Karageorgou D, Kole C. Immunotherapy-Associated Cardiotoxicity: Current Insights and Future Directions for Precision Cardio-Oncology Cancers 2025. PMID 40940933.
- Sánchez B, González P, Goveo I, Contreras P, Domínguez L, Manrique P, Vargas M. Chemotherapy-Induced Cardiotoxicity: Mechanisms, Detection and Emerging Therapies in Cardio-Oncology Discoveries (Craiova, Romania) 2025. PMID 41583362.
- Wei W, Fader ME, Kushner BH, Cardenas FI, Kramer K, Basu E, Modak S. Pulmonary Toxicity Associated With Concurrent Lorlatinib and Anti-GD2 Monoclonal Antibody Therapy in Patients With Neuroblastoma Pediatric blood & cancer 2025. PMID 40551396.
- Zhou M, Xu L, Hu J, Chen W, Hong J, Wang M, Guo Z. Metabolic reprogramming through PIM3 inhibition reverses hypoxia-induced CAR-T cell dysfunction in solid tumors Journal of translational medicine 2025. PMID 41199316.
- Camilli M, Viscovo M, Felici T, Maggio L, Ballacci F, Carella G, Bonanni A, Lamendola P, Tinti L, Di Renzo A, Coarelli G, Galli E, Liuzzo G, Burzotta F, Montone RA, Sorà F, Sica S, Hohaus S, Lanza GA, Crea F, Lombardo A, Minotti G. Inflammation and acute cardiotoxicity in adult hematological patients treated with CAR-T cells: results from a pilot proof-of-concept study Cardio-oncology (London, England) 2024. PMID 38532515.
- Karthikeyan B, Sunder SS, Puzanov I, Olejniczak SH, Pokharel S, Sharma UC. Cardiotoxic profiles of CAR-T therapy and bispecific T-cell engagers in hematological cancers Communications medicine 2024. PMID 38871977.

- Mazetto RASV, Monteiro SON, Bulhões E, Defante MLR, Antunes VLJ, Balieiro CCA, Feitoza L, Ferreira ALC, Carvalho AM, Guida C. The cardiotoxic effects of CAR-T cell therapy: An updated systematic review and meta-analysis European journal of haematology 2024. PMID 39171519.
- Patel NP, Dalal PJ, Meng Z, Baldridge AS, Cascino GJ, Sunderraj A, Sinha A, Karmali R, Feinstein MJ, Akhter N. Myocardial strain is associated with adverse cardiac events in patients treated with chimeric antigen receptor (CAR) T-cell therapy European journal of haematology 2024. PMID 37649240.
- Shouval R, Goldman A, Flynn JR, El-Moghraby A, Rehman M, Devlin SM, Corona M, Landego I, Lin RJ, Scordo M, Raj SS, Giralt SA, Palomba ML, Dahi PB, Walji M, Salles G, Nath K, Geyer MB, Park JH, Fein JA, Kosmidou I, Shah GL, Liu JE, Perales MA, Mahmood SS. Atrial arrhythmias following CAR-chimeric antigen receptor T-cell therapy: Incidence, risk factors and biomarker profile British journal of haematology 2024. PMID 38735683.
- Aurensanz-Clemente E, Pérez Casares Á, Muñoz JP, García-Canadilla P, Zuccarino F, Sanchez-de-Toledo J. Acute myocarditis with transient myocardial thickening in two oncologic patients treated with anti-GD2 immunotherapy ESC heart failure 2023. PMID 36883000.
- Camilli M, Maggio L, Tinti L, Lamendola P, Lanza GA, Crea F, Lombardo A. Chimeric antigen receptor-T cell therapy-related cardiotoxicity in adults and children cancer patients: A clinical appraisal Frontiers in cardiovascular medicine 2023. PMID 36895831.
- Gawlik M, Zimodro JM, Gąsecka A, Filipiak KJ, Szmit S. Cardiac Arrhythmias in Oncological Patients-Epidemiology, Risk Factors, and Management within the Context of the New ESC 2022 Guidelines Current oncology reports 2023. PMID 37589940.
- Guo M, Wang X, Xiao S, Liu A, Xu T, Huan C, Wu H, Hu Y, Zhou S, Zhu H, Pan D. Preliminary assessment of cardiotoxicity in chimeric antigen receptor T cell therapy: a systematic review and meta-analysis Clinical and experimental medicine 2023. PMID 36930381.
- Iris F, Michael W, Dalia W, Revel-Vilk S. A Real-life use of IV naxitamab for children with relapsed/refractory neuroblastoma Authorea Preprints 2023. DOI 10.22541/au.169339152.21980785/v1.
- Lee DH, Chandrasekhar S, Jain MD, Mhaskar R, Reid K, Lee SB, Corallo S, Hidalgo-Vargas MJ, Kumar A, Chavez J, Shah B, Lazaryan A, Khimani F, Nishihori T, Bachmeier C, Faramand R, Fradley MG, Jeong D, Oliveira GH, Locke FL, Davila ML, Alomar M. Cardiac and inflammatory biomarker differences in adverse cardiac events after chimeric antigen receptor T-Cell therapy: an exploratory study Cardio-oncology (London, England) 2023. PMID 37005652.
- Lee DH, Kumar A, Mohammed T, Peres LC, Alsina M, Bachmeier C, Blue BJ, Brayer J, Chandrasekhar S, Grajales Cruz A, De Avila G, Elmariah H, Faramand R, Freeman C, Jain M, Khadka S, Khimani F, Liu H, Nishihori T, Oswald LB, Castaneda Puglianini OA, Shain KH, Smith E, Baz RC, Locke FL, Oliveira GH, Alomar M, Hansen DK. Cardiac events after standard of care idecabtagene vicleucel for relapsed and refractory multiple myeloma Blood advances 2023. PMID 37307173.
- Mora J, Mora J, Chan GC, Morgenstern DA, Nysom K, Bear MK, Tornøe K, Kushner BH. Outpatient administration of naxitamab in combination with granulocyte-macrophage colony-stimulating factor in patients with refractory and/or relapsed high-risk neuroblastoma: Management of adverse events Cancer reports (Hoboken, N.J.) 2023. PMID 35579862.
- Varo A, Castañeda A, Chamorro S, Muñoz JP, Gorostegui M, Celma MS, Lopez S, Simao M, Perez-Jaume S, Mora J. Novel infusion strategy reduces severe adverse events caused by the anti-GD2 monoclonal antibody naxitamab Frontiers in oncology 2023. PMID 37213300.
- Briasoulis A, Chasouraki A, Sianis A, Panagiotou N, Kourek C, Ntalianis A, Paraskevaidis I. Cardiotoxicity of Non-Anthracycline Cancer Chemotherapy Agents Journal of cardiovascular development and disease 2022. PMID 35323614.
- Ganatra S, Dani SS, Yang EH, Zaha VG, Nohria A. Cardiotoxicity of T-Cell Antineoplastic Therapies: JACC: CardioOncology Primer JACC. CardioOncology 2022. PMID 36636447.
- Hanna KS, Kaur H, Alazzeh MS, Thandavaram A, Channar A, Purohit A, Shrestha B, Patel D, Shah H, Mohammed L. Cardiotoxicity Associated With Chimeric Antigen Receptor (CAR)-T Cell Therapy for Hematologic Malignancies: A Systematic Review Cureus 2022. PMID 36148204.
- Nenna A, Carpenito M, Chello C, Nappi P, Annibali O, Vincenzi B, Grigioni F, Chello M, Nappi F. Cardiotoxicity of Chimeric Antigen Receptor T-Cell (CAR-T) Therapy: Pathophysiology, Clinical Implications, and Echocardiographic Assessment International journal of molecular sciences 2022. PMID 35897819.
- Burns EA, Gentille C, Trachtenberg B, Pingali SR, Anand K. Cardiotoxicity Associated with Anti-CD19 Chimeric Antigen Receptor T-Cell (CAR-T) Therapy: Recognition, Risk Factors, and Management Diseases (Basel, Switzerland) 2021. PMID 33802788.
- Stein-Merlob AF, Rothberg MV, Holman P, Yang EH. Immunotherapy-Associated Cardiotoxicity of Immune Checkpoint Inhibitors and Chimeric Antigen Receptor T Cell Therapy: Diagnostic and Management Challenges and Strategies Current cardiology reports 2021. PMID 33483873.
- Dal'bo N, Patel R, Parikh R, Shah SP, Guha A, Dani SS, Ganatra S. Cardiotoxicity of Contemporary Anticancer Immunotherapy Current treatment options in cardiovascular medicine 2020. PMID 33162729.

- Ghosh AK, Chen DH, Guha A, Mackenzie S, Walker JM, Roddie C. CAR T Cell Therapy-Related Cardiovascular Outcomes and Management: Systemic Disease or Direct Cardiotoxicity? JACC. CardioOncology 2020. PMID 34396213.
- Gutierrez C, Brown ART, Herr MM, Kadri SS, Hill B, Rajendram P, Duggal A, Turtle CJ, Patel K, Lin Y, May HP, Gallo de Moraes A, Maus MV, Frigault MJ, Brudno JN, Athale J, Shah NN, Kochenderfer JN, Dharshan A, Beitinjaneh A, Arias AS, McEvoy C, Mead E, Stephens RS, Nates JL, Neelapu SS, Pastores SM. The chimeric antigen receptor-intensive care unit (CAR-ICU) initiative: Surveying intensive care unit practices in the management of CAR T-cell associated toxicities Journal of critical care 2020. PMID 32361219.
- Simbaqueba CC, Aponte MP, Kim P, Deswal A, Palaskas NL, Iliescu C, Jahangir E, Yang EH, Steiner RE, Lopez-Mattei J. Cardiovascular Complications of Chimeric Antigen Receptor T-Cell Therapy: The Cytokine Release Syndrome and Associated Arrhythmias Journal of immunotherapy and precision oncology 2020. PMID 35663258.
- Shusterman S, London WB, Gillies SD, Hank JA, Voss SD, Seeger RC, Reynolds CP, Kimball J, Albertini MR, Wagner B, Gan J, Eickhoff J, DeSantes KB, Cohn SL, Hecht T, Gadbaw B, Reisfeld RA, Maris JM, Sondel PM. Antitumor activity of hu14.18-IL2 in patients with relapsed/refractory neuroblastoma: a Children's Oncology Group (COG) phase II study Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2010. PMID 20921469.

hlh mas (43)

- Al-Mashdali AF, Mohamed F, Osman M, Abdelraof MO, Al Rasheed M, Salama H, Mohamed SF, Yassin MA. Real-world use of emapalumab in hemophagocytic lymphohistiocytosis: a scoping review of published evidence Frontiers in pharmacology 2026. PMID 42038293.
- Donzelli L, Zullino V, Torelli GF, De Propris MS, Piazzolla M, Ruberto F, Martelli M, Di Rocco A. Managing Treatment-Emergent Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis-Like Syndrome Following CAR-T Cell Therapy: A Case-Based Review of the use of Emapalumab Hematological oncology 2026. PMID 41311365.
- Fresa A, Cuccaro A, Bonanni M, Galati D, Ferrara G, Volzone F, Crisci S, Rivieccio M, Oro M, Morgillo E, Mele S, De Filippi R, Pinto A. Immune effector cell-associated hemophagocytic lymphohistiocytosis-like syndrome during treatment with the bivalent CD20xCD3 bispecific antibody glofitamab Haematologica 2026. PMID 41988762.
- Galletta TJ, Rubinstein JD, Dandoy CE, Khoury R, Faulhaber A, Phillips CL, Krupski C. Emapalumab for Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis-Like Syndrome Following CD19-Directed CAR-T in Two Patients With B-ALL: Clinical and Biomarker Correlates Pediatric blood & cancer 2026. PMID 42136039.
- Gower N, Houot R, Pizot C, Baruchel A, Dourthe ME, Beauvais D, Gastinne T, Brisou G, Dulery R, Guidez S, Tudesq JJ, Chauchet A, Herbreteau L, Campidelli A, Bay JO, Sesques P, Chiche E, Gros FX, Durand A, Lubnau M, Sterin A, Di Blasi R, Marquet A, Gat E, Rabian F, Leguay T, Roth-Guépin G, Boissel N, Morschhauser F, Paillassa J. Hemophagocytic lymphohistiocytosis-like syndrome after CD19-directed CAR T-cells for B-cell lymphoma and B-cell acute lymphoblastic leukemia: A LYSA, SFCE, and GRAALL study from the DESCAR-T registry HemaSphere 2026. PMID 42472032.
- Green AC, Tzankov A, Weinberg O, Traverse-Glehen A, Saft L, Leguit RJ. Reactive and therapy induced bone marrow changes linked to systemic infectious and non-infectious disorders including MAS/HLH report from the European association for haematopathology, Dubrovnik 2024 Virchows Archiv : an international journal of pathology 2026. PMID 42332261.
- Gummadi YK, Sutherland A, Rowe S, Chen LYC. Lamotrigine and other therapy-associated HLH syndromes: a review Expert review of clinical immunology 2026. PMID 42579330.
- Hines MR, Knight TE, McNerney KO, Leick MB, Jain T, Ahmed S, Frigault MJ, Hill JA, Jain MD, Johnson WT, Lin Y, Mahadeo KM, Maron GM, Marsh RA, Neelapu SS, Nikiforow S, Ombrello AK, Shah NN, Talleur AC, Turicek D, Vatsayan A, Wong SW, Maus MV, Komanduri KV, Berliner N, Henter JI, Perales MA, Frey NV, Teachey DT, Frank MJ, Shah NN. Erratum to 'Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis-Like Syndrome'[Transplant Cell Ther. 2023 Jul;29(7):438.e1-438.e16. Epub 2023 Mar 9] Transplantation and cellular therapy 2026. PMID 41067707.
- Irfan S, Ayoobkhan FS, Vojjala N, Qureshi R, Rahman R, Kamboj I, Abdallah AO, McGuirk J, Mahmoudjafari Z, Lutfi F, Ahmed N. Immune Effector Cell-Associated HLH-Like Syndrome (IEC-HS) in CAR-T and TCE-Treated Myeloma and B-Cell Malignancies: Insights from a Pharmacovigilance Study Transplantation and cellular therapy 2026. PMID 41391508.
- Johnson WT, McNerney KO, Frank MJ, Shah NN. How I treat HLH-like toxicities after immune effector cell therapy Blood 2026. PMID 41779733.
- Masuda K, Sota M, Kitamura A, Yoshida T, Tanigawa T, Takikawa K, Tozuka K, Sakamoto M, Shindo T, Fujii T, Yamaguchi K, Nakamura Y, Inaba R, Ishizuchi K, Matsushita H, Yamazaki R, Koda Y, Sakurai M, Kataoka K. Late-

onset Human Herpesvirus 6 Encephalitis with Secondary Hemophagocytic Lymphohistiocytosis after CAR T-cell Therapy Internal medicine (Tokyo, Japan) 2026. PMID 42618274.

- McNerney KO, Kwon S, Naik S, Fabrizio VA, Llaurador Caraballo G, Sanchez-Pinto LN, Talleur AC, Zeng XL, Baggott C, Vatsayan A, Verneris MR, Prabhu S, Rossoff J, Pacenta HL, Phillips CL, Talano JM, Moskop A, Myers GD, Karras NA, Bonifant CL, Qayed M, Satwani P, Winestone LE, Krupski C, Keating AK, Baumeister SHC, Egeler E, Stieglitz E, Curran KJ, Mackall CL, John S, Laetsch TW, Gardner RA, Schultz LM. HLH-like syndromes associate with increased risk of death and CD19-negative relapse post-CAR T in pediatric B-ALL Blood advances 2026. PMID 42469167.
- Mouhieddine TH, Santos JVD, Sehgal H, Salib C, Hussein S, Duffield A, Richard S, Aleman A, Lagana A, Parekh S, Jagannath S. Emapalumab and eltrombopag rescue for prolonged immune effector cell-associated hemophagocytic syndrome after CAR T-cell therapy in multiple myeloma Haematologica 2026. PMID 42389839.
- Rincón-López S, García-Muñoz R, Atencio-Matos J, Pereda-Sainz O, Larreina-Pérez J, Campeny-Najara A, Pérez PH, Hernández-Pérez M, Feliu J, Esteban-Figuerola A. Immune Effector Cell-associated Hemophagocytic Syndrome (IEC-HS) as a late immune-mediated toxicity after CAR-T therapy in Diffuse Large B-Cell Lymphoma (DLBCL): a case report Annals of hematology 2026. PMID 42360480.
- Rocco JM, Shah NN. Seeing Is Believing: IEC-HS after CAR T Cells Blood cancer discovery 2026. PMID 41636362.
- Shaforostova I, Kronig MN, Seipel K, Rovo A, Bacher U, Pabst T. Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis Following CAR T-Cell Therapy: Results of a Real-World Study Cancers 2026. PMID 42192954.
- Srinagesh HK, Kramer AM, Baird JH, Reschke A, Sahaf B, Cancilla J, Syal S, Su YJ, Agarwal N, Jensen AM, Schultz LM, Jeyakumar N, Ramakrishna S, Davis KL, Dahiya S, Feldman SA, Mackall CL, Muffly LS, Miklos DB, Frank MJ. Clinical and Cytokine Features of Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis-Like Syndrome Blood cancer discovery 2026. PMID 41411617.
- Zhang C, Wu H, Cao T, Wu C, Yan C, Qi W, Zhang Y, Ji X, Jing H, Peng Q, Zhang W. Hemophagocytic Lymphohistiocytosis After CAR T-Cell Therapy: A Systematic Review and Meta-Analysis American journal of hematology 2026. PMID 42086512.
- Zhang Y, Yang J, Xin H, Ai K, Yang M, Li Y, He Y. Chimeric antigen receptor T-cell therapies related to immune effector cell-associated hemophagocytic lymphohistiocytosis-like syndrome: Diagnosis, high-risk factors, and management Chinese medical journal 2026. PMID 41914034.
- Campodonico E, Angelillo P, Doglio M, Ugolini A, Lupo Stanghellini MT, Noviello M, Tassi E, Pedica F, Bergonzi GM, Lazzari L, Bruno A, Diral E, Ponzoni M, Corti C, Peccatori J, Ruggeri A, Ferreri AJM, Bonini C, Ciceri F, Carrabba MG, Greco R. Biomarkers for an early diagnosis of immune effector cell associated-hemophagocytic syndrome Frontiers in immunology 2025. PMID 40895549.
- Chaudhary R, Tendero S, Chaudhary A, Julio L. Hemophagocytic Lymphohistiocytosis Secondary to Macrophage Activation Syndrome in Adult-Onset Still’s Disease: A Diagnostic and Therapeutic Challenge Preprints.org 2025. DOI 10.20944/preprints202506.0350.v1.
- Lee JC, Johnson WT, Hines M, Shah NN. Immune Effector Cell-associated Hemophagocytic Lymphohistiocytosis-like Syndrome (IEC-HS) Hematology/oncology clinics of North America 2025. PMID 40158936.
- Liu Y, Ren Y, Gao L, Ahmed S, Lu X, Li B, Wang C, Yu L, Sun M, Zhuang Y, Miao Y, Ni H, Xie X, Shi X, Xu J, Zhang Y, Zhao M, Xu M, Zhuang W, Gu W, Lin G, Hua H, Zhu J, Xu M, Jia T, Liu P, Zhai L, Zhang T, Shan H, Shen Q, Fan L, Li J, Shi W, Miao Y. Endothelial Activation and Stress Index Serves as a Predictor for Survival in Lymphoma-Associated Hemophagocytic Lymphohistiocytosis: A Retrospective Multicenter Cohort Study of Jiangsu Cooperative Lymphoma Group American journal of hematology 2025. PMID 40919852.
- Manghisi B, Cotilli G, Fedele M, Perfetti P, Terruzzi E, Verga L, Borin LM, Carrer A, Fumagalli M, Ferrari MB, Moretti A, Rona R, Benini A, Vergnano B, Palumbo G, Zincone A, Maglia O, Scollo C, Steidl C, Iovino L, Balduzzi A, Piazza R, Gambacorti-Passerini C, Parma M, Aroldi A. Case Report: Successful use of emapalumab in adult B-cell acute lymphoblastic leukemia experiencing severe neurotoxicity and hemophagocytic lymphohistiocytosis-like features after CAR-T cell therapy Frontiers in immunology 2025. PMID 40255392.
- Munárriz D, Rodríguez-Lobato LG, López-Corral L, Arnaldos-Pérez C, Cabañas V, López-Muñoz N, Oliver-Caldés A, Ponce JC, Reguera JL, Martín AÁ, Martínez-Cibrián N, Tovar N, Delgado J, Guillen E, Varea S, Rodríguez-Otero P, Urbano-Ispizua Á, Moraleda JM, Martínez-López J, Martínez-López J, Mateos MV, De la Calle VG, González-Navarro EA, Fernández de Larrea C. Macrophage activation syndrome-like in multiple myeloma patients treated with the academic CAR-T against BCMA ARI0002h Frontiers in immunology 2025. PMID 41181103.
- Ortí G, Dachy G, Graham CE, Peric Z, Alarcon A, Del Bufalo F, Doglio M, Henter JI, Roddie C, Stewart O, Van de Donk WCJ, Velasco R, Aljurf M, Ruggeri A, Onida F, Sánchez-Ortega I, Yakoub-Agha I, Penack O. Less frequent complications following CAR T-cell therapies: hemophagocytic lymphohistiocytosis, graft-versus-host disease, thrombotic microangiopathy, coagulation disorders and secondary malignancies: best practice recommendations from the EBMT Practice Harmonization and Guidelines Committee Bone marrow transplantation 2025. PMID 40205032.

- Wobma H, Henderson LA, Duncan CN, Prockop SE, Randolph AG, Degar BA, Lehmann LE, Baumeister SHC. Treating and triggering hyperinflammation: tackling hemophagocytic lymphohistiocytosis and HLH-like syndromes in the pediatric cell therapy and critical care setting Frontiers in oncology 2025. PMID 41049857.
- Ye Y, Li S, Guo Z, Zhao L, Zhou H, Zhong N, He M, Cao YJ, Liu L. Novel Loop-Structure-Based CD19/CD22 Dual-Target CAR-T Therapy for High-Risk Diffuse Large B-Cell Lymphoma Presenting with Hemophagocytic Lymphohistiocytosis: A Case Report Cancer management and research 2025. PMID 40688752.
- Zhong X, Zhu X, Ma X, Zhou L, Luo X, Li P, Ding Y, Fu J, Bo J, Yang M, Liang A, Zeng Y, Xiu B. Germline mutations in B-cell non-Hodgkin lymphoma-associated hemophagocytic lymphohistiocytosis (LA-HLH) and patient outcomes Seminars in oncology 2025. PMID 40700998.
- Meireles AM, Iacoboni G, Moço LM, Ramos I, Brás G, Azevedo J, Rodrigues Â, Moreira C, Mariz M. Hemophagocytic lymphohistiocytosis in a patient with Epstein-Barr virus-positive diffuse large B-cell lymphoma treated with chimeric antigen receptor T-cell therapy Immunotherapy 2024. PMID 39417342.
- Qureshi Z, Altaf F, Jamil A, Siddique R. Optimization Strategies in CAR T-cell Therapy: A Comprehensive Evaluation of Cytopenia, HLH/MAS, and Other Adverse Events American journal of clinical oncology 2024. PMID 38907604.
- Scala JJ, Eckrich MJ, Lipak K, Yates B, Yuan C, Wang HW, Dahiya S, Henter JI, Huo JS, Frank MJ, Shah NN. Treatment strategies for progressive immune effector cell-associated hemophagocytic lymphohistiocytosis-like syndrome: case series Haematologica 2024. PMID 38813719.
- Shen H, Zhang M, Pan Y, Peng P, Wang Z, Yao C, Zhou L. Interferon-γ blockade in CD19 CART cell therapy-induced secondary macrophage activation syndrome/hemophagocytic lymphohistiocytosis Research Square 2024. DOI 10.21203/rs.3.rs-4234115/v1.
- Yan W, Xiong Y, Lv R, Du C, Yu T, Zhang S, Sui W, Deng S, Xiao J, Xu Y, Zou D, Qiu L, An G. Uncommon biphasic CAR-T expansion induces hemophagocytic lymphohistiocytosis-like syndrome and fatal multiple infections following BCMA CAR-T cell therapy: a case report Journal for immunotherapy of cancer 2024. PMID 39608976.
- Fugere T, Baltz A, Mukherjee A, Gaddam M, Varma A, Veeraputhiran M, Gentille Sanchez CG. Immune Effector Cell-Associated HLH-like Syndrome: A Review of the Literature of an Increasingly Recognized Entity Cancers 2023. PMID 37958323.
- Hines MR, Knight TE, McNerney KO, Leick MB, Jain T, Ahmed S, Frigault MJ, Hill JA, Jain MD, Johnson WT, Lin Y, Mahadeo KM, Maron GM, Marsh RA, Neelapu SS, Nikiforow S, Ombrello AK, Shah NN, Talleur AC, Turicek D, Vatsayan A, Wong SW, Maus MV, Komanduri KV, Berliner N, Henter JI, Perales MA, Frey NV, Teachey DT, Frank MJ, Shah NN. Immune Effector Cell-Associated Hemophagocytic Lymphohistiocytosis-Like Syndrome Transplantation and cellular therapy 2023. PMID 36906275.
- Rainone M, Ngo D, Baird JH, Budde LE, Htut M, Aldoss I, Pullarkat V. Interferon-γ blockade in CAR T-cell therapy-associated macrophage activation syndrome/hemophagocytic lymphohistiocytosis Blood advances 2023. PMID 35917457.
- Zu C, Wu S, Zhang M, Wei G, Xu H, Cui J, Chang AH, Huang H, Hu Y. A distinct cytokine network distinguishes chimeric antigen receptor T cell (CAR-T)-associated hemophagocytic lymphohistiocytosis-like toxicity (carHLH) from severe cytokine release syndrome following CAR-T therapy Cytotherapy 2023. PMID 37480884.
- Kim DW, Bukhari A, Lutfi F, Zafforoni F, Merechi F, Mustafa Ali MK, Gottlieb D, Lee ST, Kocoglu MH, Hardy NM, Yared J, Rapoport AP, Dahiya S, Law JY. Low utility of the H-Score and HLH-2004 criteria to identify patients with secondary hemophagocytic lymphohistiocytosis after CAR-T cell therapy for relapsed/refractory diffuse large B-Cell lymphoma Leukemia & lymphoma 2022. PMID 35045791.
- Hines MR, Keenan C, Maron Alfaro G, Cheng C, Zhou Y, Sharma A, Hurley C, Nichols KE, Gottschalk S, Triplett BM, Talleur AC. Hemophagocytic lymphohistiocytosis-like toxicity (carHLH) after CD19-specific CAR T-cell therapy British journal of haematology 2021. PMID 34263927.
- Kennedy VE, Wong C, Huang CY, Thiruvengadam SK, Wolf J, Martin TG, Shah N, Wong SW. Macrophage activation syndrome-like (MAS-L) manifestations following BCMA-directed CAR T cells in multiple myeloma Blood advances 2021. PMID 34644387.
- Lichtenstein DA, Schischlik F, Shao L, Steinberg SM, Yates B, Wang HW, Wang Y, Inglefield J, Dulau-Florea A, Ceppi F, Hermida LC, Stringaris K, Dunham K, Homan P, Jailwala P, Mirazee J, Robinson W, Chisholm KM, Yuan C, Stetler-Stevenson M, Ombrello AK, Jin J, Fry TJ, Taylor N, Highfill SL, Jin P, Gardner RA, Shalabi H, Ruppin E, Stroncek DF, Shah NN. Characterization of HLH-like manifestations as a CRS variant in patients receiving CD22 CAR T cells Blood 2021. PMID 34525183.
- Major A, Collins J, Craney C, Heitman AK, Bauer E, Zerante E, Stock W, Bishop MR, Jasielec J. Management of hemophagocytic lymphohistiocytosis (HLH) associated with chimeric antigen receptor T-cell (CAR-T) therapy using anti-cytokine therapy: an illustrative case and review of the literature Leukemia & lymphoma 2021. PMID 33559517.

gd2 cart clinical (41)

- Alliot H, Machy P, Leclair MD, Birkle S, Faraj S. Targeting GD2 with CAR-T Cell Therapy in Neuroblastoma: Updates, Challenges, and Future Perspectives Current oncology reports 2026. PMID 42154183.
- Carrion YL, Bhasin J, Anne S, Sawhney A, Ha CJ, Sharaf I, Santana J, Titkov K, Jean M, Thibault D, Pando A, Novakovic N, Parikh NS, Vojnic M, Sherman JH. Locoregional and systemic adoptive cellular therapies for pediatric brain tumors: a systematic review of CAR-T, TCR-engineered T cells, and NK cell strategies Neurosurgical review 2026. PMID 42566099.
- Ciosek A, Hofmann J, Galant K, Marinkovich MP, Wierzbowska A, Ciążyńska M, Bień N, Narbutt J, Lesiak A. Beyond Hematology-Current Insights into Chimeric Antigen Receptor (CAR) T-Cell Therapy for Skin and Connective Tissue Disorders Cells 2026. PMID 42193884.
- Habibi MA, Zand A, Robat-Jazi B, Khodadadiyan A, Dashti M, Nejati N, Lorestan P, Karimizadeh Z, Ahmadpour M, Maleki F, Ahmadpour S. The safety and efficacy of GD2-targeted CAR-T cells in patients with neuroblastoma: a systematic review and meta-analysis European journal of medical research 2026. PMID 41530803.
- Hakem Zadeh F, Shah N, Bird V. Prostate cancer and CAR-T therapy: a systematic review BMC cancer 2026. PMID 41826891.
- Sanikommu S, Santos AN, Dawoud B, Shah KH, Vazquez S, Shah AH, Lu VM. Chimeric Antigen Receptor-T Cell Therapy for Pediatric and Young Adult Primary Central Nervous System Tumors: A Systematic Review of Early Phase Clinical Trials with a Focus on Diffuse Midline Glioma Targeted oncology 2026. PMID 42295570.
- Shokouhfar M, Darzi A, Ameli F, Nami MT, Khorasani SK, Eini P, Vanan AG, Bahrami N. Immunotherapeutic advances in pediatric neuroblastoma: Overcoming resistance through biomarker-guided combinations Biomedicine & pharmacotherapy = Biomedecine & pharmacotherapie 2026. PMID 41621264.
- Srinivasan A, Abraham A, Modak AI, Pradeep A, Nimachwala NL, Chemmancheri AV, Banu A, Azad AA, Abraham A, Khorbaladze L. CAR T-cell therapy in pediatric brain tumors: a narrative review with comparative analysis of clinical trial eligibility criteria Frontiers in oncology 2026. PMID 42568366.
- Vazaios K, Tallon-Cobos AC, van Oosterhout LPJ, Waranecki P, Cornel AM, Dautzenberg NMM, van Hoesel M, Forbes C, Nierkens S, Kemp V, Hoeben RC, van der Lugt J, Calkoen FG, Hulleman E. Boosting CAR T cell functionality with oncolytic viruses for the treatment of pediatric diffuse midline gliomas Molecular therapy. Oncology 2026. PMID 42100144.
- Wannaphut C, Thirasastr P, Livingston J, Araujo D, Conley A, Keung E, Roland C, Somaiah N. The evolution of cellular therapies in sarcoma: Breakthroughs, challenges, and future directions Human vaccines & immunotherapeutics 2026. PMID 42517272.
- DuBois SG, Moreno L, Bagatell R, Cheung NK, Gray JC, Locatelli F, Reynolds CP, Rossig C, Sondel PM, Drezner N, Kholmanshikh O, Bird N, Day L, Ludwinski D, Scobie N, Pons Sanz V, Ball S, Hammer B, Alvarez Rojo I, Meager D, Sveistrup J, Berlanga P, Casanova M, Glade-Bender J, Macy M, Morgenstern DA, Owens C, Weigel BJ, Pappo A, Nysom K, Anderson J, Ligas F, Lesa G, Donoghue M, Karres D, Vassal G. Paediatric Strategy Forum for medicinal product development of agents targeting GD2 ganglioside in children and adolescents with cancer European journal of cancer (Oxford, England : 1990) 2025. PMID 41240535.
- Dupont M, de Bernardi A, Vanacker H, Dufresne A, Brahmi M, Blay JY. Cellular therapies in sarcoma and other solid tumors Current opinion in oncology 2025. PMID 40423031.
- Fradin JJ, Charlson JA. Review of Adoptive Cellular Therapies for the Treatment of Sarcoma Cancers 2025. PMID 40282478.
- Gu J, Li J, Xu Y, Zhang G, Xie J, Jia R, Chen W, Lu Z, Chang C, Wen H, Chang LJ, Ma H, Cai Q. Preliminary exploration of PSMA CAR-T combined with GD2 CAR-T for the treatment of refractory/relapsed gliomas Journal of translational medicine 2025. PMID 40420236.
- Locatelli F, Pagliara D, De Ioris MA, Becilli M, Del Baldo G, Serra A, Mastronuzzi A, Cefalo MG, Li Pira G, Leone G, Bertaina V, Fabozzi F, Di Nardo M, Rosignoli C, D'Andrea ML, Crocoli A, Vennarini S, Sinibaldi M, Di Cecca S, Guercio M, Iaffaldano L, Lucarelli B, Algeri M, Merli P, Colafati GS, De Angelis B, Quintarelli C, Del Bufalo F. GD2-targeting CAR T cells in high-risk neuroblastoma: a phase 1/2 trial Nature medicine 2025. PMID 40841488.
- Monje M, Mahdi J, Majzner R, Yeom KW, Schultz LM, Richards RM, Barsan V, Song KW, Kamens J, Baggott C, Kunicki M, Rietberg SP, Lim AS, Reschke A, Mavroukakis S, Egeler E, Moon J, Patel S, Chinnasamy H, Erickson C, Jacobs A, Duh AK, Tunuguntla R, Klysz DD, Fowler C, Green S, Beebe B, Carr C, Fujimoto M, Brown AK, Petersen AG, McIntyre C, Siddiqui A, Lepori-Bui N, Villar K, Pham K, Bove R, Musa E, Reynolds WD, Kuo A, Prabhu S, Rasmussen L, Cornell TT, Partap S, Fisher PG, Campen CJ, Grant G, Prolo L, Ye X, Sahaf B, Davis KL, Feldman SA, Ramakrishna S, Mackall C. Intravenous and intracranial GD2-CAR T cells for H3K27M+ diffuse midline gliomas Nature 2025. PMID 39537919.
- Quintarelli C, Del Bufalo F, De Ioris MA, Guercio M, Algeri M, Pagliara D, Silvestris DA, Di Nardo M, Sinibaldi M, Di Cecca S, Iaffaldano L, Manni S, Fustaino V, Garganese MC, Colafati GS, Bertaina V, Becilli M, Mastronuzzi A, Fabozzi F, Gunetti M, Iacovelli S, Bugianesi R, Macchia S, Li Pira G, Cefalo MG, Leone G, Del Baldo G, De Angelis

B, Locatelli F. Donor-derived GD2-specific CAR T cells in relapsed or refractory neuroblastoma Nature medicine 2025. PMID 39815015.

- Ciccone R, Quintarelli C, Camera A, Pezzella M, Caruso S, Manni S, Ottaviani A, Guercio M, Del Bufalo F, Quadraccia MC, Orlando D, Di Cecca S, Sinibaldi M, Aurigemma M, Iaffaldano L, Sarcinelli A, D'Amore ML, Ceccarelli M, Nazio F, Marabitti V, Giorda E, Pezzullo M, De Stefanis C, Carai A, Rossi S, Alaggio R, Del Baldo G, Becilli M, Mastronuzzi A, De Angelis B, Locatelli F. GD2-Targeting CAR T-cell Therapy for Patients with GD2+ Medulloblastoma Clinical cancer research : an official journal of the American Association for Cancer Research 2024. PMID 38551501.
- Das AK, Sinha M, Singh SK, Chaudhary A, Boro AK, Agrawal M, Bhardwaj S, Kishore S, Kumari K. CAR T-cell therapy: a potential treatment strategy for pediatric midline gliomas Acta neurologica Belgica 2024. PMID 38669002.
- Lin FY, Stuckert A, Tat C, White M, Ruggieri L, Zhang H, Mehta B, Lapteva N, Mei Z, Major A, Thakkar S, Shum T, Parikh K, Wu MF, Lindsay HB, Scherer L, Shekar M, Baxter P, Wang T, Grilley B, Moeller K, Hicks J, Roy A, Anastas J, Malbari F, Aldave G, Chintagumpala M, Blaney S, Parsons DW, Brenner MK, Heslop HE, Rooney CM, Omer B. Phase I Trial of GD2.CART Cells Augmented With Constitutive Interleukin-7 Receptor for Treatment of High-Grade Pediatric CNS Tumors Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2024. PMID 38771986.
- Monje M, Mahdi J, Majzner R, Yeom K, Schultz LM, Richards RM, Barsan V, Song K, Kamens J, Baggott C, Kunicki M, Lim AS, Reschke A, Mavroukakis S, Egeler E, Moon J, Patel S, Chinnasamy H, Erickson C, Jacobs A, Duh AK, Rietberg SP, Tunuguntla R, Klysz DD, Fowler C, Green S, Beebe B, Carr C, Fujimoto M, Brown AK, Petersen AG, McIntyre C, Siddiqui A, Lepori-Bui N, Villar K, Pham K, Bove R, Musa E, Reynolds W, Kuo A, Prabhu S, Rasmussen L, Cornell TT, Partap S, Fisher PG, Campen CJ, Grant G, Prolo L, Ye X, Sahaf B, Davis KL, Feldman SA, Ramakrishna S, Mackall C. Sequential intravenous and intracerebroventricular GD2-CAR T-cell therapy for H3K27M-mutated diffuse midline gliomas medRxiv 2024. DOI 10.1101/2024.06.25.24309146.
- [no author listed]. IMMU-18. TARGETING GD2 IN THALAMIC DIFFUSE MIDLINE GLIOMA WITH TRANSIENT MRNA CAR T CELLS Neuro-oncology 2023. abstract.
- [no author listed]. IMMU-15. PHASE I TRIAL OF GD2.CART-CELLS AUGMENTED WITH A CONSTITUTIVELY ACTIVE INTERLEUKIN-7 RECEPTOR (C7R) FOR TREATMENT OF HIGH-GRADE PEDIATRIC CNS TUMORS Neuro-oncology 2023. abstract.
- Del Bufalo F, De Angelis B, Caruana I, Del Baldo G, De Ioris MA, Serra A, Mastronuzzi A, Cefalo MG, Pagliara D, Amicucci M, Li Pira G, Leone G, Bertaina V, Sinibaldi M, Di Cecca S, Guercio M, Abbaszadeh Z, Iaffaldano L, Gunetti M, Iacovelli S, Bugianesi R, Macchia S, Algeri M, Merli P, Galaverna F, Abbas R, Garganese MC, Villani MF, Colafati GS, Bonetti F, Rabusin M, Perruccio K, Folsi V, Quintarelli C, Locatelli F, Precision Medicine Team–IRCCS Ospedale Pediatrico Bambino Gesù. GD2-CART01 for Relapsed or Refractory High-Risk Neuroblastoma The New England journal of medicine 2023. PMID 37018492.
- Halliwell E, Vitali A, Muller H, Alonso-Ferrero M, Barisa M, Gavriil A, Piapi A, Leboreiro-Babe C, Gileadi T, Yeung J, Pataillot-Meakin T, Fisher J, Tucker L, Donovan L, Chesler L, Chester K, Anderson J. Targeting of low ALK antigen density neuroblastoma using AND logic-gate engineered CAR-T cells Cytotherapy 2023. PMID 36396552.
- Heczey A, Xu X, Courtney AN, Tian G, Barragan GA, Guo L, Amador CM, Ghatwai N, Rathi P, Wood MS, Li Y, Zhang C, Demberg T, Di Pierro EJ, Sher AC, Zhang H, Mehta B, Thakkar SG, Grilley B, Wang T, Weiss BD, Montalbano A, Subramaniam M, Xu C, Sachar C, Wells DK, Dotti G, Metelitsa LS. Anti-GD2 CAR-NKT cells in relapsed or refractory neuroblastoma: updated phase 1 trial interim results Nature medicine 2023. PMID 37188782.
- [no author listed]. DIPG-15. Major tumor regressions in H3K27M-mutated diffuse midline glioma (DMG) following sequential intravenous (IV) and intracerebroventricular (ICV) delivery of GD2-CAR T-cells Neuro-oncology 2022. abstract.
- Antonucci L, Canciani G, Mastronuzzi A, Carai A, Del Baldo G, Del Bufalo F. CAR-T Therapy for Pediatric High-Grade Gliomas: Peculiarities, Current Investigations and Future Strategies Frontiers in immunology 2022. PMID 35603195.
- Majzner RG, Ramakrishna S, Yeom KW, Patel S, Chinnasamy H, Schultz LM, Richards RM, Jiang L, Barsan V, Mancusi R, Geraghty AC, Good Z, Mochizuki AY, Gillespie SM, Toland AMS, Mahdi J, Reschke A, Nie EH, Chau IJ, Rotiroti MC, Mount CW, Baggott C, Mavroukakis S, Egeler E, Moon J, Erickson C, Green S, Kunicki M, Fujimoto M, Ehlinger Z, Reynolds W, Kurra S, Warren KE, Prabhu S, Vogel H, Rasmussen L, Cornell TT, Partap S, Fisher PG, Campen CJ, Filbin MG, Grant G, Sahaf B, Davis KL, Feldman SA, Mackall CL, Mackall CL, Monje M. GD2-CAR T cell therapy for H3K27M-mutated diffuse midline gliomas Nature 2022. PMID 35130560.
- Yu L, Huang L, Lin D, Lai X, Wu L, Liao X, Liu J, Zeng Y, Liang L, Zhang G, Wang B, Wu Z, Tao S, Liu Y, Jiao C, Chang LJ, Yang L, Yang L. GD2-specific chimeric antigen receptor-modified T cells for the treatment of refractory and/or recurrent neuroblastoma in pediatric patients Journal of cancer research and clinical oncology 2022. PMID 34724115.

- Majzner R, Ramakrishna S, Mochizuki A, Patel S, Chinnasamy H, Yeom K, Schultz L, Richards R, Campen C, Reschke A, Mahdi J, Toland A, Baggott C, Mavroukakis S, Egeler E, Moon J, Landrum K, Erickson C, Rasmussen L, Barsan V, Tamaresis J, Marcy A, Kunicki M, Fujimoto M, Ehlinger Z, Kurra S, Cornell T, Partap S, Fisher P, Grant G, Vogel H, Sahaf B, Davis K, Feldman S, Mackall C, Monje M. EPCT-14. GD2 CAR T-CELLS MEDIATE CLINICAL ACTIVITY AND MANAGEABLE TOXICITY IN CHILDREN AND YOUNG ADULTS WITH H3K27M-MUTATED DIPG AND SPINAL CORD DMG Neuro-oncology 2021. abstract.
- Mikkilineni L, Yates B, Steinberg SM, Shahani SA, Molina JC, Palmore T, Lee DW, Kaplan RN, Mackall CL, Fry TJ, Gea-Banacloche J, Jerussi T, Nussenblatt V, Kochenderfer JN, Shah NN. Infectious complications of CAR T-cell therapy across novel antigen targets in the first 30 days Blood advances 2021. PMID 34619768.
- Mochizuki A, Ramakrishna S, Good Z, Patel S, Chinnasamy H, Yeom K, Schultz L, Richards R, Campen C, Reschke A, Mahdi J, Toland A, Baggot C, Mavroukakis S, Egeler E, Moon J, Landrum K, Erickson C, Rasmussen L, Barsan V, Tamaresis J, Marcy A, Kunicki M, Celones M, Ehlinger Z, Kurra S, Cornell T, Partap S, Fisher P, Grant G, Vogel H, Davis K, Feldman S, Sahaf B, Majzner R, Mackall C, Monje M. OMIC-11. SINGLE CELL RNA SEQUENCING FROM THE CSF OF SUBJECTS WITH H3K27M+ DIPG/DMG TREATED WITH GD2 CAR T-CELLULAR THERAPY Neuro-oncology 2021. abstract.
- Terry RL, Meyran D, Fleuren EDG, Mayoh C, Zhu J, Omer N, Ziegler DS, Haber M, Darcy PK, Trapani JA, Neeson PJ, Ekert PG. Chimeric Antigen Receptor T cell Therapy and the Immunosuppressive Tumor Microenvironment in Pediatric Sarcoma Cancers 2021. PMID 34572932.
- Heczey A, Courtney AN, Montalbano A, Robinson S, Liu K, Li M, Ghatwai N, Dakhova O, Liu B, Raveh-Sadka T, Chauvin-Fleurence CN, Xu X, Ngai H, Di Pierro EJ, Savoldo B, Dotti G, Metelitsa LS. Anti-GD2 CAR-NKT cells in patients with relapsed or refractory neuroblastoma: an interim analysis Nature medicine 2020. PMID 33046868.
- Huarte E, O'Connor RS, Peel MT, Nunez-Cruz S, Leferovich J, Juvekar A, Yang YO, Truong L, Huang T, Naim A, Milone MC, Smith PA. Itacitinib (INCB039110), a JAK1 Inhibitor, Reduces Cytokines Associated with Cytokine Release Syndrome Induced by CAR T-cell Therapy Clinical cancer research : an official journal of the American Association for Cancer Research 2020. PMID 32998963.
- Straathof K, Flutter B, Wallace R, Jain N, Loka T, Depani S, Wright G, Thomas S, Cheung GW, Gileadi T, Stafford S, Kokalaki E, Barton J, Marriott C, Rampling D, Ogunbiyi O, Akarca AU, Marafioti T, Inglott S, Gilmour K, Al-Hajj M, Day W, McHugh K, Biassoni L, Sizer N, Barton C, Edwards D, Dragoni I, Silvester J, Dyer K, Traub S, Elson L, Brook S, Westwood N, Robson L, Bedi A, Howe K, Barry A, Duncan C, Barone G, Pule M, Anderson J. Antitumor activity without on-target off-tumor toxicity of GD2-chimeric antigen receptor T cells in patients with neuroblastoma Science translational medicine 2020. PMID 33239386.
- Mount CW, Majzner RG, Sundaresh S, Arnold EP, Kadapakkam M, Haile S, Labanieh L, Hulleman E, Woo PJ, Rietberg SP, Vogel H, Monje M, Mackall CL. Potent antitumor efficacy of anti-GD2 CAR T cells in H3-K27M+ diffuse midline gliomas Nature medicine 2018. PMID 29662203.
- Fisher J, Abramowski P, Wisidagamage Don ND, Flutter B, Capsomidis A, Cheung GW, Gustafsson K, Anderson J. Avoidance of On-Target Off-Tumor Activation Using a Co-stimulation-Only Chimeric Antigen Receptor Molecular therapy : the journal of the American Society of Gene Therapy 2017. PMID 28341563.
- Heczey A, Louis CU, Savoldo B, Dakhova O, Durett A, Grilley B, Liu H, Wu MF, Mei Z, Gee A, Mehta B, Zhang H, Mahmood N, Tashiro H, Heslop HE, Dotti G, Rooney CM, Brenner MK. CAR T Cells Administered in Combination with Lymphodepletion and PD-1 Inhibition to Patients with Neuroblastoma Molecular therapy : the journal of the American Society of Gene Therapy 2017. PMID 28602436.
- Gargett T, Brown MP. The inducible caspase-9 suicide gene system as a "safety switch" to limit on-target, off-tumor toxicities of chimeric antigen receptor T cells Frontiers in pharmacology 2014. PMID 25389405.

renal electrolyte (31)

- AlShammari OA, Alwadi M, Abuqurayn I, Alshehri H, AlMadi E, AlBtoosh B, AlMatham KI, Alrowaie F. Acute Kidney Injury and Electrolyte Disorders Following Chimeric Antigen Receptor T-cell (CAR T-cell) Therapy in Adults With Hematologic Malignancies: A Retrospective Study Cureus 2026. PMID 41536632.
- Cacciapuoti M, Bahrainwala JZ, Weng WK, Scott SA, Bhalla V, Abra G. Hemophagocytic Lymphohistiocytosis and Fibroblast Growth Factor 23 (FGF23)-Induced Hypophosphatemia American journal of kidney diseases : the official journal of the National Kidney Foundation 2026. PMID 42379468.
- Habib A, Ahmed N, Khan AM, Chang D, Paul B, Shaikh H, Strouse C, Struble E, Vegel A, Mahmoudjafari Z, Mushtaq MU, McGuirk JP, Abdallah AO, Atrash S, Friend R. BCMA-Directed CAR T-Cell Therapy in Patients with Relapsed/Refractory Multiple Myeloma and Renal Impairment Current oncology (Toronto, Ont.) 2026. PMID 41744844.
- Jazieh H, Almashayekh A, Henson JC, Omaish R, Spears G, Vellanki S, Trikannad A, Daughdrill B, Mohan-Lal B, Pandey T, Odueke A, Vallurupalli S, Schinke C, Alshwayat A, Gentille-Sanchez C, Jain N. Acute Kidney Injury in

Recipients of Chimeric Antigen Receptor T-Cell therapy: Clinical Characteristics and Impact on Mortality Transplantation and cellular therapy 2026. PMID 42486353.

- Kansal L, Ramalingam N, Roy S, Nair D, Chaturvedi P, Tuljapurkar V. Predictors of Postoperative Hyponatremia in Patients Undergoing Head and Neck Surgery Indian journal of surgical oncology 2026. PMID 42381800.
- Muddasani A, Thanendrarajan S, Zangari M, van Rhee F, Schinke CD. Efficacy and Safety of CAR-T Cell Therapy and Bispecific Antibodies in Relapsed/Refractory Multiple Myeloma with Renal Impairment: A Propensity Score-Matched Analysis Cancers 2026. PMID 42512374.
- Renaghan AD, Ostermann M, Ronco C, Ballen K, Cosmai L, Fenoglio R, Floris M, Forni LG, Gladstone DE, Glezerman IG, Goldstein SL, Gupta S, Herrmann SM, Jaimes EA, Jhaveri KD, Karam S, Kitchlu A, Landau HJ, Latcha S, Leaf DE, Lentini P, Małyszko J, Markowitz G, Murakami N, Pani A, Perazella MA, Rashidi A, Roccatello D, Sharon E, Sprangers B, Tesař V, Torres V, Wanchoo R, Whitman A, Workeneh BT, Zepeda-Orozco D, Rosner MH. The nephrotoxic effects of anti-cancer therapies: consensus report of the 34th Acute Disease Quality Initiative workgroup Nature reviews. Nephrology 2026. PMID 41361704.
- Charkviani M, Brochero MJV, Mohan A, Vaughan LE, Sandahl TB, Corraes AMS, Lin Y, Leung N, Herrmann SM. Incidence of acute kidney injury in relapsed and refractory multiple myeloma treated with teclistamab versus CAR-T cells Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association 2025. PMID 39805729.
- Chen Z, Chen Y, Liu J, Sun Y, Zhang X, Shao L, Wang D, Wang X, Chen W, Sang W, Qi K, Li Z, Sun C, Shi M, Qiao J, Wu Q, Zeng L, Zheng J, Xu K, Li L, Cheng H, Cao J. Nephrotoxicity of CAR-T therapy in patients with relapsed and refractory multiple myeloma International urology and nephrology 2025. PMID 40310507.
- Dima D, Afrough A, Goel U, Grajales-Cruz AF, Khouri J, Julian K, Pasvolsky O, Banerjee R, Razzo B, Ferreri CJ, Vazquez-Martinez MA, Davis JA, Sannareddy A, Castaneda O, Raza S, Portuguese AJ, Gaballa MR, Rana MS, Lieberman-Cribbin A, DeJarnette S, Gonzalez R, Chen A, Herr MM, Mikkilineni L, Hosoya H, Ouchveridze E, Kaur G, Rossi A, Shune L, Anwer F, Lin Y, Richard S, Sborov DW, Baz RC, Garfall AL, Lee HC, Anderson LD, Cowan AJ, Patel KK, Voorhees PM, Sidana S, Hansen DK, Atrash S, Susanibar-Adaniya SP. Teclistamab for patients with heavily pretreated relapsed/refractory multiple myeloma and renal impairment Blood advances 2025. PMID 40198766.
- Jiang H, Ahlawat Y, Steinberg A. Using bispecific antibodies in a patient with peritoneal dialysis for relapsed multiple myeloma Journal of oncology pharmacy practice : official publication of the International Society of Oncology Pharmacy Practitioners 2025. PMID 40356496.
- Kanbay M, Mizrak B, Alper EN, Copur S, Ortiz A. Acute kidney injury following CAR-T cell therapy: a nephrologist's perspective Clinical kidney journal 2025. PMID 39781479.
- Sadowski K, Ploch W, Downar A, Giza W, Szcześ D, Olejarz W, Jędrzejczak WW, Małyszko J, Basak G. Nephrotoxicity in CAR-T Cell Therapy Transplantation and cellular therapy 2025. PMID 40107382.
- Sharp J, Zhao Q, Voorhees TJ, Bond DA, Sawalha Y, Reneau J, Sigmund A, Hanel W, Sehgal L, Alinari L, Baiocchi R, Brammer JE, Maddocks K, Christian B, Bezerra E, Denlinger N, Vasu S, De Lima M, Prosek J, Epperla N. Acute kidney injury after chimeric antigen receptor T-cell therapy is associated with inferior survival in patients with relapsed/refractory large B-cell lymphoma British journal of haematology 2025. PMID 40589093.
- Swamy V, Kelly K, Koyner JL, Derman BA, Bonilla M. Chimeric Antigen Receptor T-Cell Therapy in Dialysis-Dependent Patients With Multiple Myeloma: Insights From a 5-Case Series Kidney medicine 2025. PMID 41342049.
- Chinese Hematology Association, Chinese Geriatrics Association, Society of Hematology, Chinese Research Hospital Association, Society of Nephrology. [Guideline for the diagnosis and management of multiple myeloma-related renal impairment (2024 version)] Zhonghua nei ke za zhi 2024. PMID 38561279.
- León-Román J, Iacoboni G, Bermejo S, Carpio C, Bolufer M, García-Carro C, Sánchez-Salinas M, Alonso-Martínez C, Bestard O, Barba P, Soler MJ. Transient acute kidney injury after chimeric antigen receptor T-cell therapy in patients with hematological malignancies Clinical kidney journal 2024. PMID 38500492.
- Petgrave Y, Selukar S, Epperly R, Naik S, Santos ND, Triplett BM, Gottschalk S, Bissler J, Talleur AC. Acute kidney injury following treatment with CD19-specific CAR T-cell therapy in children, adolescent, and young adult patients with B-cell acute lymphoblastic leukemia Pediatric nephrology (Berlin, Germany) 2024. PMID 38507119.
- Rousseau A, Zafrani L. Acute kidney injury after CAR-T cell infusion Bulletin du cancer 2024. PMID 36220698.
- Russo E, Gambella M, Raiola AM, Beltrametti E, Zanetti V, Chirco G, Viazzi F, Angelucci E, Esposito P. Acute kidney injury in hematological patients treated with CAR-T cells: risk factors, clinical presentation and impact on outcomes Scientific reports 2024. PMID 39506012.
- Sidana S, Peres LC, Hashmi H, Hosoya H, Ferreri C, Khouri J, Dima D, Atrash S, Voorhees P, Simmons G, Sborov DW, Kalariya N, Hovanky V, Bharadwaj S, Miklos D, Wagner C, Kocoglu MH, Kaur G, Davis JA, Midha S, Janakiram M, Freeman C, Alsina M, Locke F, Gonzalez R, Lin Y, McGuirk J, Afrough A, Shune L, Patel KK, Hansen DK. Idecabtagene vicleucel chimeric antigen receptor T-cell therapy for relapsed/refractory multiple myeloma with renal impairment Haematologica 2024. PMID 37731379.

- Vincendeau M, Joseph A, Thieblemont C, Rabian F, Harel S, Valade S, Zafrani L. Acute kidney injury after CAR-T cell therapy: exploring clinical patterns, management, and outcomes Clinical kidney journal 2024. PMID 38915438.
- Khan I, Khan N, Wolfson N, Djebabria K, Rehman MEU, Anwer F. Safety of CAR-T Cell Therapy in Patients With Renal Failure/Acute Kidney Injury: Focused Review Clinical hematology international 2023. PMID 37010812.
- Marquard F, Langebrake C, Janson D, Mahmud M, Dadkhah A, Kröger N, Ayuk F. Case Report: Lymphodepletion followed by CAR-T cell therapy with Idecabtagen vicleucel in a patient with severe renal impairment Frontiers in oncology 2023. PMID 38033500.
- Farooqui N, Sy-Go JPT, Miao J, Mehta R, Vaughan LE, Bennani NN, Wang Y, Bansal R, Hathcock MA, Hayman SR, Johnston PB, Villasboas JC, Paludo J, Ansell SM, Leung N, Lin Y, Herrmann SM. Incidence and Risk Factors for Acute Kidney Injury After Chimeric Antigen Receptor T-Cell Therapy Mayo Clinic proceedings 2022. PMID 35787856.
- Joseph A, Lafarge A, Azoulay E, Zafrani L. Acute Kidney Injury in Cancer Immunotherapy Recipients Cells 2022. PMID 36552755.
- Uppal NN, Workeneh BT, Rondon-Berrios H, Rondon-Berrios H, Jhaveri KD. Electrolyte and Acid-Base Disorders Associated with Cancer Immunotherapy Clinical journal of the American Society of Nephrology : CJASN 2022. PMID 35063968.
- Wood AC, Perez AP, Arciola B, Patel K, Johnson G, DiMaggio E, Bachmeier CA, Reid K, Carallo S, Vargas MH, Faramand R, Chavez JC, Shah B, Gaballa S, Khimani F, Elmariah H, Nishihori T, Lazaryan A, Freeman C, Davila ML, Locke FL, Mhaskar R, Bassil C, Jain MD. Outcomes of CD19-Targeted Chimeric Antigen Receptor T Cell Therapy for Patients with Reduced Renal Function Including Dialysis Transplantation and cellular therapy 2022. PMID 36174934.
- Acharya R, Horn B, Zeng X, Upadhyay K. Collapsing Focal Segmental Glomerulosclerosis and Acute Kidney Injury Associated With Chimeric Antigen Receptor T-Cell (CAR-T) Therapy: A Case Report Kidney medicine 2021. PMID 34939018.
- Melilli E, Mussetti A, Linares GS, Ruella M, La Salette C, Savchenko A, Taco MDR, Montero N, Grinyo J, Fava A, Gomà M, Meneghini M, Manonelles A, Cruzado J, Sureda A, Bestard O. Acute Kidney Injury Following Chimeric Antigen Receptor T-Cell Therapy for B-Cell Lymphoma in a Kidney Transplant Recipient Kidney medicine 2021. PMID 34401733.
- Gupta S, Seethapathy H, Strohbehn IA, Frigault MJ, O'Donnell EK, Jacobson CA, Motwani SS, Parikh SM, Curhan GC, Reynolds KL, Leaf DE, Sise ME. Acute Kidney Injury and Electrolyte Abnormalities After Chimeric Antigen Receptor T-Cell (CAR-T) Therapy for Diffuse Large B-Cell Lymphoma American journal of kidney diseases : the official journal of the National Kidney Foundation 2020. PMID 31973908.

cns locoregional delivery (21)

- Cevering CK, Abdel-Azim H, Khazal SJ, Casassa C. Immune Effector Cell-Associated Neurotoxicity Syndrome: A Practical Overview for the General Neurologist Neurology. Clinical practice 2026. PMID 41445934.
- Hou Z, Jin Y, Ruan M, Jiang Y, Fu Y, Yang J, Ge J, Long Z. ICANS After CAR-T Therapy: Mechanisms and Management With a Focus on Corticosteroid-Refractory ICANS Immunology 2026. PMID 42484496.
- Ronsley R, Choe M, Wright J, Seidel K, Lee A, Wendler J, Annesley C, Jensen MC, Park JR, Vitanza NA, Gust J. Tumor inflammation-associated neurotoxicity in children with diffuse intrinsic pontine glioma receiving B7-H3-targeting CAR T cells on BrainChild-03 Neuro-oncology practice 2026. PMID 41798119.
- Ronsley R, Huang W, Rohlf E, Seidel K, Brown C, Beebe A, Rawlings-Rhea SD, Lindgren C, Sad T, Crotty EE, Leary SES, Thomsen A, Beckstrom AR, Holtzclaw SL, Hoeppner C, Hurst D, MacQuivey M, Goldstein HE, Browd SR, Hauptman JS, Lee A, Ojemann JG, Wright JN, Albert CM, Pinto N, Gustafson JA, Ho O, Pattabhi S, Summers C, Annesley C, Wendler JN, Gust J, Gardner RA, Park JR, Jensen MC, Vitanza NA. Intracerebroventricular B7-H3-targeting CAR T cells for non-pontine DMG and recurrent/refractory pediatric CNS tumors: a phase 1 trial Neuro-oncology 2026. PMID 42503899.
- Yin Q, Mei X, Ma Y, Zheng M. Central nervous system infections caused by carbapenem-resistant klebsiella pneumoniae after CAR T-cell therapy in a patient with preexisting colonization: a case report and literature review European journal of clinical microbiology & infectious diseases : official publication of the European Society of Clinical Microbiology 2026. PMID 41511674.
- Zhang Y, Chi X, Feng R, Xian N, Huang N, Sun S, Zhao X, Zhang P, Liu K, Ma Y, Zhang Y, Zhang K, Wang S, Chen X, Zhang J, Huang W, Huang B, Su TT, Yeung J, He J, Chen L, Ji N, Huang G. Intracranial delivery of B7-H3-targeting CAR-T cells for recurrent glioblastoma: a phase 1 trial Nature medicine 2026. PMID 42562965.
- Bagley SJ, Desai AS, Fraietta JA, Silverbush D, Chafamo D, Freeburg NF, Gopikrishna GK, Rech AJ, Nabavizadeh A, Bagley LJ, Park J, Jarocha D, Martins R, Sarmiento N, Maloney E, Lledo L, Stein C, Marshall A, Leskowitz RM, Jadlowsky JK, Mackey S, Christensen S, Oner BS, Plesa G, Brennan A, Gonzalez V, Chen F, Barrett D, Colbourn R,

Nasrallah MP, Mourelatos Z, Hwang WT, Alanio C, Siegel DL, June CH, Hexner EO, Binder ZA, O'Rourke DM. Intracerebroventricular bivalent CAR T cells targeting EGFR and IL-13Rα2 in recurrent glioblastoma: a phase 1 trial Nature medicine 2025. PMID 40451950.

- D Souza A, Patel C, Vijayaraghavan KN, Wang QA, Correa A. A Unique Presentation of Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS) Presenting as Foot Drop in B-cell Acute Lymphoblastic Leukemia: A Case Report Cureus 2025. PMID 40831812.
- Gust J, Cole BL, Ronsley R, Wilson AL, Seidel K, Wendler J, Pattabhi S, Brown C, Rawlings-Rhea SD, Shtanukhina N, Browd SR, Hauptman JS, Lee A, Ojemann JG, Crotty EE, Leary SES, Perez FA, Wright JN, Albert CM, Pinto N, Gardner RA, Vitanza NA, Jensen MC, Park JR. Locoregional infusion of EGFR806-CAR T cells for recurrent or refractory pediatric CNS tumors: Results of the completed BrainChild02 phase 1 clinical trial Neuro-oncology 2025. PMID 40070357.
- Hadjigeorgiou AG, Munn LL, Stylianopoulos T, Jain RK. Physiologically-based pharmacokinetic model for CAR-T cells delivery and efficacy in solid tumors bioRxiv 2025. DOI 10.1101/2025.08.20.671229.
- Mahdi J, Gust JA, Vitanza NA, Scott B, Monje M, Ronsley R. Neurotoxicity in central nervous system tumors treated with CAR T cell therapy: a review Journal of neuro-oncology 2025. PMID 41171480.
- Meng J, Lyu H, Wang Z, Bai X, Zhu H, Pu Y, He X, Xiao X, Zhao M. Early intrathecal dexamethasone and methotrexate as an effective approach for immune effector cell-associated neurotoxicity syndrome after CAR-T cell therapies Frontiers in immunology 2025. PMID 41479887.
- Vitanza NA, Ronsley R, Choe M, Seidel K, Huang W, Rawlings-Rhea SD, Beam M, Steinmetzer L, Wilson AL, Brown C, Beebe A, Lindgren C, Gustafson JA, Wein A, Holtzclaw S, Hoeppner C, Goldstein HE, Browd SR, Hauptman JS, Lee A, Ojemann JG, Crotty EE, Leary SES, Perez FA, Wright JN, Alonso MM, Dun MD, Foster JB, Hurst D, Kong A, Thomsen A, Orentas RJ, Albert CM, Pinto N, Annesley C, Gardner RA, Ho O, Pattabhi S, Gust J, Wendler JP, Park JR, Jensen MC. Intracerebroventricular B7-H3-targeting CAR T cells for diffuse intrinsic pontine glioma: a phase 1 trial Nature medicine 2025. PMID 39775044.
- Bagley SJ, Logun M, Fraietta JA, Wang X, Desai AS, Bagley LJ, Nabavizadeh A, Jarocha D, Martins R, Maloney E, Lledo L, Stein C, Marshall A, Leskowitz R, Jadlowsky JK, Christensen S, Oner BS, Plesa G, Brennan A, Gonzalez V, Chen F, Sun Y, Gladney W, Barrett D, Nasrallah MP, Hwang WT, Ming GL, Song H, Siegel DL, June CH, Hexner EO, Binder ZA, O'Rourke DM. Intrathecal bivalent CAR T cells targeting EGFR and IL13Rα2 in recurrent glioblastoma: phase 1 trial interim results Nature medicine 2024. PMID 38480922.
- Ju A, Choi S, Jeon Y, Kim K. Lymphodepletion in Chimeric Antigen Receptor T-Cell Therapy for Solid Tumors: A Focus on Brain Tumors Brain tumor research and treatment 2024. PMID 39542517.
- Vitanza N, Ronsley R, Choe M, Huang W, Seidel K, Rawlings-Rhea S, Beam M, Steinmetzer L, Wilson A, Brown C, Beebe A, Lindgren C, Gustafson J, Wein A, Holtzclaw S, Hoeppner C, Goldstein H, Browd S, Hauptman J, Lee A, Ojemann J, Crotty E, Leary S, Perez F, Wright J, Alonso M, Dun M, Foster J, Hurst D, Kong A, Thomsen A, Orentas R, Albert C, Pinto N, Annesley C, Gardner R, Ho O, Pattabhi S, Wendler J, Gust J, Park J, Jensen M. Intracerebroventricular B7-H3 CAR T cells for children and young adults with diffuse intrinsic pontine glioma Research Square 2024. DOI 10.21203/rs.3.rs-5248718/v1.
- [no author listed]. TRLS-05. PRELIMINARY RESULTS FROM A PHASE I TRIAL TO EVALUATE INTRAVENTRICULAR DELIVERY OF IL13RA2-TARGETING CAR T CELLS AFTER LYMPHODEPLETION IN PEDIATRIC BRAIN TUMORS PATIENTS Neuro-oncology 2023. abstract.
- [no author listed]. TRLS-12. INTRAVENTRICULAR B7-H3 CAR T CELLS FOR DIFFUSE INTRINSIC PONTINE GLIOMA: INTERIM ANALYSIS OF BRAINCHILD-03 ARM C Neuro-oncology 2023. abstract.
- Wang L, Oill AT, Blanchard M, Wu M, Hibbard J, Sepulveda S, Peter L, Kilpatrick J, Munoz M, Stiller T, Shulkin N, Wagner J, Dolatabadi A, Nisis M, Shepphird J, Sanchez G, Lingaraju C, Manchanda M, Natri H, Kouakanou L, Sun G, Oliver-Cervantes C, Georges J, Aftabizadeh M, Forman S, Priceman S, Ressler J, Arvanitis L, Cotter J, D'Apuzzo M, Tamrazi B, Badie B, Davidson T, Banovich N, Brown C. Expansion of endogenous T cells in CSF of pediatric CNS tumor patients undergoing locoregional delivery of IL13Rα2-targeting CAR T cells: an interim analysis Research Square 2023. DOI 10.21203/rs.3.rs-3454977/v1.
- Donovan L, Bielamowicz K, Joseph S, Delaidelli A, Sorenson P, Ahmed N, Taylor M. IMMU-05. COMBINATIONAL CAR T-CELL AND EPIGENETIC MODIFIER THERAPY TO TARGET POSTERIOR FOSSA TUMORS Neuro-oncology 2019. abstract.
- Shalabi H, Wolters PL, Martin S, Toledo-Tamula MA, Roderick MC, Struemph K, Kane E, Yates B, Delbrook C, Mackall CL, Lee DW, Fry TJ, Shah NN. Systematic Evaluation of Neurotoxicity in Children and Young Adults Undergoing CD22 Chimeric Antigen Receptor T-Cell Therapy Journal of immunotherapy (Hagerstown, Md. : 1997) 2018. PMID 30048343.

steroids immunomodulation (18)

- Ahmed N, Ali A, Kim S, Oloyede T, Bye M, Gowda L, Kamble RT, Mirza AS, Nadiminti KV, Pophali PA, Rodriguez-Bonilla C, Sieg A, Strouse C, Abid MB, Afrough A, Aljurf M, Beitinjaneh AM, Chhabra S, Choe M, Hilal T, Iqbal M, Jain T, Kharfan-Dabaja MA, Modi D, Orti G, Pawarode A, Rao U, Riedell PA, Saad A, Seo S, Balari AS, Wagner JL, Wudhikarn K, Turtle CJ, Phillips CL, Pasquini MC, Ahmed S, Ganguly S, Moskop A. Comparison of Fludarabine Versus Bendamustine as a Lymphodepleting Chemotherapy Prior to CAR-T for Large Cell Lymphoma Transplantation and cellular therapy 2026. PMID 42341926.
- Amatya C, Weissler KA, Lam N, Natrakul DA, Brudno JN, Cutmore LC, Mikkilineni L, Kochenderfer JN. Corticosteroids ameliorate CAR T-cell-induced cytokine-release syndrome without inhibiting multiple myeloma treatment Journal for immunotherapy of cancer 2026. PMID 41702648.
- Forsberg PA, Turner JA, Meyer M, Abbott D, Nicholson S, Schade H, Matous J, Gregory T. Dexamethasone prophylaxis for excessive lymphocyte expansion after cilta-cel in multiple myeloma Blood advances 2026. PMID 41894686.
- Grieb N, Wiemers T, Born P, Fandrei D, Fischer L, Ferle M, Franke S, Keller J, Wang SY, Jentzsch M, Herling C, Metzeler KH, Herling M, Heyn S, Neumuth T, Platzbecker U, Vučinić V, Franke GN, Merz M. Impact of Renal Impairment and Lymphodepletion Regimen on Outcomes after CAR T Cell Therapy in Relapsed/Refractory Multiple Myeloma Transplantation and cellular therapy 2026. PMID 42144192.
- Huo BN, Shu L, Zhang Y, Yin NG, Ji HH, Jia YT, Song L. Adverse events related to drug-drug interactions in tocilizumab combination therapy: a retrospective analysis and clinical implications Frontiers in medicine 2026. PMID 41836963.
- Liang EC, Kungwankiattichai S, Wu X, Voutsinas J, Macero D, Khouderchah C, Qiao Y, Jeon Y, Basom R, Huang JJ, Portuguese AJ, Kimble EL, Hirayama AV, Shadman M, Chen AI, Desai A, Hayes-Lattin B, Meyers G, Bastola S, Rai M, Leonard J, Trussell J, Stomberg C, Spurgeon SE, Okada C, Shree T, Silbermann R, Medvedova E, Ratterree B, Bixler R, Williamson S, Phan C, Ahrenkiel R, Wu Q, Maziarz RT, Gauthier J. Early neurologic response predicts outcomes after anakinra for immune effector cell-associated neurotoxicity syndrome Blood advances 2026. PMID 42530978.
- Xu W, Qiao X, Zhang L, Xing L, Zhang J, Guo X, Qiao S. Lisaftoclax combined with ixazomib and dexamethasone after CAR-T for maintenance therapy in transplant-ineligible relapsed/refractory ultra-high-risk multiple myeloma: two case reports and literature review Frontiers in oncology 2026. PMID 42130633.
- Zhang N, Yang F, Liu R, Fu Z, Li D, Guo Y, Ma L, Cao M, Deng B, Gan S, Chen C, Ke X, Hu K. Baseline immunonutritional status drives CAR T-cell efficacy, survival, and safety in R/R lymphoma: validation of a pre-lymphodepletion PNI threshold Annals of hematology 2026. PMID 42420673.
- Coleman GT, Wang Y. CART-induced eosinophilic colitis with good response to corticosteroids and infliximab: a case report Therapeutic advances in medical oncology 2025. PMID 40017900.
- Dos Santos MH, de Azevedo JTC, da Silva Januário ME, Schmidt DF, Tirapelle MC, Biggi AFB, Ebrahimabadi S, Silvestre RN, Covas DT, Calado RT, Picanço-Castro V. 2B4 co-stimulation and dasatinib modulation enhance anti-CD19 CAR-NK-92 cell cytotoxicity Frontiers in immunology 2025. PMID 41459521.
- Mantion CF, Biichlé S, Roussel X, Rolin G, Lejeune A, Labaigt T, Bôle-Richard E, Daguindau E, Royer B, Renosi F, Adotevi O, Loyon R, Fredon M, Garnache-Ottou F. Evaluation of dasatinib and ponatinib for the control of CD123 CAR-T cell functionalities Molecular therapy. Oncology 2025. PMID 41477551.
- Panetta JC, Talleur AC, Naik S, Gottschalk S, Leggas M. Age-adjusted dosing of fludarabine for lymphodepletion in CAR T-cell therapy: a clinical trial simulation study Blood advances 2025. PMID 40609087.
- Zhang M, Fu S, Feng J, Hong R, Wei G, Zhao H, Zhao M, Xu H, Cui J, Huang S, Wu X, Liu L, Sun J, Wu W, Zhu Y, He J, Zhao Y, Cai Z, Zheng W, Ye X, Shi J, Luo Y, Wang D, Chang AH, Hu Y, Huang H. Dasatinib and CAR T-Cell Therapy in Newly Diagnosed Philadelphia Chromosome-Positive Acute Lymphoblastic Leukemia: A Nonrandomized Clinical Trial JAMA oncology 2025. PMID 40244598.
- Frame DG, Geer M, Kasha S, Markstrom D, Scappaticci G, Feeney T, Hayduk A, Mansoor HM, Oberfeld A, D'Antonio H, Anand S, Choi SW, Maciejewski J, Pawarode A, Riwes MM, Tewari M, Magenau J, Ghosh M. Comparing 2-day vs 3-day flu-CY lymphodepleting regimens for CD19 CAR T-cell therapy in patients with non-hodgkin's lymphoma Frontiers in immunology 2024. PMID 38947326.
- Hegde M, Navai S, DeRenzo C, Joseph SK, Sanber K, Wu M, Gad AZ, Janeway KA, Campbell M, Mullikin D, Nawas Z, Robertson C, Mathew PR, Zhang H, Mehta B, Bhat RR, Major A, Shree A, Gerken C, Kalra M, Chakraborty R, Thakkar SG, Dakhova O, Salsman VS, Grilley B, Lapteva N, Gee A, Dotti G, Bao R, Salem AH, Wang T, Brenner MK, Heslop HE, Wels WS, Hicks MJ, Gottschalk S, Ahmed N. Autologous HER2-specific CAR T cells after lymphodepletion for advanced sarcoma: a phase 1 trial Nature cancer 2024. PMID 38658775.
- Yang J, Zhang J, Wan X, Cai J, Wang T, Yang X, Li W, Ding L, Song L, Miao Y, Wang X, Ma Y, Luo C, Tang J, Gu L, Chen J, Lu J, Tang Y, Li B. Impact of corticosteroids on the efficacy of CD19/22 CAR-T cell therapy in pediatric patients with B-ALL: a single-center study Frontiers in pediatrics 2024. PMID 39872915.

- Hashmi H, McGann M, Greenwell BI. Use of long-term corticosteroids in patients treated with CAR T-cell therapy Journal of oncology pharmacy practice : official publication of the International Society of Oncology Pharmacy Practitioners 2023. PMID 35635259.
- Owen K, Ghaly R, Shohdy KS, Thistlethwaite F. Lymphodepleting chemotherapy practices and effect on safety and efficacy outcomes in patients with solid tumours undergoing T cell receptor-engineered T cell (TCR-T) Therapy: a systematic review and meta-analysis Cancer immunology, immunotherapy : CII 2023. PMID 36315268.

gd2 cart preclinical (13)

- Barragán Bravo GA, de la Cerda DA, Landoni E, Dholakia K, Humeniuk P, Caraballo Galva LD, Wang Y, Tian G, Yang B, Guo L, Wood MS, Rios X, Xu X, Courtney AN, Di Pierro EJ, Jacob J, Li YH, Adaikkalavan A, Watanabe N, Mao S, Miles G, Dotti G, Metelitsa LS. IL-18 metabolically reprograms CAR-expressing natural killer T cells and enhances their antitumor activity Molecular therapy : the journal of the American Society of Gene Therapy 2026. PMID 41520178.
- Liu L, He P, Wang Y, Ma F, Li D, Bai Z, Qu Y, Zhu L, Yoon CW, Yu X, Huang Y, Liang Z, Zhang Y, Liu K, Guo T, Zeng Y, Zhou Q, Chung HK, Fan R, Wang Y. Engineering sonogenetic EchoBack-CAR T cells Cell 2025. PMID 40179881.
- Ni H, Reitman ZJ, Zou W, Akhtar MN, Paul R, Huang M, Zhang D, Zheng H, Zhang R, Ma R, Ngo G, Zhang L, Diffenderfer ES, Motlagh SAO, Kim MM, Minn AJ, Dorsey JF, Foster JB, Metz J, Koumenis C, Kirsch DG, Gong Y, Fan Y. FLASH radiation reprograms lipid metabolism and macrophage immunity and sensitizes medulloblastoma to CAR-T cell therapy Nature cancer 2025. PMID 39910249.
- Vogt KC, Silberman PC, Lin Q, Han JE, Laflin A, Gellineau HA, Heller DA, Scheinberg DA. Microenvironment actuated CAR T cells improve solid tumor efficacy without toxicity Science advances 2025. PMID 39841845.
- Zhou RW, Purohit PK, Kim JH, Lee SU, Burshteyn N, Tifrea D, Cordon A, Grigorian A, Newton BL, Edwards RA, Demetriou M. Safe immunosuppression-resistant pan-cancer immunotherapeutics by velcro-like density-dependent targeting of tumor-associated carbohydrate antigens Cell 2025. PMID 41005308.
- Pezzella M, Quintarelli C, Quadraccia MC, Sarcinelli A, Manni S, Iaffaldano L, Ottaviani A, Ciccone R, Camera A, D'Amore ML, Di Cecca S, Sinibaldi M, Guercio M, Aurigemma M, De Falco P, Fustaino V, Rota R, Pomella S, Cassandri M, Di Giannatale A, Agrati C, Bordoni V, Guarracino F, Massa M, Del Baldo G, Becilli M, Milano GM, Del Bufalo F, Locatelli F, De Angelis B. Tumor-derived G-CSF induces an immunosuppressive microenvironment in an osteosarcoma model, reducing response to CAR.GD2 T-cells Journal of hematology & oncology 2024. PMID 39695851.
- [no author listed]. IMMU-22. Safely targeting GD2 in thalamic diffuse midline glioma with mRNA CAR T cells Neuro-oncology 2022. abstract.
- Reppel L, Tsahouridis O, Akulian J, Davis IJ, Lee H, Fucà G, Weiss J, Dotti G, Pecot CV, Savoldo B. Targeting disialoganglioside GD2 with chimeric antigen receptor-redirected T cells in lung cancer Journal for immunotherapy of cancer 2022. PMID 35022195.
- Zhang J, Siller-Farfán JA. Current and future perspectives of chimeric antigen receptors against glioblastoma Immunotherapy advances 2022. PMID 36284838.
- Duan Y, Chen R, Huang Y, Meng X, Chen J, Liao C, Tang Y, Zhou C, Gao X, Sun J. Tuning the ignition of CAR: optimizing the affinity of scFv to improve CAR-T therapy Cellular and molecular life sciences : CMLS 2021. PMID 34966954.
- Chen Y, Sun C, Landoni E, Metelitsa L, Dotti G, Savoldo B. Eradication of Neuroblastoma by T Cells Redirected with an Optimized GD2-Specific Chimeric Antigen Receptor and Interleukin-15 Clinical cancer research : an official journal of the American Association for Cancer Research 2019. PMID 30617136.
- Menard M, McHenry L, Nasholm N, Wong-Michalak S, Lu E, Simonds E, Gustafson W, Weiss W. IMMU-06. INVESTIGATING IMMUNOSUPPRESSIVE MECHANISMS IN THE TUMOR MICROENVIRONMENT OF HIGH-RISK NEUROBLASTOMA; AN IMMUNOCOMPETENT, MYCN-DRIVEN, NON-GERMLINE GEM MODEL Neuro-oncology 2019. abstract.
- Richman SA, Nunez-Cruz S, Moghimi B, Li LZ, Gershenson ZT, Mourelatos Z, Barrett DM, Grupp SA, Milone MC. High-Affinity GD2-Specific CAR T Cells Induce Fatal Encephalitis in a Preclinical Neuroblastoma Model Cancer immunology research 2018. PMID 29180536.

on target off tumor (8)

- Maji M, Mandal S, Dhali A, Sharma A. Real-World Dermatologic Adverse Events of CAR T-Cell Therapy: A Decade-Wide Disproportionality Analysis of the FDA Adverse Event Reporting System Cancers 2026. PMID 42449670.

- Birocchi F, Almazan AJ, Parker A, Bouffard AA, Goncalves S, Kelly C, Frank J, Leick MB, Haradhvala NJ, Kagawa S, Getz G, Escobar G, Salas-Benito D, Mucci A, Berger TR, Maus MV. On-target off-tumor toxicity of claudin18.2-directed CAR-T cells in preclinical models Nature communications 2025. PMID 41176533.
- Stip MC, Evers M, Nederend M, Chan C, Reiding KR, Damen MJ, Heck AJR, Koustoulidou S, Ramakers R, Krijger GC, de Roos R, Souteyrand E, Cornel AM, Dierselhuis MP, Jansen M, de Boer M, Valerius T, van Tetering G, Leusen JHW, Meyer-Wentrup F. IgA antibody immunotherapy targeting GD2 is effective in preclinical neuroblastoma models Journal for immunotherapy of cancer 2023. PMID 37479484.
- Liu XY, Chen YL, Liu GJ, Deng XN, Cui Y, Tan J, Dong XC, Li HY, Chen GJ, Ou ZM, Wang CH. Development of a variant of dinutuximab with enhanced antitumor efficacy and reduced induction of neuropathic pain FEBS open bio 2022. PMID 35792784.
- Mastrangelo S, Rivetti S, Triarico S, Romano A, Attinà G, Maurizi P, Ruggiero A. Mechanisms, Characteristics, and Treatment of Neuropathic Pain and Peripheral Neuropathy Associated with Dinutuximab in Neuroblastoma Patients International journal of molecular sciences 2021. PMID 34884452.
- Davies AJ, Rinaldi S, Costigan M, Oh SB. Cytotoxic Immunity in Peripheral Nerve Injury and Pain Frontiers in neuroscience 2020. PMID 32153361.
- Diccianni MB, Kempińska K, Gangoti JA, Yu AL, Sorkin LS. Anti-GD2 induced allodynia in rats can be reduced by pretreatment with DFMO PloS one 2020. PMID 32697811.
- Kushner BH, Cheung IY, Modak S, Basu EM, Roberts SS, Cheung NK. Humanized 3F8 Anti-GD2 Monoclonal Antibody Dosing With Granulocyte-Macrophage Colony-Stimulating Factor in Patients With Resistant Neuroblastoma: A Phase 1 Clinical Trial JAMA oncology 2018. PMID 30326045.

mitigation engineering (7)

- Wells J, Gupta A. CAR-T cell therapy for pediatric solid tumors: armored CAR-T cells and beyond Cancer metastasis reviews 2026. PMID 41770377.
- Garcia-Robledo JE, Cabrera-Salcedo S, Brandauer AM, Romano F, Rengifo-Martinez J, Toro-Pedroza A, Victoria JS, Rios-Serna LJ, Loukanov A, Cardona AF, Genovese P, Baena JC. Engineering the next generation of CAR T- cells: precision modifications, logic gates and universal strategies to overcome exhaustion and tumor resistance Frontiers in oncology 2025. PMID 41584599.
- Ercilla-Rodríguez P, Sánchez-Díez M, Alegría-Aravena N, Quiroz-Troncoso J, Gavira-O'Neill CE, González-Martos R, Ramírez-Castillejo C. CAR-T lymphocyte-based cell therapies; mechanistic substantiation, applications and biosafety enhancement with suicide genes: new opportunities to melt side effects Frontiers in immunology 2024. PMID 39091493.
- Stein MN, Dumbrava EE, Teply BA, Gergis US, Guiterrez ME, Reshef R, Subudhi SK, Jacquemont CF, Senesac JH, Bayle JH, Scripture CD, Chatwal MS, Bilen MA, Stadler WM, Becerra CR. PSCA-targeted BPX-601 CAR T cells with pharmacological activation by rimiducid in metastatic pancreatic and prostate cancer: a phase 1 dose escalation trial Nature communications 2024. PMID 39737899.
- Ventin M, Cattaneo G, Arya S, Jia J, Gelmi MC, Sun Y, Maggs L, Ksander BR, Verdijk RM, Boland GM, Jenkins RW, Haq R, Jager MJ, Wang X, Ryeom S, Ferrone CR. Chimeric Antigen Receptor T Cell with an Inducible Caspase-9 Suicide Gene Eradicates Uveal Melanoma Liver Metastases via B7-H3 Targeting Clinical cancer research : an official journal of the American Association for Cancer Research 2024. PMID 38767611.
- Xiao W, Xu L, Wang J, Yu K, Xu B, Que Y, Zhao J, Pan Q, Gao C, Zhou P, Zhang X. FGFR4-specific CAR-T cells with inducible caspase-9 suicide gene as an approach to treat rhabdomyosarcoma Cancer gene therapy 2024. PMID 39183354.
- Tey SK. Adoptive T-cell therapy: adverse events and safety switches Clinical & translational immunology 2014. PMID 25505965.

grading management (7)

- Banerjee R, Amonoo HL, Barata A, Bhatt NS, Espinoza-Gutarra MR, Jayani-Kosarzycki RV, Katz H, Kennedy VE, Nawas M, Steineck A, Wanjiku C, Costanzo E, Cusatis RN, Knight JM, Schoemans H, Sidana S, Wood WA, Sung AD, Lee CJ, Hamilton BK. Assessing Quality of Life and Symptoms in Transplantation and CAR-T Recipients: Expert Panel Recommendations from the Survivorship Special Interest Group of ASTCT Transplantation and cellular therapy 2025. PMID 40609744.
- Betof Warner A, Hamid O, Komanduri K, Amaria R, Butler MO, Haanen J, Nikiforow S, Puzanov I, Sarnaik A, Bishop MR, Schoenfeld AJ. Expert consensus guidelines on management and best practices for tumor-infiltrating lymphocyte cell therapy Journal for immunotherapy of cancer 2024. PMID 38423748.

- Zweegman S, van de Donk NWCJ. Supportive care in myeloma-when treating the clone alone is not enough Hematology. American Society of Hematology. Education Program 2024. PMID 39643986.
- Granroth G, Rosenthal A, McCallen M, Coughlin C, Benson H, Palmer J, Castro JE, Munoz J. Supportive Care for Patients with Lymphoma Undergoing CAR-T-cell Therapy: the Advanced Practice Provider's Perspective Current oncology reports 2022. PMID 36336769.
- Moghanloo E, Mollanoori H, Talebi M, Pashangzadeh S, Faraji F, Hadjilooei F, Mahmoodzadeh H. Remote controlling of CAR-T cells and toxicity management: Molecular switches and next generation CARs Translational oncology 2021. PMID 33789222.
- Lee DW, Santomasso BD, Locke FL, Ghobadi A, Turtle CJ, Brudno JN, Maus MV, Park JH, Mead E, Pavletic S, Go WY, Eldjerou L, Gardner RA, Frey N, Curran KJ, Peggs K, Pasquini M, DiPersio JF, van den Brink MRM, Komanduri KV, Grupp SA, Neelapu SS. ASTCT Consensus Grading for Cytokine Release Syndrome and Neurologic Toxicity Associated with Immune Effector Cells Biology of blood and marrow transplantation : journal of the American Society for Blood and Marrow Transplantation 2019. PMID 30592986.
- Mahadeo KM, Khazal SJ, Abdel-Azim H, Fitzgerald JC, Taraseviciute A, Bollard CM, Tewari P, Duncan C, Traube C, McCall D, Steiner ME, Cheifetz IM, Lehmann LE, Mejia R, Slopis JM, Bajwa R, Kebriaei P, Martin PL, Moffet J, McArthur J, Petropoulos D, O'Hanlon Curry J, Featherston S, Foglesong J, Shoberu B, Gulbis A, Mireles ME, Hafemeister L, Nguyen C, Kapoor N, Rezvani K, Neelapu SS, Shpall EJ, Pediatric Acute Lung Injury and Sepsis Investigators (PALISI) Network. Management guidelines for paediatric patients receiving chimeric antigen receptor T cell therapy Nature reviews. Clinical oncology 2019. PMID 30082906.

hepatic (6)

- Fan Y, Guo X, Lyu J, Wang X, Xu F, Yan J, He Y. Hepatotoxicity in the Era of Precision Medicine: Toxicological Mechanisms and Management of Immune and Cell Therapy-Induced Liver Injury Medicinal research reviews 2026. PMID 42544021.
- Fettiplace A, Regev A, Kiazand A, Heinzel-Pleines U, Bahjat M, Garg A, Kikuchi L, Li H, Hamidi A, Alpers DH, Tillmann H, Larrey D, Di Bisceglie AM, Lewis JH. Hepatic Effects, Potential Drug-Induced Liver Injury, and Other Liver Safety Considerations of Chimeric Antigen Receptor T-Cell (CAR-T) Therapy in the New Era of Expanding Non-oncology Indications: Literature Review and Expert Consensus Drug safety 2026. PMID 41345813.
- Cash T, Marachelian A, DuBois SG, Chi YY, Baregamyan A, Groshen SG, Jonus HC, Shamirian A, Crowley M, Goodarzian F, Acharya PT, Pawel B, Erbe AK, Shahi A, Zaborek J, Kennedy E, Asgharzadeh S, Villablanca JG, Pinto N, Weiss BD, Mosse YP, Desai AV, Macy ME, Granger M, Vo KT, Sondel PM, Matthay KK, Park JR, Goldsmith KC. Phase I Study of 131I-Metaiodobenzylguanidine With Dinutuximab ± Vorinostat for Patients With Relapsed or Refractory Neuroblastoma: A New Approaches to Neuroblastoma Therapy Trial Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2025. PMID 40549985.
- Wang Y, Gu J, Chen G, Jiang Y, Xu Y, Huang X, Sun W, Gan J. Roles of inflammatory cytokines in the pathogenesis of hepatitis B virus-related acute-on-chronic liver failure and CAR-T therapy Virology journal 2025. PMID 40819081.
- Wang G, Wang J, Du R, Wang Y, Li Z. Toxicity Spectrum of Anti-GD2 Immunotherapy: A Real-World Study Leveraging the US Food and Drug Administration Adverse Event Reporting System Paediatric drugs 2024. PMID 38153627.
- Cupit-Link M, Federico SM. Treatment of High-Risk Neuroblastoma with Dinutuximab and Chemotherapy Administered in all Cycles of Induction Cancers 2023. PMID 37760578.